Elliptic Curves

— *EYNTKA* Series —

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June 29, 2025

Contents

0.0.1 Silverman Influence Note that currently this book is very heavily influenced by Silverman *Arithmetic's of Elliptic Curves*. It is my goal to eventually evolve these notes into something more personal that would better server my EYTNKA series. However, as these notes usually are also how I initially learn the subject I am going to unabashedly and heavily be influenced by Silverman until I have a firmer grasp of my intuitions on the field.

A *Diophantine equation* is a polynomial equation with integer (or rational) coefficients. A major problem posed by mathematicians is when such equations have integer/rational solutions. As seen in [5], the study of these equations in one variable $p(x)$ (of all degree's) is the study of (classical) algebraic number theory, and in [2] it was shown the study of the solutions over such polynomials culminates in *Artin's Reciprocity* which shows that the information about these equations is captured using reciprocity laws and the analysis over local fields.

The next step is to now consider question in two variables $p(x, y)$. The simplest example would be degree 1 curves:

$$ax + by = c \qquad a, b, c \in \mathbb{Z}\ (a \text{ or } b \neq 0)$$

Using linear algebra, such equation are quick to analyze to conclude what type of integer/rational solution they may poses. The next level is degree 2 equations, that is quadratic equations, which up to linear isomorphism (with rational coefficients) can be classed up to[1]:

$$\begin{aligned} ax^2 + by^2 &= c \text{ elliptic} \\ ax^2 - by^2 &= c \text{ hyperbolic} \\ ax + by^2 &= 0 \text{ parabolic} \end{aligned}$$

See [5, chapter 21] for more details. In [2], we showed that a quadratic polynomial has a rational solution if and only if it has a solution $(x, y) \in \mathbb{Q}_p^2$ for every prime p and has a solution over $\mathbb{R}$ (i.e. localization at the single infinite place). It shall be shown in ref:HERE that $3x^3 + 4y^3 + 5z^3 = 0$ has solution over each localization $\mathbb{Q}_p$ and $\mathbb{R}$, but no solutions over $\mathbb{Q}$ giving a counter-example to the Hasse Principle when increasing the genus. We shall show in chapter 5 we shall show that the rational points form a finitely generated abelian group $\mathbb{Z}^r \times T$ where T is a torsion group. For degree ≥ 2 (or more specifically, for genus ≥ 2), it shall be shown (Faltings's theorem, ref:HERE) that the set of rational points is *finite*

Once we get to degree 3 equations, we start running into complication. By [1, chapter 7], we see that these are the genus 1 k-varieties of dimension 1; the genus is the key impediment to the simpler categorization of the solutions. These curves are the topic of this book.

For historic reasons, genus 1 curves are called elliptic curves. The term "elliptic curve" has a somewhat misleading name, as these curves when embedded in $\mathbb{R}^2$ are not actually ellipses. The name originates from their connection to elliptic integrals, which were studied in the 18th and 19th centuries. Elliptic integrals arose when mathematicians attempted to calculate the arc length of an ellipse (hence the name elliptic integral), leading to integrals of the form $\int \frac{dx}{\sqrt{P(x)}}$ where $P(x)$ is a polynomial of degree 3 or 4 with no repeated roots[2]. Later, mathematicians like Abel and Jacobi discovered that these integrals could be inverted to give functions with remarkable properties. The resulting theory led to curves defined by equations of the form $y^2 = x^3 + ax + b$, which we now call elliptic curves (where it can be seen that integrating over a degree 2 curves gives a degree 3 curves). This naming convention persisted even as elliptic curves developed, and the naming convention has now cemented itself as a permanent historical quirk.

[1] The term elliptic in the processing should not be confused with the elliptic curves we shall be covering! The two share a name due to historical reasons

[2] To arrive at this, start with the arclength formula $\sqrt{1 + \frac{dy}{dx}^2}$, then substitute the equation for an ellipse with the major/minor axis on the x/y axis, then substitute $x = a\sin(\theta)$ (which will eliminate the fraction) and then $t = \sin(\theta)$ (which eliminate the trigonometric function), leaving a polynomial

Varieties and Curves

(here, will be a quick overview from other notes)

0.1 Varieties

This is a *very* speedy reminder of how varieties are defined in modern algebraic geometry.

Let $k = \overline{k}$ be an algebraically closed field. Then the set of points k^n will be denoted $\mathbb{A}_k^n$ (so that there is no confusing $\mathbb{A}_k^n$ with a vector-space). Any polynomial $f(x_1, ..., x_n)$ in n-variables can be defined as a function $\mathbb{A}_k^n \to \mathbb{A}_k^1$. The zero-set of these polynomials are closed under arbitrary intersection and finite unions, and $\emptyset$ as well as $\mathbb{A}_k^n$ are in this collection (given by $0, 1 \in k[x_1, ..., x_n]$) and hence forms a topology known as the *Zariski topology*. The closed sets in this space are called the *algebraic sets*. The Zariski topology has non-trivial irreducible sets. The irreducible sets are called *varieties*. See [5] for the important notions about varieties, including the global sections that correspond to them. By the Nullstellensatz, every maximal ideal of a global section ring of a variety corresponds to a point in the variety. This lays the foundation for considering more general rings as geometric objects. It was observed by Grothendeick that we should consider the collection of *prime ideals* as they operate better functorially (namely, the pre-image of a prime ideal is always prime). The collection of all prime ideals is the spectrum of a ring R. The Zariski topology is defined similarly. We then keep track of the global and local sections on open subsets via a sheaf known as the *structure sheaf*; a spectrum along with the structure sheaf is called an *affine scheme*. A *Scheme* is a (locally) ringed space $(X, \mathcal{O}_X)$ that is covered by open sets that are isomorphic to affine schemes. The dimension of a scheme is defined as the maximum chain of irreducible subsets (starting the indexing at 0). A curve and a surface is a one and two dimensional (irreducible) scheme respectively. A k-variety is a separated integral reduced k-scheme of finite type. Varieties defined in this way correspond to quasi-projective varieties in classical algebraic geometry. A proper variety is called a *complete variety*, where classical projective varieties (when interpreted as schemes) make up most examples of such varieties (though there do exist non-projective complete varieties, see for example the blow up example in [1]). This book focuses on 1-dimensional complete varieties with genus 1.

Let us now set some conventions, and for simplicity stick to classical algebraic geometry language. Let K be given (and recall assumed to be perfect). Then we shall call *affine n-space*:

$$\mathbb{A}^n = \mathbb{A}(\overline{K}) = \left\{P = (x_1, .., x_n) \;\middle|\; x_i \in \overline{K}\right\}$$

Notice that we are taking its algebraic closure; we can think of this as the more *geometric setting*. The set of K-rational points of $\mathbb{A}^n$ will be denoted:

$$\mathbb{A}^n(K) = \{P = (x_1, ..., x_n) \mid x_i \in K\}$$

There is a natural way of defining $\mathbb{A}^n(K)$: if $G_{\overline{K}/K} = \mathrm{Gal}_K(\overline{K})$ is the absolute galois group with respect to K, then there is a natural $G_{\overline{K}/K}$ action on $\mathbb{A}^n$

$$G_{\overline{K}/K} \curvearrowright \mathbb{A}^n \qquad \sigma \cdot P = P^\sigma = (x_1^\sigma, ..., x_n^\sigma)$$

We shall let $V \subseteq \mathbb{A}^n$ be our varieties (in particular, the coefficients of the defining polynomials belong to $\overline{K}$).Let V/K be a variety defined over the field K, and $V(K)$ be the set:

$$V(K) = V \cap \mathbb{A}^n(K)$$

Note the difference between the two: V/K takes on coefficients in K, while $V(K)$ allows for "solutions" over K. We shall often be looking at $V/\mathbb{Q}$ and ask for $V(K)$ for various fields (ex. $\mathbb{C}$, local fields, global fields, etc.) Note that $V(K)$ could also be defined as the points of V which are invariant under the $G_{\overline{K}/K}$ action:

$$V(K) = \left\{P \in V \;\middle|\; P^\sigma = P, \quad \text{for all } \sigma \in G_{\overline{K}/K}\right\}$$

0.2 Curves

We shall assume k is a perfect field (that every algebraic extension is separable), unless stated otherwise.

We summarize the important concepts from [1] and [5], with some extra details for the material that is covered later in [1] to alleviate some of the burden of pre-requisite knowledge. We shall consider a curve to be a one-dimensional k-scheme that is

- Noetherian (the natural finiteness condition)
- Integral (hence it is irreducible and reduced)
- Separated (no points are arbitrarily close to each other)

It was shown in [1] that all such schemes *must* be projective, and so we can simply our notion of curve by saying a curve is a projective variety of dimension 1 (we shall provide a proof in section 1.4 for elliptic curves). As C is irreducible, the rational field for C is well-defined, and is denoted $k(C)$. The notation C/k shall often be written to represent the curve over the field k, as this may be easier to reader as compact to $C(k)$ which is similar to the $k(C)$ notation.

If furthermore all local rings are regular, the curve is said to be a *regular curve* or *smooth curve*. In one dimension, regular local rings are DVRs. If C is a curve and $\mathcal{O}_p(C)$ is the local ring at a regular point p, we can define the discrete valuation ν or ord as:

$$\nu_p : \mathcal{O}_p(C) \to \mathbb{N} \cup \{\infty\} \qquad \nu_p(f) = \sup\left\{d \in \mathbb{Z} \;\middle|\; f \in \mathfrak{m}_p^d\right\}$$

This can be extended to $k(C)$ by defining $\nu_p(f/g) = \nu_p(f) - \nu_p(g)$, giving us:

$$\nu_p : k(C) \to \mathbb{Z} \cup \{\infty\}$$

Recall that the *uniformizer* for C at p is a function whose order of vanishing is 1, and it is unique up to multiplication by a unit. Using the discrete valuation, we can identify the zero's and poles of a function $f \in k(C)$, define the regular points when $\nu_p(f) \geq 0$, and prove that all but finitely many points are regular[3].

When working over 1-dimensional irreducible projective varieties, we have enough restriction to be able to define smooth curves by taking the Zariski topology on $k(C)$ on the set of DVR's that dominate k. Hence, smooth curves are characterized by the collection DVR's, and can even be defined in this way, with morphisms given by the map $\varphi^* : k(C_2) \to k(C_1)$ where $f \mapsto f \circ \varphi$ ([4])

For the purposes of working with curves over finite characteristic, the following lemma will prove useful:

Lemma 0.2.1: Curves and Uniformizers

Let C/k be a curve, $p \in C$ a point, and $t \in k(C)$ a uniformizer for the in p. Then $k(C)/k(t)$ is a finite separable field extension.

***Proof* :**
Let $p = \text{char}(k)$.

First, the extension is certainly finite by Noether's Normalization theorem as it is finitely generated over k and has transcendence degree 1 over k (as C is a curve) and $t \notin k$. What's let to show is that any $a \in k(C)$ is separable over $k(t)$.

As it is algebraic over $k(t)$, there is a polynomial equation $\varphi(t, x)$ such that $\varphi(t, a) = 0$. Without loss of generality we may assume φ is a minimal polynomial for a. Now, if φ contains a monomial $a_{ij}t^i x^j$ where $j \not\equiv 0 \bmod p$, then the x-derivative of φ is non-zero, hence x is separable over $k(t)$.

For the sake of contradiction, let's say this is not the case. Then we may re-write $\varphi(t, x) = \psi(t, x^p)$. Then as $\text{char}(k) = p$, we have that $\varphi(t, x)^p = \varphi(t^p, x^p)$. Note how it was important that k is a separable extension so that each coefficient of $\varphi(t, x)$ can be represented as the pth power of an element in an algebraic extension. Thus, if $a_{ij} = b_{ij}^p$, we can write:

$$\varphi(t, x) = \psi(t, x^p) = \sum_{k=0}^{p-1} \left(\sum_{i,j} b_{ijk} t^{ip} x^{jp} \right) t^k = \sum_{k=0}^{p-1} \varphi_k(t, x)^p t^k$$

Now, as φ is the minimal polynomial, $\varphi(t, a) = 0$. On the other hand, as t is the uniformizer at p, we have:

$$\nu_p(\varphi_k(t, a)^p t^k) = p\nu_p(\varphi(t, a)) + k\nu_p(t) \equiv k \bmod p$$

Thus, each term $\varphi_k(t, a)^p t^k$ *must* have a distinct order at p, and hence every term must vanish:

$$\varphi_0(t, a) = \varphi_1(t, a) = \cdots = \varphi_{p-1}(t, a) = 0$$

[3] A very simple but elegant proof of this fact is that the stalk at the generic point is locally free in some open neighbourhood and is isomorphic to the differential of the generic point, which means it is regular on an open dense subset, and in dimension 1 such a set is cofinite

[4] Though this characterization does not generalize to higher dimensions as the local valuation do not form a scheme, see (this link – commented because not compiling pdf) for more details

However, at least one of the terms $\varphi_k(t,x)$ must have an x term, which will then be a polynomial of degree *smaller* than the minimal polynomial ($\deg_x \varphi_k(t,x) \leq \frac{1}{p}\deg_x \varphi(t,x)$), contradicting minimality of $\varphi(t,x)$, as we sought to show.

Let us now look at maps between curves. Recall that a rational map $\varphi : C_1 \to V \subseteq \mathbb{P}^n_k$ is of the form $[f_0, f_1, ..., f_n]$ where the polynomials are subject to the restrictions given by the defining polynomials for V. Most of the time, we shall consider the case where the codomain is a curve C_2. Let $\deg\varphi = [k(C_1) : \varphi^*(k(C_2))]$ be the *degree* of φ. If we must keep track of the separable and inseparable extensions and their corresponding degree, we shall write $\deg_s \varphi$ and $\deg_i \varphi$ respectively. One thing that we left as an exercise in [1] that is worth pointing out is that we can define a map $\varphi_* : k(C_1) \to k(C_2)$ by doing the following: let $L = k(C_1)$ and $K = k(C_2)$, and $N_{L/K} : L \to K$ be the field norm. Then:

$$\varphi_* = (\varphi^*)^{-1} \circ N_{L/K}$$

This is the *pushforward*, as compared to φ^* which is the *pullback*.

If $V = \mathbb{P}^1_k$, then we have $[f_0, f_1]$ with no defining restrictions. If $f_1 \neq 0$, then we can write this map a $[f_0/f_1 : 1]$. If $f_1 = 0$, then $[f_0 : 0] = [1 : 0]$ and hence it is a constant function at infinity. Thus, all rational maps $C \to \mathbb{P}^1_k$ are in bijective correspondence with $k(C) \cup \{\infty\}$. When working with rational maps over curves, most points will be regular; indeed show that if $p \in C$ is smooth then φ is regular at p (exercise

Next, it was shown in [5] that maps to a projective variety is either surjective or constant, hence a regular morphism $C_1 \to C_2$ is either surjective or constant. When φ is non-constant, φ induce a map $\varphi^* : k(C_2) \to k(C_1)$. These maps between function fields and rational maps between polynomials have the following properties:

Proposition 0.2.2: Properties of Pullbacks of Morphisms

1. $k(C_1)$ is a finite extension of $\varphi^*(k(C_1))$
2. if $\iota : k(C_2) \to k(C_1)$ is a k-homomorphism, there is a unique non-constant rational map $\varphi : C_1 \to C_2$ such that $\varphi^* = \iota$
3. If $L \subseteq k(C_1)$ is a subfield of finite index containing k, then there exists a unique smooth curve C'/k up to k-isomorphism and a non-constant rational map $\varphi : C_1 \to C'$ such that $\varphi^*(k(C')) = L$
4. If $\varphi : C_1 \to C_2$ is a map of degree 1 (so $[k(C_1) : \varphi^*(k(C_2))] = 1$), then φ^* is an isomorphism

***Proof* :**
see [5]

These results that the category of smooth curves over k with morphisms being non-constant rational morphisms (which as smooth curves must be regular, can equivalently be characterized as surjective regular morphisms) is dual to the category whose objects are finitely generated extensions K/k of transcendence degree 1 where $K \cap \overline{k} = k$ whose morphisms are k-field homomorphisms.

An important notion for curves is the ramification at a point:

Definition 0.2.3: Ramification Index

Let $\varphi : C_1 \to C_2$ be a non-constant map of smooth curves, and $p \in C_1$ a point. Then the *ramification index* of φ at p is given by:

$$e_\varphi(p) = \nu_p(\varphi^*(t_{\varphi(p)}))$$

where $t_{\varphi(p)} \in k(C_2)$ is the uniformizer at $\varphi(p)$. If $e_\varphi(p) = 1$, we say that p is unramified, and ramified otherwise (i.e. if $e_\varphi(p) > 1$[a].). If φ is unramified at every point, we shall say that φ is *unramified*.

[a]The ramification will always be positive

The following facts about the ramification can be found in exercises in [5] or in [1, chapter 7]:

Proposition 0.2.4: Properties of Ramification Index

Let $\varphi : C_1 \to C_2$ be a non-constant map of smooth curves. Then:

- for every $q \in C_2$,
$$\sum_{p \in \varphi^{-1}(q)} e_\varphi(p) = \deg \varphi$$
- For all but finitely many $q \in C_2$,
$$\#\varphi^{-1}(q) = \deg_s \varphi$$
- If $\psi : C_2 \to C_2$ is another non-constant map of smooth curves, then for each $p \in C_1$
$$e_{\psi \circ \varphi}(p) = e_\varphi(p) \cdot e_\psi(\varphi(p))$$

Proof **:**
see [1]

By combining the first two bullet points, we get that a map $\varphi : C_1 \to C_2$ is unramified if and only if for all $q \in C_2$:

$$\#\varphi^{-1}(q) = \deg(\varphi)$$

Example 0.2.1: Ramification Index

Consider the map

$$\varphi : \mathbb{P}^1_k \to \mathbb{P}^1_k \qquad [x : y] \mapsto [x^3(x-y)^2 : y^5]$$

Verify that φ is ramified at $[0 : 1]$ and $[1 : 1]$ where:

$$e_\varphi([0:1]) = 3 \qquad \text{and} \qquad e_\varphi([1:1]) = 2$$

We see that $\deg \varphi = 5$.

0.2.1 Frobenius Map

The Frobenius map is a natural map in finite characteristic (the galois group of finite extensions of finite fields is generated by the Frobenius automorphism, but this is not necessarily the case for non-finite fields! It is at least always injective). Let $\text{char}(k) > 0$ and $q = p^r$. Then for any $f \in k[x]$, let $f^{(q)} \in k[x]$ represent the polynomial where each coefficient of f was raised to the power of q (which i still in $k[x]$ as the Frobenius automorphism was applied to each element). Consider then $C^{(q)}/k$ be the curve generated by:

$$I(C^{(q)}) = \langle f^{(q)} \; : \; f \in I(C) \rangle$$

Then the natural map:

$$\varphi : C \to C^{(q)} \qquad \varphi([x_0 : \cdots x_n]) = [x_0^q : \cdots : x_n^q]$$

is a regular morphism, since for every $\varphi(P)$:

$$\begin{aligned} f^{(q)}(\varphi(P)) &= f^{(q)}(x_0^q, ..., x_n^q) \\ &= (f(x_0, ..., x_n))^q && \text{char}(k) = p \\ &= 0^q = 0 && f(P) = 0 \end{aligned}$$

Proposition 0.2.5: Properties of Frobenius Morphism

Let k be a (perfect) field of characteristic $p > 0$, $q = p^r$, and C/k a curve. Let $\varphi : C \to C^{(q)}$ be the qth power Frobenius morphism. Then[a]

1. $\varphi^*(K^{(q)} = K(C)^q = \{f^q \mid f \in K(C)\}$
2. φ is purely inseparable
3. $\deg \varphi = q$

[a] Note that if k were not perfect, then (2) and (3) remain unchanged, but (1) requires modification

***Proof* :**

1. $$\varphi^* \left(\frac{f}{g} \right) = \frac{f(x_0^q, ..., x_n^q)}{g(x_0^q, ..., x_n^q)}$$

   Similarly, elements of $K(C)^q \subseteq K(C)$ are of the form:

   $$\frac{f(x_0, ..., x_n)^q}{g(x_0, ..., x_n)^q}$$

   As k is perfect, every of k is a qth power, hence:

   $$(k[x_0, ..., x_n])^q = k[x_0^q, ..., x_n^q]$$

   giving equality

2. immediate from above and field theory (see [5, chapter 17])

3. Taking a finite extension if necessary[a] take a smooth point $P \in C(k)$. Then let $t \in K(C)$ be a uniformized at P. Then $K(C)$ is separable over $K(t)$. This gives a tower:

$$\begin{tikzcd}
 & & K(C & & \\
 & & K(C)^q(t) \arrow[u] & & \\
K(t) \arrow[uurr, "\text{separable}"] \arrow[urr] & & & & K(C)^q \arrow[uull, "\text{purely inseparable}"'] \arrow[ull]
\end{tikzcd}$$

Recall that if L/F is a separable extension over a field of characteristic q, then $L = F(L^q)$ ([5]). We thus get that $k(C) = k(C)^q(t)$, thus by (1):

$$\deg \varphi = [k(C)^q(t) : k(C)^q]$$

Now, $t^q \in k(C)^q$, so in order to prove $\deg(\varphi) = q$, we must show $t^{q/p} \notin k(C)^q$. If $t^{q/p} = f^q$ for some $f \in k(C)$, then

$$\frac{p}{q} = \text{ord}_P(t^{q/p}) = q\text{ord}_P(f)$$

which is impossible: $\text{ord}_P(f)$ must be an integer, completing the proof.

[a] Or using the fact that smooth points form an open dense subset

Corollary 0.2.6: Factoring Morphisms Over Positive Characteristic

Let $\psi : C_1 \to C_2$ be a map of (smooth) curves over a field of characteristic $p > 0$. Then φ factors as

$$C_1 \xrightarrow{\varphi} C_1^{(q)} \xrightarrow{\lambda} C_2$$

where $q = \deg_i(\psi)$, φ is the q-th power Frobenius map, and λ is a separable map.

***Proof* :**
let K be the separable closure of $\psi^*(k(C_2)) \subseteq k(C_1)$. Then $k(C_1)/K$ is purely inserparble of degree q, hence $k(C_1)^1 \subseteq K$. Then by proposition 0.2.5

$$k(C_1)^q = \varphi^*(k(C_1^{(q)})) \qquad \text{and} \qquad k[k(C_1) : \varphi^*(k(C_1)^{(q)}] = q$$

By comparing degrees, it is immediate that $K\varphi^*(C_1^{(q)})$. Thus, there is a tower of fields:

$$k(C_1)/\varphi^*(k(C_1)^{(q)})/\psi^*(k(C_2))$$

and hence the correspondence morphisms

$$\begin{tikzcd}
C_1 \arrow[r, "\varphi"'] \arrow[rr, bend left, "\psi"] & C_1^{(q)} \arrow[r, "\lambda"'] & C_2
\end{tikzcd}$$

as we sought to show.

0.2.2 Divisors

Let C/k be a curve. Then its divisor group $\mathrm{Div}(C)$ is the free-abelian group generated by the points of C, i.e. any $D \in \mathrm{Div}(C)$ is a finite sum:

$$\sum_{p \in C} n_p(p)$$

for some finite choices of points $p \in C$ and integers $n_p \in \mathbb{Z}$. The degree of $D \in \mathrm{Div}(C)$ is the sum of the n_p. The divisors of degree zero over a curve C, denoted $\mathrm{Div}^0(C)$, are all divisors whose degree is 0. Note that $\mathrm{Gal}_k(\overline{k})$ acts on $\mathrm{Div}(C)$ (resp. $\mathrm{Div}^0(C)$) in the natural way:

$$D^\sigma = \sum_{p \in C} n_p(p^\sigma)$$

We would say that D is defined over k if $D^\sigma = D$ for all $\sigma \in \mathrm{Gal}_k(\overline{k})$. The group of divisors over k is defined to be $\mathrm{Div}_k(C)$ (and $\mathrm{Div}_k^0(C)$ for the degree 0 terms).

Assume now that C is smooth. Then for each $f \in \overline{k}(C)^*$, we can define at each point $p \in C$ the order $\mathrm{ord}_p(f)$. We thus can define:

$$\mathrm{div}(f) = \sum_{p \in C} \mathrm{ord}_p(f)(p)$$

which is a finite sum as f would have zero on all but finitely many points. Such as divisor is called a *principal divisor*. It should be checked that if $\sigma \in \mathrm{Gal}_k(\overline{k})$, then:

$$\mathrm{div}(f^\sigma) = (\mathrm{div}(f))^\sigma$$

and hence $\mathrm{div}(f) \in \mathrm{Div}_k(C)$. As ν_p is a valuation, we can define the group homomorphism:

$$\mathrm{Div} : \overline{k}(C) \to \mathrm{Div}(C)$$

This can be thought as being analogous as sending an element $a \in F/\mathbb{Q}$ of a number field to it's principal fractional ideal $(a) \in \mathcal{I}_{F/\mathbb{Q}}$. The principal divisors form an abelian subgroup, label it $\mathcal{P}(C)$, and hence we can define the *divisor class group* or *Picard Group* as:

$$\mathrm{Pic}(C) = \frac{\mathrm{Div}(C)}{\mathcal{P}(C)}$$

We shall denote $\mathrm{Pic}_K(C) \leqslant \mathrm{Pic}(C)$ the subgroup fixed by $\mathrm{Gal}_k(\overline{k})$. As we are assuming C is smooth, we have the following from [1]:

- $\mathrm{div}(f) = 0$ if and only if $f \in \overline{k}^*$: if we assume $\mathrm{div}(f) = 0$, it has no poles so $f : C \to \mathbb{P}^1_{\overline{k}}$ is not surjective, hence constant, hence $f \in \overline{k}$. The converse is immediate.
- $\deg(\mathrm{div}(f)) = 0$: This is a classical result stating that the number of poles and roots must match.
- If C is a curve, $\mathrm{Pic}(C) \cong \mathbb{Z}$ if and only if $C \to \mathbb{P}^1$ is an isomorphism. Indeed, the if $D = 0$, then it must be that $D = \mathrm{div}(f)$ namely since $D = \sum n_p(p)$, writing $p = [a_p : b_p]$ we get that:

$$D = \mathrm{div}\left(\prod (b_p x - a_p y)^{n_p}\right)$$

which from this we see the degree map gives $\deg : \mathrm{Pic}(C) \cong \mathbb{Z}$. Conversely, if $\mathrm{Pic}(C) \cong \mathbb{Z}$ (this has to due with the decomposition of $\mathrm{Pic}(C) \cong \mathrm{Pic}^0(C) \oplus \mathbb{Z}$ where we thus deduce $\mathrm{Pic}^0(C) \cong 0$ and get that the genus must be zero giving us a rational equivalence, which we may raise to an isomorphism)

Example 0.2.2: Divisor of Elliptic Curve

For an elliptic curve $y^2 = (x-e_1)(x-e_2)(x-e_3)$, let $p_i = (e_i, 0)$, and let p_∞ be the point at infinity. Then:

$$\mathrm{div}(x-e_i) = 2(p_i) - 2(p_\infty)$$
$$\mathrm{div}(y) = (p_1) + (p_2) + (p_3) - 3(p_\infty)$$

This shall soon tie to a natural group structure we can put on divisors, see chapter 1.

We would be remiss the connection between the Picard group and Class group from number theory was not mentioned. Let $\mathrm{Div}^0(C) \subseteq \mathrm{Div}(C)$ be the collection of degree 0 divisors. Then define $\mathrm{Pic}^0(C)$ to be the quotient of $\mathrm{Div}^0(C)$ by the degree 0 principal divisors, which as pointed out earlier is given by $\overline{k}(C)$. We thus get a natural exact sequence:

$$1 \to \overline{k} \hookrightarrow \overline{k}(C) \xrightarrow{\mathrm{Div}} \mathrm{Div}^0(C) \twoheadrightarrow \mathrm{Pic}^0(C) \to 0$$

which mirrors the following canonical exact sequence in number theory:

$$1 \to (\text{units of number ring}) \to k^* \to (\text{ frac. ideals}) \to (\text{ideal class group}) \to 1$$

Moving onto morphisms, if $\varphi : C_1 \to C_2$ is a non-constant map of smooth curves, from φ^* and φ_* we can define:

$$\begin{aligned} \varphi^* : \mathrm{Div}(C_2) &\to \mathrm{Div}(C_1) \varphi_* : \mathrm{Div}(C_1) \to \mathrm{Div}(C_2) \\ (q) &\mapsto \sum_{p \in \varphi^{-1}(q)} (e_\varphi(p))\,(p)(p) \mapsto (\varphi(p)) \end{aligned}$$

If $f \in \overline{k}(C)^*$ which is associated to $f : C \to \mathbb{P}^1$, then we get that:

$$\mathrm{div}(f) = f^*\left((0) - (\infty)\right)$$

The following basic properties from [1] ought to be known or reviewed:

Proposition 0.2.7: Properties of Divisors

1. $\deg(\varphi^* D) = (\deg \varphi)(\deg D) \quad$ for all $D \in \mathrm{Div}(C_2)$.
2. $\varphi^*(\mathrm{div}\, f) = \mathrm{div}(\varphi^* f) \quad$ for all $f \in \overline{K}(C_2)^*$.
3. $\deg(\varphi_* D) = \deg D \quad$ for all $D \in \mathrm{Div}(C_1)$.
4. $\varphi_*(\mathrm{div}\, f) = \mathrm{div}(\varphi_* f) \quad$ for all $f \in \overline{K}(C_1)^*$.
5. $\varphi_* \circ \varphi^*$ acts as multiplication by $\deg \varphi$ on $\mathrm{Div}(C_2)$.
6. φ^* and φ_* induce maps on the degree 0 Picard Groups.
7. If $\psi : C_2 \to C_3$ is another such map, then
$$(\psi \circ \varphi)^* = \varphi^* \circ \psi^* \quad \text{and} \quad (\psi \circ \varphi)_* = \psi_* \circ \varphi_*.$$

0.2.3 Differentials

Differentials on curves shall play the same role as differential forms from calculus, further be useful for determining when an algebraic map is separable (recall that there is a derivative condition to check when a polynomial creates a separable field extension?). The main difference between the differential-geometric case and the current algebraic case is that we may define differentials on non-smooth curves. This shall be a review of [1, chapter 5.5]

Definition 0.2.8: Differential Forms On A Curve

Let C be a (not necessarily smooth) curve. Then the (meromorphic) *differential forms* on C, denoted Ω_C, is the $\overline{k}(C)$-vector-space generated by symbols of the form dx for each $x \in \overline{k}(C)$ such that:

1. $d(x + y) = dx + dy$
2. $d(xy) = xdy + dx \cdot y$
3. $da = 0$ for all $a \in \overline{k}$

If $\varphi : C_1 \to C_2$ is a non-constant map of curves, then φ^* induces a map:

$$\varphi^* : \Omega_{C_2} \to \Omega_{C_1} \qquad \varphi^* \left(\sum_i f_i dx_i \right) = \sum_i (\varphi^* f_i)\, d(\varphi^* x_i)$$

Proposition 0.2.9: Properties of Differentials

Let C be a curve.

1. Ω_C is a 1-dimensional $\overline{K}(C)$-vector space.
2. Let $x \in \overline{K}(C)$. Then dx is a $\overline{K}(C)$-basis for Ω_C if and only if $\overline{K}(C)/\overline{K}(x)$ is a finite separable extension.
3. Let $\varphi : C_1 \to C_2$ be a nonconstant map of curves. Then φ is separable if and only if the map
$$\varphi^* : \Omega_{C_2} \longrightarrow \Omega_{C_1}$$
is injective (equivalently, nonzero).

***Proof* :**

1. See [1]
2. First, note that for dx to be a basis, x must be transcendental over $\overline{K}$ (otherwise $dx = 0$).

   For the $\Rightarrow$ direction, suppose dx is a $\overline{K}(C)$-basis for Ω_C. This means every differential form can be written as $f \cdot dx$ for some $f \in \overline{K}(C)$. If $\overline{K}(C)/\overline{K}(x)$ were inseparable, then there would exist an element $y \in \overline{K}(C)$ such that $dy = 0$ (in characteristic $p > 0$, this happens when $y = z^p$ for some z). But then dy cannot be expressed as $f \cdot dx$ for any f, contradicting our assumption that dx is a basis. Therefore, $\overline{K}(C)/\overline{K}(x)$ must be separable.

   Conversely, suppose $\overline{K}(C)/\overline{K}(x)$ is a finite separable extension. Let $\overline{K}(C) = \overline{K}(x)[y]/(P(y))$ where P is the minimal polynomial of y over $\overline{K}(x)$. Since the extension is separable, $P'(y) \neq 0$. For any element $f \in \overline{K}(C)$, we can write $df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy$. But from the relation $P(y) = 0$, we get $P'(y)dy + \sum_i \frac{\partial P}{\partial a_i}da_i = 0$ where a_i are the coefficients of P. Since $P'(y) \neq 0$, we can solve for dy in terms of dx. This means any differential form can be expressed as $f \cdot dx$ for some $f \in \overline{K}(C)$, making dx a basis for Ω_C
3. Using (1) and (2), choose $y \in \overline{K}(C_2)$ such that $\Omega_{C_2} = \overline{K}(C_2)\, dy$ and such that $\overline{K}(C_2)/\overline{K}(y)$ is a separable extension. Note that $\varphi^*\overline{K}(C_2)$ is then separable over $\varphi^*\overline{K}(y) = \overline{K}(\varphi^* y)$. Then:
$$\begin{aligned}\varphi^* \text{ is injective} &\iff d(\varphi^* y) \neq 0 \\ &\iff d(\varphi^* y) \text{ is a basis for } \Omega_{C_1} \text{ (from (a))}, \\ &\iff \overline{K}(C_1)/\overline{K}(\varphi^* y) \text{ is separable (from (b))}, \\ &\iff \overline{K}(C_1)/\varphi^*\overline{K}(C_2) \text{ is separable},\end{aligned}$$
where the last equivalence follows because $\varphi^*\overline{K}(C_2)/\overline{K}(\varphi^* y)$ is separable.

Example 0.2.3: Differential On Curves

1. For the projective line $\mathbb{P}^1_k$, the function field is $k(t)$ where t is a coordinate on the affine line. Any rational function on $\mathbb{P}^1_k$ can be written as $f(t) = \frac{p(t)}{q(t)}$ where p and q are polynomials. The differential forms on $\mathbb{P}^1_k$ are of the form $\omega = f(t)dt$ where $f(t) \in k(t)$.

   Since every differential form can be expressed in terms of dt, it forms a basis for $\Omega_{\mathbb{P}^1_k}$. Therefore, $\Omega_{\mathbb{P}^1_k}$ is a 1-dimensional $k(t)$-vector space.

   Consider the differential form $\omega = \frac{t^2+1}{t-3}dt$ on $\mathbb{P}^1_k$. This is expressed in terms of the basis element dt multiplied by the rational function $\frac{t^2+1}{t-3} \in k(t)$.

   For the circle defined by the equation $x^2 + y^2 = 1$, we examine its function field and differential forms.

2. From the equation $x^2 + y^2 = 1$, we can parameterize the circle using the rational parameterization:

$$x = \frac{1-t^2}{1+t^2}$$
$$y = \frac{2t}{1+t^2}$$

   where $t = \tan(\theta/2)$ and θ is the angle parameter.

   The function field $k(C) = k(x, y)$ with the relation $x^2 + y^2 = 1$ is isomorphic to $k(t)$.

   Taking differentials of the equation $x^2 + y^2 = 1$, we get:

$$2x\,dx + 2y\,dy = 0$$
$$\Rightarrow dy = -\frac{x}{y}dx \quad (\text{when } y \neq 0)$$

   This means any differential form can be expressed in terms of dx alone. Therefore, Ω_C is a 1-dimensional $k(C)$-vector space with basis $\{dx\}$.

   Using the parameterization, we can express the basic differential forms in terms of dt:

$$dx = \frac{d}{dt}\left(\frac{1-t^2}{1+t^2}\right)dt = \frac{-4t}{(1+t^2)^2}dt$$
$$dy = \frac{d}{dt}\left(\frac{2t}{1+t^2}\right)dt = \frac{2(1-t^2)}{(1+t^2)^2}dt$$

   Any differential form $\omega = f(x, y)dx + g(x, y)dy$ can be rewritten as $\omega = h(t)dt$ for some rational function $h(t)$, confirming that Ω_C is 1-dimensional with basis $\{dt\}$.

3. For the singular curve defined by the equation $x^2 - y^3 = 0$ (known as a cusp),the curve C has a singularity at the origin, but we can still analyze its function field. From the equation $x^2 = y^3$, we can parameterize the curve as:

$$x = t^3$$
$$y = t^2$$

where t is a parameter.

Using our parameterization, we can express the basic differential forms in terms of dt:

$$dx = \frac{d}{dt}(t^3)dt = 3t^2 dt$$
$$dy = \frac{d}{dt}(t^2)dt = 2tdt$$

This shows that any differential form can be expressed in terms of dt, making $\{dt\}$ a basis for Ω_C over $k(t) \cong k(C)$.

Consider the differential form $\omega = xdx + ydy$. Using our parameterization and the relations above:

$$\omega = t^3(3t^2 dt) + t^2(2tdt)$$
$$= (3t^5 + 2t^3)dt$$

demonstrating how any differential form can be written as $h(t)dt$ for some rational function $h(t)$.

Proposition 0.2.10: Differentials and Order

Let C be a curve, let $p \in C$, and let $t \in \bar{k}(C)$ be a uniformizer at p.

1. For every $\omega \in \Omega_C$ there exists a unique function $g \in \bar{k}(C)$, depending on ω and t, satisfying
$$\omega = gdt.$$
We denote g by ω/dt.
2. Let $f \in \bar{k}(C)$ be regular at p. Then df/dt is also regular at p.
3. Let $\omega \in \Omega_C$ with $\omega \neq 0$. The quantity
$$\operatorname{ord}_p(\omega/dt)$$
depends only on ω and p, independent of the choice of uniformizer t. We call this value the order of ω at p and denote it by $\operatorname{ord}_p(\omega)$.
4. Let $x, f \in \bar{k}(C)$ with $x(p) = 0$, and let $P = \operatorname{char} k$. Then
$$\operatorname{ord}_p(fdx) = \operatorname{ord}_p(f) + \operatorname{ord}_p(x) - 1, \qquad \text{if } P = 0 \text{ or } P \nmid \operatorname{ord}_p(x)$$
$$\operatorname{ord}_p(fdx) \geq \operatorname{ord}_p(f) + \operatorname{ord}_p(x), \qquad \text{if } P > 0 \text{ and } P \big| \operatorname{ord}_p(x)$$
5. Let $\omega \in \Omega_C$ with $\omega \neq 0$. Then
$$\operatorname{ord}_p(\omega) = 0 \quad \text{for all but finitely many } p \in C.$$

***Proof* :**
see [8]

From this, we can define:

Definition 0.2.11: Divisor of Differential

Let $\omega \in \Omega_C$> Then the *divisor associated to* ω is:

$$\operatorname{div} = \sum_{p \in C} \operatorname{ord}_p(\omega)(p) \in \operatorname{Div}(C)$$

the differential $\omega \in \Omega_C$ is *regular* (or *holomorphic*) if:

$$\operatorname{ord}_p(\omega) \geq 0 \qquad \forall p \in C$$

and *non-vanishing* if

$$\operatorname{ord}_p(\omega) \leq 0 \qquad \forall p \in C$$

and thus it is holomorphic and non-vanishing if $\operatorname{ord}_p(\omega) = 0$ for all $p \in C$.

Note that if $\omega_1, \omega_2 \in \Omega_C$ are nonzero differentials, then proposition 0.2.9(1) implies that there must exist a $f \in \overline{k}(C)^*$ such that $\omega_1 = f\omega_2$. By proposition 0.2.10 this implies:

$$\operatorname{div}(\omega_1) = \operatorname{div}(f) + \operatorname{div}(\omega_2)$$

In particular:

$$\operatorname{div}(\omega_1) - \operatorname{div}(\omega_2) = \operatorname{div}(f)$$

showing that divisor differ up to a principal divisor! We thus have the following well-defined notion:

Definition 0.2.12: Canonical Divisor

Let C be a curve. Then the *canonical divisor* on C is the image $\operatorname{div}(\omega)$ in $\operatorname{Pic}(C)$ for any nonzero differential $\omega \in \Omega_C$. Any divisor in this class is called the *canonical divisor*.

Example 0.2.4: Canonical Divisors

1. Let $C = \mathbb{P}^1$, and let t be a coordinate function. Then for all $\alpha \in \overline{k}$ the function $t - \alpha$ is a uniformizer at α and so:
$$\operatorname{ord}_\alpha(dt) = \operatorname{ord}_\alpha(d(t-\alpha)) = 0$$
However, at $\infty \in \mathbb{P}^1$, we need to use a function like $1/t$ as our uniformizer, hence:
$$\operatorname{ord}_\infty(dt) = \operatorname{ord}_\infty\left(-t^2 d\left(\frac{1}{t}\right)\right) = -2$$
Thus,
$$\operatorname{div}(dt) = -2(\infty)$$
Note that this shows that dt is *not* holomorphic. Then, for any $\omega \in \Omega_{\mathbb{P}^1}$ by proposition 0.2.10 we have:
$$\deg \operatorname{div}(\omega) = \deg \operatorname{div}(dt) = -2$$

and so no ω can be holomorphic

2. Let $C = y^2 = (x - e_1)(x - e_2)(x - e_3)$. Letting $p_i = (e_i, 0)$, we can compute to get:

$$\operatorname{div}(dx) = (p_1) + (p_2) + (p_3) - 3(p_\infty)$$

where the $(p_1), (p_2), (p_3)$ come from how $dx = d(x - e_i)$ and the $-3(p_\infty)$ comes form:

$$dx = -x^2 d\left(\frac{1}{x}\right) \qquad \Rightarrow \qquad \operatorname{ord}_{p_\infty}(dx) = -2 - 1 = -3$$

Thus, combining withdiv$(y) = (p_1) + (p_2) + (p_3) - 3(p_\infty)$, we can construct a non-vanishing holomorphic form:

$$\operatorname{div}\left(\frac{dx}{y}\right) = 0$$

0.2.4 Riemann-Roch Theorem

So what is the invariant information that the canonical divisor provides? The *Riemann-Roch* theorem provides this information: its *degree* shall be invariant, and shall be deterministically be tied to the *genus* of the underlying curve. The Riemann Roch Theorem gives us a way to use algebraic tools to measure the genus of a curve.

Definition 0.2.13: Effective Divisor

Let C be a curve and $D = \sum_p n_p(p)$ be a divisor. Then D is an *effective divisor* or *positive divisor* if $n_p \geq 0$ for every $p \in C$. In this case, we write

$$D \geq 0$$

If D_1, D_2 are two divisors, we write $D_1 \geq D_2$ is the divisor $(D_1 - D_2)$ is positive.

Intuitively, every $f \in \mathcal{L}(D)$ can have poles only at points where D has negative coefficients (that is, there is a bound on the poles). As an example, let $f \in \overline{k}(C)^*$ be a function that is regular at all points except one, where it has a pole of degree at most n at p. Then we can write:

$$\operatorname{div}(f) \geq -n(p)$$

Definition 0.2.14: Riemann-Roch Space For Divisor

Let $D \in \operatorname{Div}(C)$ be a divisor. Then the *Riemann Roch space* associated to D is the space:

$$\mathcal{L}(D) = \left\{ f \in \overline{k}(C)^* \;\middle|\; \operatorname{div}(f) \geq -D \right\} \cup \{0\}$$

Furthermore, label:

$$\ell(D) = \dim_{\overline{k}} \mathcal{L}(D)$$

We can think of $\mathcal{L}(D)$ has being the collection of (meromorphic) functions that are allowed to have poles at the divisor D.

Proposition 0.2.15: Properties of Riemann-Roch Space

Let $D \in \mathrm{Div}(C)$.

1. If $\deg D < 0$, then
$$\mathcal{L}(D) = \{0\} \quad \text{and} \quad \ell(D) = 0.$$
2. $\mathcal{L}(D)$ is a finite-dimensional $\bar{K}$-vector space.
3. If $D' \in \mathrm{Div}(C)$ is linearly equivalent to D, then
$$\mathcal{L}(D) \cong \mathcal{L}(D'), \quad \text{and so} \quad \ell(D) = \ell(D').$$

Proof **:**
see [8]

From these, we can interpret $\mathcal{L}(K_C)$ where $K_C \in \mathrm{Div}(C)$ is the canonical divisor (so $K_C = \mathrm{div}(\omega)$ for some appropriate choice of ω). By definition we have that $f \in \mathcal{L}(K_C)$ is:

$$\mathrm{div}(f) \geq -\mathrm{div}(\omega)$$

Thus $\mathrm{div}(f\omega) \geq 0$, i.e. $f\omega$ is holomorphic. Then certainly by definition if $f\omega$ is holomorphic then $\mathrm{div}(f\omega) \geq 0$ so $f \in \mathcal{L}(K_C)$. As K_C is canonical, every differential form on C can be represented as $f\omega$, hence we have a $\bar{k}$-vector space isomorphism:

$$\mathcal{L}(K_C) \cong \{\omega \in \Omega_C \mid \omega \text{ is holomorphic}\}$$

The dimension $\ell(K_C)$ forms an important invariant of our curve in the *Riemann Roch equation*:

Theorem 0.2.16: Riemann-Roch Theorem

let C be a curve and K_C the canonical divisor on C. Then there exists an integer $g \geq 0$, called the *genus* of C such that for every divisor $D \in \mathrm{Div}(C)$:

$$\ell(D) - \ell(K_C - D) = \deg(D) - g + 1$$

Proof **:**
see [1, chapter 7.1]

Corollary 0.2.17: Canonical Divisor and Genus

Let K_C be the canonical divisor for a curve C. Then:

1. $\ell(K_C) = g$
2. $\deg(K_C) = 2g - 2$
3. if $\deg D > 2g - 2$ then $\ell(D) = \deg(D) - g + 1$

***Proof* :**
immediate

Example 0.2.5: Riemann-Roch Application

1. Let $C = \mathbb{P}^1_k$. Then we saw in example 0.2.4 that there are no holomorphic differentials on C, and hence $\ell(K_C) = 0$. Thus, we get from the Riemann Roch Theorem that:

$$\ell(D) - \ell(-2(\infty) - D) = \deg(D) + 1$$

   if $D \geq -1$, then:

$$\ell(D) = \deg(D) + 1$$

2. Let $C = y^2 = (x - e_1)(x - e_2)(x - e_3)$. Then in example 0.2.4 we had:

$$\operatorname{div}\left(\frac{dx}{y}\right) = 0$$

   and hence by proposition 0.2.10 we have that $K_C = 0$. Thus, we can compute:

$$g = \ell(K_C) = \ell(0) = 1$$

   Hence, C has genus 1. Given $D \geq 1$, we further get that:

$$\ell(D) = \deg(D)$$

   Let us use this to compute some special cases of D:

   (a) Let $p \in C$ be some point. Then by the above it must be that $\ell((p)) = 1$, however $\mathcal{L}((p))$ certainly contains the constant functions, but these have no poles, hence no functions on C have a single simple point

   (b) If we take the point at infinity p_∞, then $\ell(2(p_\infty)) = 2$. The set $\{1, x\}$ forms a basis for this set.

   (c) Similarly, for $\ell(3(p_\infty))$ and $\ell(4(p_\infty))$, we have a basis $\{1, x, y\}$ and $\{1, x, y, x^2\}$.

   (d) For $\mathcal{L}(6(p_\infty))$, note that $\{1, x, y, x^2, xy, x^3, y^2\}$ is a subset, but by our above formula we have $\ell(6(p_\infty)) = 6$. Thus, that set is *linearly dependent*, which we can also see by using the defining elliptic equation.

Next, we should take a look at non-algebraically closed fields.

Proposition 0.2.18: Riemann Roch over a field K

Let C/K be a smooth curve and let $D \in \operatorname{Div}_K(C)$. Then $\mathcal{L}(D)$ has a basis consisting of functions in $K(C)$.

***Proof* :**
Since D is defined over K, we have

$$f^\sigma \in \mathcal{L}(D^\sigma) = \mathcal{L}(D) \quad \text{for all } f \in \mathcal{L}(D) \text{ and all } \sigma \in G_{\bar{K}/K}$$

Thus $G_{\bar{K}/K}$ acts on $\mathcal{L}(D)$. Then by the following lemma (lemma 0.2.19), we get the desired result, completing the proof.

Lemma 0.2.19: Galois Action and Invariant Basis

Let C/K be a smooth curve and let $D \in Div_k(C)$. Then $\mathcal{L}(D)$ has a basis consisting of functions in $k(C)$

***Proof* :**

Finally, we shall take advantage of the following theorem from [1] for the relations between maps between curves and genus:

Theorem 0.2.20: Hurwitz Theorem

Let $\varphi : C_1 \to C_2$ be a non-constant separable map of smooth curves of genus g_1, g_2. Then:

$$2g_1 - 2 \geq \deg(\varphi)(2g_2 - 2) + \sum_{p \in C_1} (e_\varphi(p) - 1)$$

where equality holds if and only if either:

1. $\mathrm{char}(k) = 0$, or
2. $\mathrm{char}(k) = p > 0$ and $p \nmid e_\varphi(p)$ for all $p \in C_1$

***Proof* :**
see [1]. (or look at [8, p.37])

0.2.5 A word on hyper-elliptic curves

Silverman p.22

Exercise 0.2.1

1. Let C be a curve and $\varphi : C \to V \subseteq \mathbb{P}^n_k$. Show that if $p \in C$ is smooth then φ is regular at p. Thus if C is smooth, φ must be regular.

1

Geometry of Elliptic Curves

By the Riemann-Roch theorem, we can classify curves up to homeomorphism[1] by their genus. Genus 0 curves (i.e. rational curves) are readily understood, with their global geometric structure, their classification, and finding solutions over number fields. This chapter introduces the basic properties when we consider the next genus, genus 1 curves.

The chapter starts with a better understanding of the coordinate representation of elliptic curves, known as the *Weierstrass equations*, which give concrete information such as classifying isomorphism classes using the j-invariant, using the discriminant to find singular points, classifying singular points, and so forth. We then take a moment to represent elliptic curves geometrically by representing their construction within complex geometry. Then, using the fact that $\mathrm{Pic}^0(E) \cong E$ (ref:HERE), we can define a natural group on elliptic curves. From this, we tie together the equivalence between smooth Weierstrass equations and smooth genus 1 curves.

1.1 Weierstrass Equation

Let us write down an elliptic curve in coordinates. As shown in [1], any smooth curve is projective and can embed in $\mathbb{P}^n$. In [1, chapter 4] it was shown that a degree 3 curve can embed in $\mathbb{P}^2$. Thus, let us consider for a moment a general cubic in $\mathbb{P}^2$. Let us say that after an appropriate projective translation, our choice of base point is $[0 : 1 : 0]$. Then any elliptic curve can be written in coordinates as:

$$y^2z + a_1xyz + a_3yz^2 = x^3 + a_2x^2z + a_4xz^2 + a_6z^3 \tag{1.1}$$

[1] And sometimes isomophrism, like smooth genus 0 curves

For $a_1, ..., a_6 \in \overline{k}$. The above equation will be called the *Weierstrass equation* over $\overline{k}$. When dehomogenizing at z, we get:

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$$

where we appropriately re-labeled the terms. Given $\mathrm{char}(\overline{k}) \neq 2$, we can apply the transformation:

$$y \mapsto \frac{1}{2}(y - a_1, x - a_3)$$

giving us the familiar:

$$y^2 = 4x^3 + b_2x^2 + 2b_4x + b_6 \tag{1.2}$$

where:

$$b_2 = a_1^2 + 4a_2 \qquad b_4 = 2a_4 + a_1a_3 \qquad b_6 = a_3^2 + 4a_6$$

Given the above transformed curve and $\mathrm{char}(\overline{k}) \neq 2, 3$, we can do the transformation:

$$(x, y) \mapsto \left(\frac{x - 3b_2}{36}, \frac{y}{108}\right)$$

giving us a simpler equation of:

$$E: \ y^2 = x^3 - 27c_4x + 54c_6 \tag{1.3}$$

this is the simplest representation of a cubic curve (and thus of an elliptic curve) and hence we shall often write the above when considering an arbitrary cubic. Note it is only possible when the characteristic is not 2 or 3; if $\mathrm{char}(\overline{k}) \neq 2$ then equation (1.2) is the simplest, and if $\mathrm{char}(\overline{k}) = 2$ then there is little simplification that can be made. However, we shall still need the case of characteristic 2 or 3 when reducing the Weierstrass equation over all primes p. In most proofs, we shall assume characteristic $\neq 2, 3$ unless it is key to the proof. Thus, we can write an elliptic equation as:

$$y^2 = x^3 + Ax + B$$

We shall often label such a curve curve E. We want to define invariants of this curve up to isomorphism based on their coefficients. For the details, again see [1], for now here is an overview: The most general admissible transformation preserving the Weierstrass form and keeping the base point $[0 : 1 : 0]$ fixed is:

$$x = u^2x' + r \qquad y = u^3y' + su^2x' + t$$

where $u \neq 0$ and r, s, t are constants. When we apply these transformations, the coefficients b_2, b_4, and b_6 change. However, certain more complex combinations exhibit interesting behavior. If the reader has done all the computations for the transformation, they would get that:

$$c_4 = b_2^2 - 24b_4 \qquad c_6 = -b_2^3 + 36b_2b_4 - 216b_6$$

Under the above transformations, c_4 and c_6 transform as follows:

$$c_4' = u^4c_4 \qquad c_6' = u^6c_6$$

We shall take advantage of these relations soon; to take full advantage of them we will need the discriminant. If $f(x)$ represents the right hand side of the equation, recall the *discriminant* of a curve is given by

$$\mathrm{Res}(f, f') = \begin{vmatrix} 4 & b_2 & 2b_4 & b_6 & 0 \\ 0 & 4 & b_2 & 2b_4 & b_6 \\ 12 & 2b_2 & 2b_4 & 0 & 0 \\ 0 & 12 & 2b_2 & 2b_4 & 0 \\ 0 & 0 & 12 & 2b_2 & 2b_4 \end{vmatrix}$$

Expanding the determinant, we get:

$$\Delta = -b_2^2 b_8 - 8b_4^3 - 27b_6^2 + 9b_2 b_4 b_6$$

Under our transformations, the discriminant transforms as:

$$\Delta' = u^{12}\Delta$$

Naturally, a zero discriminant means the curve has a singularity. A negative discriminant shall represent a “detached” component (in the euclidean topology) when embedding $E(\mathbb{R})$, a zero discriminant shall represent having a node or cusp when it , and a positive discriminant and shall be “one connected piece” (in the euclidean topology). Here are some examples from Silverman:

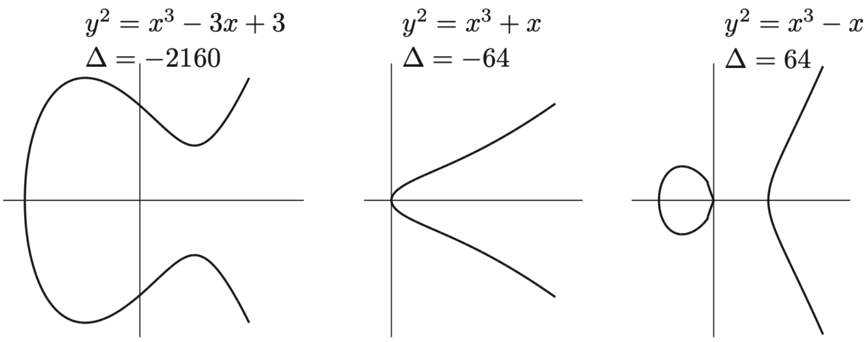


Figure 1.1: Suleiman: Three Elliptic Curves

If the discriminant is zero, the curve is singular and hence not elliptic. It is an exercise to show that any singular cubic curve is projectively equivalent to one of the following:

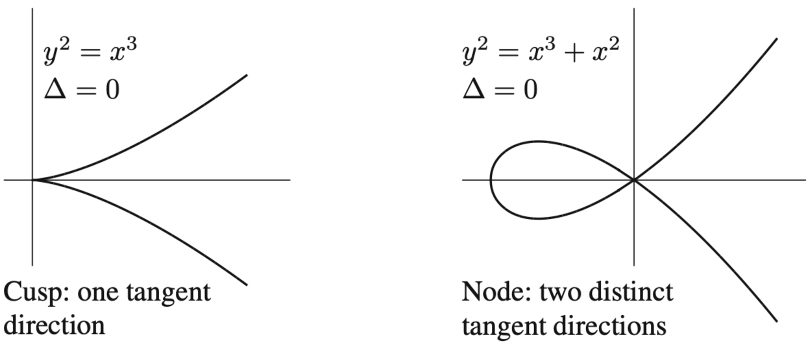


Figure 1.2: Silverman: singular Cubics

Let’s say the discriminant is zero, indicating we have a singularity. Let $f(x, y)$ equal the equation of moving everything to the left hand side of equation (1.1):

$$f(x,y) = y^2z + a_1xyz + a_3yz^2 - x^3 - a_2x^2z - a_4xz^2 - a_6z^3 = 0$$

Then if $p = (x_0, y_0)$ is a singular point, we must have:

$$\frac{\partial f}{\partial x}(p) = \frac{\partial f}{\partial y}(p) = 0$$

Then there exists $\alpha, \beta \in \overline{k}$ such that the Taylor series expansion around p has the form:

$$f(x,y) - f(x_0,y_0) = (y - y_0) - \alpha(x - x_0))\,(y - y_0) - \beta(x - x_0)) - (x - x_0)^3$$

Definition 1.1.1: Node and Cusp of Cubic

Let f be a singular cubic with singular point p, and α, β be as above. Then if $\alpha \neq \beta$, p is called a *node*, and if $\alpha = \beta$ then p is called a *cusp*. In the former case, the tangent lines are given by:

$$y - y_0 = \alpha(x - x_0) \qquad y - y_0 = \beta(x - x_0)$$

and in the latter case, it is a double-tangent line given by these two equations

Now, If the discriminant nonzero the transformation properties of c_4, c_6, and Δ suggest a natural invariant:

Definition 1.1.2: j-invariant

The j-invariant of an elliptic curve is defined as:

$$j = \frac{c_4^3}{\Delta}$$

where Δ is the discriminant.

given the equation $y^2 = x^3 + Ax + B$, we can write these quantities as:

$$\Delta = -16(4A^3 + 27B^2) \qquad j = -1728\frac{(4A)^3}{\Delta}$$

Then the only change of variable that makes this form invariant is:

$$x = u^2x' \qquad y = u^3y'$$

where $u \in \overline{k}^*$, from which we get:

$$u^4A' = A \qquad u^6B' = B \qquad u^12\Delta' = \Delta$$

Proposition 1.1.3: Properties of Discriminant and j-invariance

1. Let W be a Weierstrass equation. Then:
   (a) It is nonsingular if and only if $\Delta \neq 0$
   (b) it has a node if and only if $\Delta = 0$ and $c_4 \neq 0$
   (c) it has a cusp if and only if $\Delta = 0$ and $c_4 = 0$

   In the singular cases, there is only one singular point
2. Two elliptic curves are isomorphic over $\overline{k}$ if and only if they have the same j-invariant
3. For every $j_0 \in \overline{k}$[a], there exists an elliptic curve E_{j_0} (or to be precised $E_{j_0}(k(j_0))$) whose j-invariant is j_0

[a] Note this *includes* $j = 0$

Note that we are assuming that the base-point is $[0 : 1 : 0]$, i.e. that the only point that intersects the line at infinity $z = 0$ is this point.

Proof **:**

1. see exercise 1.11.1(1)
2. This is a lot of computation. Given

$$x = u^2x' + r \qquad y = u^3y' + su^2x' + t$$

Then:

$$\begin{aligned}
ua_1' &= a_1 + 2s\\
u^2a_2' &= a_2 - sa_1 + 3r - s^2\\
u^3a_3' &= a_3 + ra_1 + 2t\\
u^4a_4' &= a_4 - sa_3 + 2ra_2 - (t+rs)a_1 + 3r^2 - 2st\\
u^6a_6' &= a_6 + ra_4 + r^2a_2 + r^3 - ta_3 - t^2 - rta_1\\
\hline
u^2b_2' &= b_2 + 12r\\
u^4b_4' &= b_4 + rb_2 + 6r^2\\
u^6b_6' &= b_6 + 2rb_4 + r^2b_2 + 4r^3\\
u^8b_8' &= b_8 + 3rb_6 + 3r^2b_4 + r^3b_2 + 3r^4\\
\hline
u^4c_4' &= c_4\\
u^6c_6' &= c_6\\
u^{12}\Delta' &= \Delta\\
j' &= j\\
u^{-1}\omega' &= \omega
\end{aligned}$$

Figure 1.3: j-invariant computation

Or, if we already have c_4':

$$j' = \frac{(c_4')^3}{\Delta'} = \frac{(u^4c_4)^3}{u^{12}\Delta} = \frac{c_4^3}{\Delta} = j$$

For the converse, we shall assume $\mathrm{char}(\overline{k}) \geq 5$. Let E, E' be two elliptic curves with the same j-invariant with Weierstrass equations:

$$E : y^2 = x^3 + Ax + B$$
$$E' : y^2 = x^3 + A'x + B'$$

Then by assumption:

$$j(E) = \frac{(4A)^3}{4A^3 + 27B^2} = \frac{(4A')^3}{4(A')^3 + 27(B')^2} = j(E')$$

which simplifying gives us:

$$A^3(B')^2 = (A')^3B^2$$

Consider then an isomorphism $(x, y) = (u^2x', u^3y')$. Then:

(a) Case 1. $A = 0$ ($j = 0$). Then $B \neq 0$, since $\Delta \neq 0$, so $A' = 0$, and we obtain an isomorphism using $u = (B/B')^{1/6}$.

(b) Case 2. $B = 0$ ($j = 1728$). Then $A \neq 0$, so $B' = 0$, and we take $u = (A/A')^{1/4}$.

(c) Case 3. $AB \neq 0$ ($j \neq 0, 1728$). Then $A'B' \neq 0$, since if one of them were 0, then both of them would be 0, contradicting $\Delta \neq 0$. Taking $u = (A/A')^{1/4} = (B/B')^{1/6}$ gives the desired isomorphism.

3. Let $j_0 \in \overline{k}$. For a moment, assume $j \notin \{0, 1728\}$. Then take the curve:

$$E : y^2 + xy = x^3 - \frac{36}{j_0 - 1728}x - \frac{1}{j_0 - 1728}$$

Then:

$$\Delta = \frac{j_0^3}{(j_0 - 1728)^3} \qquad \text{and} \qquad j = j_0$$

Giving us the desired result when $j \notin \{0, 1728\}$. If $j = 0$ or $j = 1728$, take:

$$E : y^2 + y = x^3 \qquad \Delta = -27,\ j = 0$$
$$E : y^2 + y = x^3 + x \qquad \Delta = -64,\ j = 1728$$

Note these are both non-singular. In characteristic 2 or 3, $1728 = 2^6 \cdot 3^3 = 0$; in these cases one of these two curves will be nonsingular and fill in the missing j-value.

Furthermore, we can find a j for each element of $\overline{k}^*$, and hence $\overline{k}^*$ is the moduli space classifying elliptic curves.

Next, let's look at differential forms on Weierstrass equations. When in the form on equation (1.3), we can represent the canonical differential form as:

$$\omega = \frac{dx}{2y}$$

Naturally in this case, we require $\mathrm{char}\overline{k} \neq 2, 3$. For a general Weierstrass equation, we can represented it as:

$$\omega = \frac{dx}{2y + a_1x + a_3} = \frac{dy}{3x^2 + 2a_2x + a_4 - a_1y}$$

The key property of this form is that it is *holomorphic* and *nowhere-vanishing* (contrast this to the canonical form on $\mathbb{P}^1$ which is meromorphic with a pole of order 2 at ∞, and forms on curves of genus > 1 which will always vanish at at-least one point!). We shall prove this in after boxing the definition:

Definition 1.1.4: Invariant Differential Form

Let ω be defined as above. Then this form is called the *invariant differential form.*

We have the following important property:

Proposition 1.1.5: Invariant Form Holomorphic and Nonvanishing

Let E be an elliptic curve. Then the invariant differential form ω associated to the Weierstrass equation of E is holomorphic and nonvanishing, namely $\mathrm{div}(\omega) = 0$

***Proof* :**
Let $p = (x_0, y_0) \in E$ and

$$E : F(x, y) = y^2 + a_1xy + a_3y - x^3 - a_2x^2 - a_4x - a_6 = 0,$$

so

$$\omega = \frac{d(x - x_0)}{F_y(x, y)} = -\frac{d(y - y_0)}{F_x(x, y)}.$$

Thus p cannot be a pole of ω, since otherwise $F_y(p) = F_x(p) = 0$, which would say that p is a singular point of E. The map

$$E \longrightarrow \mathbb{P}^1, \quad [x, y, 1] \longmapsto [x, 1],$$

is of degree 2, so $\mathrm{ord}_p(x - x_0) \leq 2$, and we have equality $\mathrm{ord}_p(x - x_0) = 2$ if and only if the quadratic polynomial $F(x_0, y)$ has a double root. In other words, either $\mathrm{ord}_p(x - x_0) = 1$, or else $\mathrm{ord}_p(x - x_0) = 2$ and $F_y(x_0, y_0) = 0$. Thus in both cases, by proposition 0.2.10 we can compute:

$$\mathrm{ord}_p(\omega) = \mathrm{ord}_p(x - x_0) - \mathrm{ord}_p(F_y) - 1 = 0.$$

This shows that ω has no poles or zeros of the form (x_0, y_0), so it remains to check what happens at O.

Let t be a uniformizer at O. Since $\mathrm{ord}_O(x) = -2$ and $\mathrm{ord}_O(y) = -3$, we see that $x = t^{-2}f$ and $y = t^{-3}g$ for functions f and g satisfying $f(O) \neq 0, \infty$ and $g(O) \neq 0, \infty$. Now

$$\omega = \frac{dx}{F_y(x, y)} = \frac{-2t^{-3}f + t^{-2}f'}{2t^{-3}g + a_1t^{-2}f + a_3}dt = \frac{-2f + tf'}{2g + a_1tf + a_3t^3}dt.$$

Here we are writing $f' = df/dt$; cf. (II.4.3). In particular, (II.4.3b) tells us that f' is regular at O. Hence assuming that $\mathrm{char}(K) \neq 2$, the function

$$\frac{-2f + tf'}{2g + a_1tf + a_3t^3}$$

is regular and nonvanishing at O, and thus

$$\mathrm{ord}_O(\omega) = 0.$$

Finally, if $\mathrm{char}(K) = 2$, then the same result follows from a similar calculation using $\omega = -dy/F_x(x,y)$. We leave the details to the reader.

Note that from this we get a very simple proof that a singular Weierstrass equation is in fact rational:

Proposition 1.1.6: Singular Weierstrass then Rational Curve

Let E be the curve given by a Weierstrass equation that is singular. Then there eixsts a rational map $\varphi : E \dashrightarrow \mathbb{P}^1$ of degree 1 (i.e a birational morphism).

***Proof* :**
Making a linear change of variable, we may assume $(x,y) = (0,0)$ is singular. Then chekcing the partial derivatives, we get that the Weierstrass equation has the form:

$$y^2 + a_1xy = x^3 + a_2x^2$$

Then define:

$$E \to \mathbb{P}^1 \qquad (x,y) \mapsto [x:y]$$

which has degree one. Then it's rational inverse is given by:

$$\mathbb{P}^1 \to E \qquad [1:t] \mapsto (t^2 + a_1t - a_2,^3 + a_1t^2 - a_2t)$$

which is derived by taking $t = y/x$ and by dividing the Weierstrass equation of E by x^2 (which gives $t^2 + a_1t = x + a_2$). Then it is a matter of computation to check these are indeed inverses.

Next, the following is a very important representation of a Weierstrass equation:

Definition 1.1.7: Legendre Form

Let E be a Weierstrass equation. Then the *Legendre form* of the equation (if it exists) is:

$$y^2 = x(x-1)(x-\lambda)$$

Proposition 1.1.8: Properties of Legendre Form

Assume that $\operatorname{char}(k) \neq 2$. Then:

1. every elliptic curve $E/\overline{k}$ is isomorphic to an elliptic curve in Legendre form for some $\lambda \in \overline{k}$ where $\lambda \neq 0, 1$; denote the elliptic curve in Legendre from E_λ.
2. The j-invariant of E_λ is:
$$j(E_\lambda) = 2^8 \frac{(\lambda^2 - \lambda + 1)^2}{\lambda^2(\lambda - 1)^2}$$
3. The map:
$$\overline{k} \setminus \{0, 1\} \to \overline{k} \qquad \lambda \mapsto j(E_\lambda)$$
is surjective, in particular it is six-to-one ever where except $j = 0, 1728$, where it is two-to-one and three-to-one respectively. (unless $\operatorname{char}(k) = 3$, in which case $0 = 1728$ and the map is one-to-one).

Proof **:**

1. As $\operatorname{char}(k) \neq 2$, we can write the Weierstrass equation as:
$$y^2 = 4x^3 + b_2 x^2 + 2b_4 x + b_6$$
Doing the coordinate change $(x, y) = (x, 2y)$ and factoring the cubic, we get:
$$y^2 = (x - e_1)(x - e_2)(x - e_3)$$
for appropriate $e_1, e_2, e_3 \in \overline{k}$. Then as :
$$\Delta = 16(e_1 - e_2)^2(e_1 - e_3)^2(e_2 - e_3)^2 \neq 0$$
we see that each e_i must be distinct (or else we would get a singular point and a zero-discriminant). Now, do the substitution:
$$x = (e_2 - e_1)x' + e_1 \qquad y = (e_2 - e_1)^{3/2} y'$$
which puts the equation in Legendre form, where:
$$\lambda = \frac{e_3 - e_1}{e_2 - e_1} \in \overline{k} \qquad \lambda \neq 0, 1$$
2. This is a matter of (tedious) computation, and is left to do once in the readers life.
3. Let $j \in \overline{k}$. Then we know there exists an elliptic curve E such that $j = j(E)$. As elliptic curves have a Legendre form, we know that $j = j(E_\lambda)$ for some λ. To show it is 6-to-1 when $j \neq 0, 1728$, consider when $j(E_\mu) = j(E_\lambda)$ so that $E_\mu \cong E_\lambda$. Then their Weierstrass equation (and Legendre form) are related by a change of variables:
$$x = u^2 x' + r \qquad y = u^3 y'$$

Then equating the two:

$$x(x-1)(x-\mu) = \left(x + \frac{r}{u^2}\right)\left(x + \frac{r-1}{u^2}\right)\left(x + \frac{r-\lambda}{u^2}\right)$$

We see there are six ways of assigning the linear terms to one another, namely we can see that the six values for μ in terms of λ are:

$$\lambda,\ \frac{1}{\lambda},\ 1-\lambda,\ \frac{1}{1-\lambda},\ \frac{\lambda}{\lambda-1},\ \frac{\lambda-1}{\lambda}$$

Note that some of these values coincide when $j = 1728$ and $j = 0$, namely:

$$\lambda \in \left\{-1, 2, \frac{1}{2}\right\}$$

$$\lambda^2 - \lambda + 1 = 0$$

If k had characteristic 3, then the above values of λ would coincide and the equation $j(\lambda) = 0 = 1728$ would have the unique solution of $\lambda = -1$.

1.2 Weierstrass $\wp$-Function

We know elliptic curves are genus 1 complex curves. When taking it's analytification, we get a genus one complex manifold. Let us take a moment to examine elliptic curves from this perspective. The required complex analysis background can be found in [3], and an exposition on elliptic curves can be found in [3, chapter 6].

A *elliptic function* on $\mathbb{C}$ is a meromorphic function where $f(z + ae_i + be_j) = f(z)$ where e_i, e_j is some $\mathbb{Z}$-basis for a lattice in $\mathbb{C}$. The existence of such functions can be found in the above textbook, and these functions are generated by the *Weierstrass $\wp$-function*:

$$\wp(p) = \frac{1}{z^2} \sum_{\omega \in \Omega \setminus \{0\}} \left(\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2}\right)$$

Theorem 1.2.1: Elliptic Curve Of Weiestrass Function

Let $\wp$ be a Weiestrass $\wp$-function. Then there exists constants g_2 and g_3 such that

$$(\wp')^2 = 4\wp^3 - g_2\wp - g_3$$

This can be intuitively seen like so: since $\wp$ has a pole of order 2, $\wp'$ has a pole of order 3. Hence, there is some relation between $(\wp')^2$ and $\wp^3$ up to some massaging:

***Proof* :**
Recall that two doubly-periodic functions are equal up to a constant if they have the same poles and principal parts, and if then it may be shown that the constant is zero, the two functions are equal. This essentially means that we can construct many important identities by "cancelling

out the poles" so to speak. Thus, start with:

$$\wp(z) = \frac{1}{z^2} + a_2 z^2 + a_4 z^4 + \cdots$$

Then

$$\begin{aligned}
\wp(z) &= -2z^{-3} + 2a_2 z + 4a_4 z^3 + \cdots \\
\wp'(z) &= -2z^{-3} + 2a_2 z + 4a_4 z^3 + \cdots \\
\wp'(z)^2 &= 4z^{-6} + 8a_2 z^2 - 16a_4 + \cdots \\
\wp(z)^3 &= z^{-6} + 3a_2 z^{-2} + 3a_4
\end{aligned}$$

Thus:

$$\wp'(z)^2 - 4\wp(z)^3 = -20a_2 z^{-2} - 28a_4 + z^2(\cdots)$$

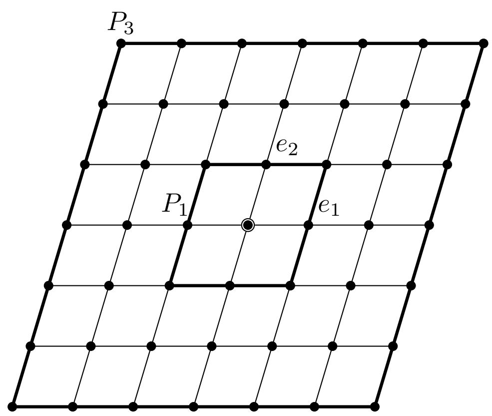
Thus, we get that the function$\wp'(z)^2 - 4\wp(z)^3 + 20a_2\wp(z) + 28a_4$ is holomorphic in some neighbourhood of the origin and hence has no poles and is must be holomorphic and hence constant as it is bounded. As it is zero at the origin, it must be zero. We thus get:

$$\wp'(z)^2 = 4\wp(z)^3 - 20a_2\wp(z) - 28a_4$$

which can be thought of as plugging in $(\wp, \wp')$ into the algebraic curve:

$$y^2 = 4x^3 - 20a_2 x - 28a_4$$

As all elliptic functions are doubly periodic, they create a repeating tiling of $\mathbb{C}$ called the *fundamental domain*:



The fundamental domain can be glued tougher long it's edges, showing that $\wp$ (an elliptic functions in general) are defined on the tori.

1.3 Group Law on Cubics

I want to now move this discussion to 1.4.

Let E be an elliptic curve with distinguished point O, which we think at the moment in abstract terms: it is a smooth genus-1 dimension-1 $\overline{k}$-scheme. It can be represented as a fundamental domain, namely in E^{an} we can think of E as being defined over $\mathbb{C}$. Given a choice of two points p, q in the fundamental domain, we can associated them in $\text{Pic}^0(E)$ to the divisors $[p] - [O]$ and $[q] - [O]$. In fact, *every* element of $\text{Pic}^0(E)$ is associated to a point on E (see [1, chapter 5.3]). Furthermore, $[p] - [O]$ for any point $p \in E$ is not a principal divisor (any principal divisor would need a pole of degree at least 2).

Thus, we see that the points of E and $\text{Pic}^0(E)$ are in bijective correspondence. But $\text{Pic}^0(E)$ is a group! In particular, it can be shown that we get a linearly equivalent:

$$([p] - [O]) + ([q] - [O]) \sim [p] + [q] - [O] - [P + Q]$$

sort this out more properly. Notice that the right hand side can also be given by a principal divisor.

Now, if we put it into coordinates, consider an elliptic curve in $\mathbb{P}^2$. Let ℓ be a line in $\mathbb{P}^2$. By Bézout's theorem, this line must intersect 3 points. Consider the de-homogenized line, then:

$$\text{Div}\left(\frac{\ell}{z}\right) = 3[O] - [Q_1] - [Q_1] - [Q_1]$$

We thus see a concrete representation of the group law on elliptic curves:

Definition 1.3.1: Group Law Via Lines

Let E be an elliptic curve and $p, q \in E$. Then the line L passing through p, q must intersect a third point R (if $p = q$, take the tangent line). Let L' be the line through R and O. Then L' intersects E at R, o, and a third point. this point is denoted $P \oplus Q$.

The following visuals are an example of the group law:

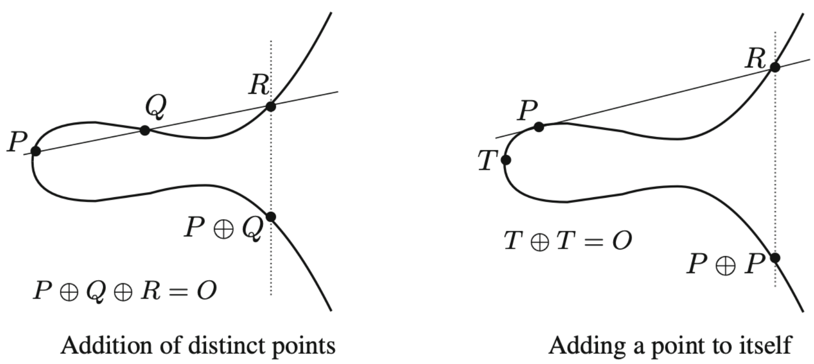


Figure 1.4: Silverman: group law examples

Proposition 1.3.2: $\oplus$ operation forms a Group

Let E be an elliptic curve with distinguished point O and with the group law given by $\oplus$. Then:

1. If a line L intersects E at the (not necessarily distinct) points P, Q, R, then:
$$(P \oplus Q) \oplus R = O$$
2. $P \oplus O = P$ for all $P \in E$
3. $P \oplus Q = Q \oplus P$ for all $P, Q \in E$
4. Let $P, Q, R \in E$. Then:
$$(P \oplus Q) \oplus R = P \oplus (Q \oplus R)$$
5. Suppose E is defined over K. Then E/K (along with the point at infinity) is a subgroup of E.

Proof **:**
see [5] for the full details.

Due to this, we shall just write $+$ and $-$ instead of $\oplus$ or $\ominus$. Furthermore, for any $m \in \mathbb{Z}$, we shall write (depending on whether m is positive, negative, or zero):

$$[m]P = P + \cdots P \qquad [m]P = -P - \cdots - P \qquad [0]P = O$$

here, Silverman derives an explicit construction for the group operation on a general elliptic curve, but I will omit this for now. There is one important corollary that I will put down but not prove yet.

Proposition 1.3.3: Group Law Constructed Explicitly

Silverman p. 53

Corollary 1.3.4: Even Functions On Elliptic Curves

A function $f \in \overline{k}(E)$ is even if $f(p) = f(-p)$ for all $p \in E$. Then f is even if and only if $f \in \overline{k}(x)$.

Proof **:**
Silverman p.54

Example 1.3.1: Group Law On Elliptic Curves
Silverman does $E/\mathbb{Q}$ with $E : y^2 = x^3 + 17$. You can practice this too with some explicit points. He first finds some points integer solutions by plugging in low values (less than 30) and find as few more via compeer search. He then shows some relations between then and finds some rational solutions.

He also claims that $E(\mathbb{Q})$ is isomorphic to $\mathbb{Z} \times \mathbb{Z}$ and that it is hard to prove, and that there are only 16 integral points (the 8 that he are listed and their negatives).

Proposition 1.3.5: Group Law On Nonsingular Points

here

Proof **:**
here

1.3.1 Singular Weierstrass Equations

The group law required defining the arithmetic properties of a genus 1 curve, and it is not immediately obvious that the group law exists on a singular curve representable by a singular Weierstrass equation. However, it still does exist given that we ignore the singular points. Singular Weierstrass equations will be important when we reduce elliptic curves $\bmod p$, i.e. over $\mathbb{F}_p$. If we have that $p \big| \Delta$, then the resulting elliptic curve will be singular, but we still want to be able to analyze it.

Definition 1.3.6: Nonsingular Part of Elliptic Curve

Let E be a (possibly singular) curve given by a Weierstrass equation. Then the *nonsingular part of* E, denoted E_{ns}, is the set of nonsingular points of E. Similarly, if E is defined over k, the n$E_{\text{ns}}(k)$ is the set of nonsingular points of $E(k)$.

Recall that if E is singular, then there is one singularity that is either a node or a cusp which is determined on whether $c_4 \neq 0$ or $c_4 = 0$.

Proposition 1.3.7: Nonsingular Points Form Abelian Group

Let D be a curve given by a Weierstrass equation with $\Delta = 0$. Then the composition law makes E_{ns} into an abelian group given the following conditions:

1. Suppose that E has a node ($c_4 \neq 0$) and let:

$$y = \alpha_1 x + \beta_q \qquad y = \alpha_2 x + \beta_2$$

be the distinct tangent lines of E at the singular point S. Then the map:

$$E_{\text{ns}} \to \overline{k}^* \qquad (x, y) \mapsto \frac{y - \alpha_1 x - \beta_1}{y = \alpha_2 x = \beta_2}$$

is an isomorphism of abelian groups

2. Suppose that E has a cusp ($c_4 = 0$) and let:

$$y = \alpha x = \beta$$

be the tangent line at the singular point $S = (x_0, y_0)$. Then the map:

$$E_{\text{ns}} \to \overline{k}^+ \qquad (x, y) \mapsto \frac{x - x_0}{y - \alpha x - \beta}$$

is an isomorphism of abelian groups.

Proof **:**
I'm leaving the proof alone for now, see p.56 of Silverman.

1.4 Elliptic Curves in Coordinates, and Group law on Elliptic Curves

Definition 1.4.1: Elliptic Curve

An *elliptic curve* is a pair (E, O) where E is a nonsingular curve of genus one and $O \in E$. The elliptic curve E is *defined over* k, denoted E/k if E is defined over K and $O \in E(K)$.

Proposition 1.4.2: Elliptic Curve and Weierstrass Equation

Let (E, O) be an elliptic curve over k. Then:

1. there exists $x, y \in k(E)$ such that the map:
$$\varphi : E \to \mathbb{P}^2 \qquad \varphi = [x : y : 1]$$
gives an isomorphism from E/k onto a curve given by a Weierstrass equation:
$$C : Y^2 + a_1XY + a_3Y = X^3a_2X^2 + a_4X + a_6$$
where $a_1, ..., a_6 \in k$ and $\varphi(O) = [0 : 1 : 0]$. The functions x, y are called the *Weierstrass coordinates* for E/K.

2. Any two Weierstrass equations from (1) are equivalent up to a linear change of variables
$$X = u^2X' + r \qquad Y = u^3Y' + su^2X' + t$$
where $u \in k^*$ and $r, s, t \in k$.

3. Every smooth curve curve C/k given by a Weierstrass equation is an elliptic curve E/k with base point $O = [0 : 1 : 0]$.

***Proof* :**

1. Consider $\mathcal{L}(n(O))$ for $n \in \mathbb{N}_{>0}$. As $g = 1$, by corollary 0.2.17(3):
$$\ell(n(O)) = \dim \mathcal{L}(n(O)) = n \qquad \forall n \geq 1$$
For example, for $\mathcal{L}(2(O))$, we can choose a basis $\{1, x\}$ where $x \in \overline{k}(E)$ is some element that has a pole of order 2 at O, and for $\mathcal{L}(3(O))$ we can choose a basis $\{1, x, y\}$ where $y \in \overline{k}(E)$ is some other element of order 3 at O; we know these exists because we know the dimension of $\mathcal{L}(n(O))$.

   Now, $\mathcal{L}(6(O))$ is 6 dimensional, so the subset $\{1, x, y, x^2, xy, y^2, x^3\}$ with 7 elements must be linearly dependent:
$$A_1 + A_2x + A_3y + A_4x^2 + A_5xy + A_6y^2 + A_7x^3 = 0$$
where $A_1, ..., A_7 \in k$. Now, first notice that $A_6A_7 \neq 0$ or else every term would have a pole of different order at O, and then all the A_i's would vanish contradicting linear dependence (see exercise 1.11.1(2)). Thus, we can first replace x, y with $-A_6A_7x, A_6A_7^2y$ respectively and divide by $A_6^3A_7^4$ to deduce the cubic Weierstrass form from this equation. But then the image of the map
$$\varphi : E \to \mathbb{P}^2 \qquad \varphi = [x, y, 1]$$
lies in the locus of the Weierstrass equation; thus write $\varphi : E \to C$. Then certainly φ is non-constant, so by section 0.2 φ is a surjective morphism and $\varphi(O) = [0 : 1 : 0]$ (when normalizing in y, we get $(x/y, y/y, 1/y) = (0, 1, 0)$ as y has a higher order pole than x).

   To show φ is an isomorphism, we must show the maps is degree 1 and smooth. To show it's degree 1, it is equivalent to show that $k(E) = k(x, y)$. To that end, we take advantage of the tower property of composition. Take the map $[x:1]: E \to \mathbb{P}^1$. Then as x has a double pole at O and no other poles, this map has degree 2; hence $[k(E):k(x)] = 2$. Similarly for $[y:1] \to \mathbb{P}^2$ which has degree 3; hence $[k(E):k(y)] = 3$. Then $[k(E):k(x,y)]$ must divide 2 and 3, necessarily that implies $[k(E):k(x,y)] = 1$. For smoothness, let's show C is not singular. Suppose it is. Then by 1.1.6 there is a degree 1 rational map $\psi : C \to \mathbb{P}^1$. Then the composition $\psi \circ \varphi : E \to \mathbb{P}^1$ is a degree 1 map between smooth curves, and hence is an isomorphism. But that contradicts the fact that E has genus 1 and $\mathbb{P}^1$ has genus 0, hence C must be smooth, and as the map is degree 1 we have that $\varphi : E \to C$ is an isomorphism.

2. Consider $\{x, y\}$ and $\{x', y'\}$ to be two coordinate function for E where x, x' have poles of order 2 at O and y, y' have poles of order 3 at O. Then $\{1, x\}$ and $\{1, x'\}$ form a basis for $\mathcal{L}(2(O))$ and $\{1, x, y\}$ and $\{1, x', y'\}$ form a basis for $\mathcal{L}(3(O))$. Thus, we must have constants

$$u_1, u_2 \in k^* \qquad r, s_2, t \in k$$

for the change of basis equations:

$$x = u_1 x' + r \qquad y u_2 y' + s_2 x' + t$$

Now, as (x, y) and (x', y') satisfy the Weierstrass equation (where the Y^2 and X^3 have terms with coefficient 1), we must that that:

$$u_1^3 = u_2^2$$

letting $u = u_2/u_1$ and $s = s_2/u^2$ gives the desires change of basis form.

3. Let E be a given smooth Weierstrass equation. Then by 1.1.5 we have the differential (in coordinates):

$$\omega = \frac{dx}{2y + a_1 x + a_3} \in \Omega_E$$

where $\operatorname{div}(\omega) = 0$. Then by the Riemann-Roch theorem, if g is the genus of E, we have:

$$2g - 2 = \deg(\operatorname{div}(\omega)) = 0$$

showing that E must have genus 1. Taking $[0:1:0]$ as the base point now makes E an elliptic curve, completing the proof.

1.4.1 Change of Basis over different fields Not all change of basis as described in proposition 1.4.2 must come from the same ground-field. Consider $y^2 = x^3 - x$. This has coefficients over $\mathbb{Q}$, but contains maps to itself with coefficients that are not in $\mathbb{Q}$: $x = -x'$ and $y = \sqrt{-1}y'$.

Note that as $k(E) = k(x, y)$, we can write any rational function on E in $\{x, y\}$ coordinates. Furthermore, we can think of E as being symmetric across the x axis when thinking of $U_1 \subseteq \mathbb{P}^2$ where $[x:y:1] \in U_1$, which leads credence to the concrete group law on elliptic curves. In fact, we can now define the group law abstractly on an elliptic curve:

Lemma 1.4.3: Picard 0 Group on elliptic Curves

Let C be a curve of genus 1 and let $p, q \in C$. Then:

$$(p) \sim (q) \qquad \text{if and only if} \qquad p = q$$

Proof **:**
If $p = q$ then $(p) - (q) = (p) - (p) = 0\text{div}(1)$, and hence $(p) \sim (q)$. Conversely, as $(p) \sim (q)$ then there exists a $f \in \overline{k}(C)^*$ such that

$$\text{div}(f) = (p) - (q)$$

Then by definition $f \in \mathcal{L}((q))$. By the Riemann-Roch theorem we must have:

$$\dim \mathcal{L}((q)) = 1$$

Hence $\mathcal{L}((q))$ is generated by constant functions, hence $f \in \overline{k}$, hence $\text{div}(f) = 0$, hence $p = q$ as we sought to show.

Proposition 1.4.4: Properties of Degree 0 Picard Group On Elliptic Curves

Let (E, O) be an elliptic curve. Then:

1. For every degree 0 divisor $D \in \text{Div}^0(E)$, there exists a unique point $p \in E$ such that

$$D \sim (p) - (O)$$

which gives the map:

$$\sigma : \text{Div}^0(E) \to E \qquad D \sim \text{“}(p) - (O)\text{''} \mapsto p$$

2. The map σ is surjective and induces a bijective map $\text{Pic}^0(E) \to E$.
3. Given a coordinate representation of E as a Weierstrass equation and the group law for cubics, then σ is a group isomorphism onto $\text{Pic}^0(E)$[a]

[a] $\text{Pic}^0(E)$ is naturally a group, being a subgroup of an abelian quotient group

Proof **:**

1. As E has genus 1, By Riemann Roch we deduce that:

$$\ell(D + (O)) = 1$$

Now consider $0 \neq f \in \overline{k}(E)$ where $f \in \mathcal{L}(D + (O))$; by the above f must be a basis for the vector space. As $f \in \mathcal{L}(D + (O))$:

$$\text{div}(f) \geq -D - (O) \qquad \text{and} \qquad \deg(\text{div}(f)) = 0$$

Hence HERE

$$\mathrm{div}(f) - D - (O) + (P) \qquad \text{i.e.} \qquad D \sim (P) - (O)$$

To show uniqueness, suppose $P' \in E$ such that

$$(P) \sim D + (O) \sim (P')$$

Then lemma 1.4.3 gives $P = P'$, giving uniqueness. Thus, we can define a map $\sigma : \mathrm{Div}^0(E) \to E$.

2. Let's show surjectivity, then injectivity up to degree 0 principal divisor. It is certainly surjective as for any $P \in E$,

$$\sigma((P) - (O)) = P$$

Next, if $D_1, D_2 \in \mathrm{Div}^0(E)$ and

$$P_1 = \sigma(D_1) = \sigma(D_2) = P_2$$

then

$$(P_1) - (P_2) \sim D_1 - D_2$$

giving us injectivity on Pic^0.

3. Let E be given by a Weierstrass equation. We must show that for any $P, Q \in E$. Define

$$\kappa : E \to \mathrm{Pic}^0(E) \qquad P \mapsto (\text{divisor class of}(P) - (O))$$

This map is a bijection by part (2); what's left to show is

$$\kappa(P+Q) = \kappa(P) + \kappa(Q) \qquad \text{equiv.} \qquad \kappa(P+Q) - \kappa(P) - \kappa(Q) = 0$$

Consider any line that goes through P, Q:

$$f(X,Y,Z) = \alpha X = \beta Y + \gamma Z = 0$$

By Bézouts theorem, it goes through a third point R. Now let:

$$f'(X,Y,Z) = \alpha' X + \beta' Y + \gamma' Z = 0$$

be the line passing through R and O. Now, by definition of the cubic group law and that the line $Z = 0$ intersects E with multiplicity 3, we get:

$$\mathrm{div}(f/Z) = (P) + (Q) + (R) - 3(O)$$
$$\mathrm{div}(f'/Z) = (R) + (P+Q) - 2(O)$$

Thus:

$$(P+Q) - (P) - (Q) + (O) = \mathrm{div}\left(\frac{f'}{f}\right) \sim 0$$

Thus:

$$\kappa(P+Q) - \kappa(P) - \kappa(Q) = 0 \qquad \text{i.e.} \qquad \kappa(P+Q) = \kappa(P) + \kappa(Q)$$

hence κ is a homomorphism, and with bijectivity we get the isomorphism, completing the proof.

Corollary 1.4.5: Elliptic Curve and Principal Divisors

Let E be an elliptic curve and $D = \sum n_P(P) \in \mathrm{Div}(E)$. Then D is a principal divisor if and only if

$$\sum_{P \in E} n_P = 0 \qquad \text{and} \qquad \sum_{P \in E} [n_P]P = O$$

***Proof* :**

$$D \sim 0 \Leftrightarrow \sigma(D) = O \Leftrightarrow \sum_{P \in E} \sigma((P) - (O)) \stackrel{!}{=} \sum_{P \in E} P = O$$

where $\stackrel{!}{=}$ comes from $\sigma((P) - (O)) = \sigma((P))$.

From proposition 1.4.4 we can define the exact sequence:

$$1 \to \overline{k}^* \to \overline{k}(E)^* \xrightarrow{\mathrm{div}} \mathrm{Div}^0(E) \xrightarrow{\sigma} E \to 0$$

which (by ref:HERE) is invariant under the galois action:

$$1 \to k^* \to k(E) \xrightarrow{\mathrm{div}} \mathrm{div}_k^0(E) \xrightarrow{\sigma} E/K \to 0$$

We shall return to this in section ref:HERE.

The next fundamentally important result is to show that the group homomorphism is furthermore a *regular morphism*; namely the addition map $+ : E \times E \to E$ is a morphism.

Theorem 1.4.6: Elliptic Curve Addition is A Morphism

Let E/k be an elliptic curve. Then the addition and subtraction maps

$$\begin{aligned} + : E \times E &\to E \\ (P_1, P_2) &\mapsto P_1 + P_1 \end{aligned} \qquad\qquad \begin{aligned} - : E &\to E \\ P &\mapsto -P \end{aligned}$$

are morphisms.

***Proof* :**

The negation is immediate: given $- : E \to E$ where $(x, y) \mapsto (x, -y - a_1 x - a_3)$ we see that it is a rational smooth map between smooth curves, and hence is a morphism.

For the addition, for any particular $Q \in E$ consider the map

$$\tau_Q : E \to E \qquad \tau(P) = P + Q$$

This is certainly a rational degree 1 map between smooth curves, hence a morphism (and even an isomorphism with inverse $P \mapsto P - Q$). For $+ : E \times E \to E$, we can readily see that the map is a morphism for all points except potentially

$$(P, P) \qquad (P, -P) \qquad (P, O) \qquad (O, P)$$

since for points *not* of the above form, we have that addition is a composition of the slope formula and the reflection formula:

$$\lambda = \frac{y_2 - y_1}{x_2 - x - 1} \qquad \text{and} \qquad \nu = \frac{y_1 x - 2 - y_2 x_1}{x_2 - x_1}$$

The exceptional cases can either be explicitly computed (exercise ref:HERE or [8, remark 3.6.1]), or we can take advantage of the property of the group law. Let τ_1 and τ_2 be translation maps for the points Q_1 and Q_2 respectively and take the composition of maps

$$\varphi : E \times E \xrightarrow{\tau_1 \times \tau_2} E \times E \xrightarrow{+} E \xrightarrow{\tau_1^{-1}} E \xrightarrow{\tau_2^{-1}} E$$

Since the group law on E is associative and commutative, the net effect of the above maps is as follows:

$$\begin{aligned} (P_1, P_2) &\xrightarrow{\tau_1 \times \tau_2} (P_1 + Q_1, P_2 + Q_2) \\ &\xrightarrow{+} P_1 + Q_1 + P_2 + Q_2 \\ &\xrightarrow{\tau_1^{-1}} P_1 + P_2 + Q_2 \\ &\xrightarrow{\tau_2^{-1}} P_1 + P_2 \end{aligned}$$

Thus, the rational map φ agrees with the addition map wherever they are both defined. Furthermore, as the τ_i's are isomorphisms, it follows that φ is a morphism except possibly at pairs of points of the form

$$(P - Q_1, P - Q_2), \quad (P - Q_1, -P - Q_2), \quad (P - Q_1, -Q_2), \quad (-Q_1, P - Q_2)$$

But Q_1 and Q_2 are arbitrary points! Hence, by varying Q_1 and Q_2, we can find a finite set of rational maps

$$\varphi_1, \varphi_2, \ldots, \varphi_n : E \times E \longrightarrow E$$

with the following properties:

1. φ_1 is the addition map
2. For each $(P_1, P_2) \in E \times E$, some φ_i is defined at (P_1, P_2).
3. If φ_i and φ_j are both defined at (P_1, P_2), then $\varphi_i(P_1, P_2) = \varphi_j(P_1, P_2)$.

It follows that addition is defined on all of $E \times E$, so it is a morphism, as we sought to show.

1.5 Elliptic Curve Morphisms: Isogenies

As the group law respects the morphism properties, we have that elliptic curves are within the category of *abelian varieties.* Morphisms of abelian varieties shall be given a name:

Definition 1.5.1: Isogeny

Let E_1, E_2 be elliptic curves (over $\overline{k}$). Then an *isogeny* $\varphi : E_1 \to E_2$ is a morphism satisfying $\varphi(O) = O$. Two elliptic curves are *isogenous* if there is an isogeny $\varphi : E_1 \to E_2$ where $\varphi(E_1) \neq \{O\}$.

The collection of all Isogenies between elliptic curve E_1, E_2 shall be denoted $\mathrm{Hom}_{\mathbf{Iso}[\overline{k}]}(E_1, E_2)$, or just $\mathrm{Hom}(E_1, E_2)$ when the morphisms are clear from context.

We have not explicitly defined the map to be a group homomorphism. Theorem 1.5.2 will show this suffices to prove it is a group homomorphism; an intriguing enough result that merits it being proven as a theorem rather than being part of the definition. As elliptic curves are curves, either φ maps to O or is surjective, hence besides the zero isogeny $[0](P) = O$, every other isogeny is a finite map of curves.

1.5.1 Isogenies on Abelian Varieties When working over curves, all non-trivial morphisms are finite maps. When working over general abelian varieties, it shall be necessary to specify that the *kernel* is finite.

Given the non-trivial case, we get the dual:

$$\varphi^* : \overline{k}(E_2) \to \overline{k}(E_1)$$

Note that

$$\deg(\psi \circ \varphi) = \deg(\psi)\deg(\varphi)$$

is preserved.

As Elliptic curves are abelian groups, $\mathrm{Hom}(E_1, E_2)$ is an abelian group, and $(\mathrm{End}(E), +, \circ)$ is a (not necessarily commutative) ring, and the invertible isogenies are denoted $\mathrm{Aut}(E)$. If we are working with elliptic curves over k, we can look only at isogenies over k. We shall denoted these isogenies as:

$$\mathrm{Hom}_k(E_1, E_2) \qquad \mathrm{End}_k(E) \qquad \mathrm{Aut}_k(E)$$

Theorem 1.5.2: Isogeny is Group Homomorphism

Let $\varphi : E_1 \to E_2$ be an isogeny. Then for all $P, Q \in E$

$$\varphi(P + Q) = \varphi(P) + \varphi(Q)$$

***Proof* :**
if φ is the zero isogeny we're done, so assume φ is nonzero. Then it is a finite map and it induces a map:

$$\varphi_* : \mathrm{Pic}^0(E_1) \to \mathrm{Pic}^0(E_2)$$
$$\varphi_*\left(\left[\sum n_i(P_i)\right]\right) = \left[\sum n_i(\varphi(P_i))\right]$$

On the other hand, we have natural group homomorphisms:

$$\kappa_i : E_i \to \mathrm{Pic}^0(E_i) \qquad P \mapsto [(P) - (O)]$$

As $\varphi(O) = O$, we get the commutative diagram:

$$\begin{tikzcd} E_1 \arrow[r, "\cong"] \arrow[d, "\varphi"'] & \mathrm{Pic}^0(E_1) \arrow[d, "\varphi_*"] \\ E_2 \arrow[r, "\cong"] & \mathrm{Pic}^0(E_2) \end{tikzcd}$$

Now, as κ_1, κ_2 and φ_* are all group homomorphism and κ_2 is injective, φ is a homomorphism, as we sought to show.

Corollary 1.5.3: Kernel of Isogeny

Let $\varphi : E_1 \to E_2$ be a nonzero isogeny. Then $\ker \varphi$ is a finite group

Proof **:**
as $\deg(\varphi)$ is always finite, the order of the kernel is always at most $\deg(\varphi)$.

Example 1.5.1: Isogenies

1. Consider any $m \in \mathbb{Z}$ and any elliptic curve E. Define

$$[m] : E \to E \qquad P \mapsto \underbrace{P + \cdots + P}_{n \text{ times}}$$

Then as $+$ is a morphism, using induction the above map is certainly a morphism, and as $O \mapsto O$ it is an isogeny.

In ref:HERE, we shall see that when $\mathrm{char}(k) = 0$, in most cases the maps $[m]$ for $m \in \mathbb{Z}$ compromise all the endomorphisms, namely $[-] : \mathbb{Z} \xrightarrow{\cong} \mathrm{End}(E)$ is an isomorphism. If $\mathbb{Z} \subsetneq \mathrm{End}(E)$ then we shall say that E has *complex multiplication*, or CM (see ref:HERE for why). If k is a finite field, $\mathrm{End}(E)$ will always be larger than $\mathbb{Z}$ (see ref:HERE).

2. Let $\mathrm{char}(E) \neq 2$ and let $i \in \overline{k}$ represent the 4th root of unity. Then as seen in exercise ref:HERE (see p. 69 of Silverman, example 4.4, III.32) we have that the curve E/k given by

$$E : y^2 = x^3 - x$$

has an endomorphism ring $\mathrm{End}(E)$ strictly larger than $\mathbb{Z}$, namely there is the map:

$$[i] : (x, y) \mapsto (-x, iy)$$

Thus, we say that E has CM (complex multiplication). Naturally such a map is only defined over k if and only if $i \in k$, hence it is possible that $\mathrm{End}_k(e)$ is strictly smaller than $\mathrm{End}(E)$.

Note furthermore that

$$[i] \circ [i](x, y) = [i](-x, iy) = (x, -y) = -(x, y)$$

Hence, $[i] \circ [i] = [-1]$. From this we get the ring homomorphism:

$$\mathbb{Z}[i] \to \mathrm{End}(E) \qquad m + ni \mapsto [m] + [n] \circ [i]$$

which is an isomorphism when $\mathrm{char}(k) = 0$. In that case:

$$\mathrm{Aut}(E) \cong \mathbb{Z}[i]^* = \{\pm 1, \pm i\}$$

showing it is a cyclic group of order 4.

3. Assume $\mathrm{char}(k) \neq 2$ and $a, b \in k$ where $b \neq 0$ and $r = a^2 - 5b \neq 0$. Then define:

$$\begin{aligned} E_1 :& y^2 = x^3 + ax2 + bx \\ E_2 :& Y^2 = X3 - 2aX^2 + rX \end{aligned}$$

Then there are two degree 2 isogenies connecting these curves:

$$\begin{aligned} \varphi : E_1 &\longrightarrow E_2, & \hat{\varphi} : E_2 &\longrightarrow E_1, \\ (x, y) &\longmapsto \left(\tfrac{y^2}{x^2}, \tfrac{y(b-x^2)}{x^2}\right), & (X, Y) &\longmapsto \left(\tfrac{Y^2}{4X^2}, \tfrac{Y(r-X^2)}{8X^2}\right). \end{aligned}$$

By directly computing, we get $\hat{\varphi} \circ \varphi = [2]$ on E_1 and $\varphi \circ \hat{\varphi} = [2]$ on E_2. We shall show that these maps are considered *dual isogenies* (see section 1.7).

4. Let $\mathrm{char}(k) = p > 0$ and $q = p^r$. Then given an elliptic curve E/k, there is a well-defined curve $E^{(q)}$

$$\varphi_q : E \to E^{(q)} \qquad (x, y) \mapsto (x^q, y^q)$$

given some coordinate representation as a Weierstrass equation. As $E^{(q)}$ is the zero set of a Weierstrass equation, if it is nonsingular it will be an elliptic curve. Indeed, it is not a hard computation to verify that:

$$\Delta(E^{(q)}) = \Delta(E)^q \qquad \text{and} \qquad j(E^{(q)}) = j(E)^q$$

and as the $(-)^q$ is an automorphism, the discriminant is non-singular, and so $E^{(q)}$ is an elliptic curve.

Suppose now that $k = \mathbb{F}_q$ is a finite field. Then the qth-power map on k is the identity, and so:

$$E^{(q)} = E$$

In particular, the set of points fixed by φ_q is exactly $E(\mathbb{F}_q)$ (this shall come back in theorem 3.1.1), and φ_q is now an *endomorphism*.

5. The translation map $\tau_Q : E \to E$ given by $P \mapsto P + Q$ is not an isogeny, unless of course $Q = O$.

6. Let $F : E_1 \to E_2$ be an arbitrary morphism of elliptic curves. Then the composition:

$$\varphi := \tau_{-F(O)} \circ F$$

is an isogeny as $\varphi(O) = O$. As translations are invertible, any morphism F between elliptic curves can be written as:

$$F = \tau_{F(O)} \circ \varphi$$

that is, maps between elliptic curves are isogenies and a translation.

Proposition 1.5.4: Properties of Isogenies

Let E/k be an elliptic curve and $m \in \mathbb{Z}$. Then:

1. the map
$$[m] : E \to E$$
is nonconstant, hence surjective
2. $\mathrm{Hom}(E_1, E_2)$ is a torsion-free $\mathbb{Z}$-module
3. $\mathrm{End}(E)$ is a (not necessarily commutative) ring of characteristic 0 with no zero divisors.

***Proof* :**

1. Let's first show $[2] \neq [0]$. If $P = (x, y) \in E$ has order 2, then with some manipulation of equations we see it must satisfy:
$$4x^3 + b - 2x^2 + 2b_4x + b_6 = 0$$
Now, if $\mathrm{char}(k) \neq 2$, we immediately see that there are only finitely many such points (Fund. thm. of algebra). If $\mathrm{char}(k) = 2$, then we have $[2] = [0]$ only if the cubic polynomial is identically zero, meaning $b_2 = b_6 = 0$, which would imply $\Delta = 0$. But elliptic curves are smooth, hence in all cases $[2] \neq [0]$. Then certainly $[mn] = [m] \circ [n]$, this shows all even cases, hence what's left is the case where m is odd.
To that end, assume again that $\mathrm{char}(k) \neq 2$. The by long division, we can verify that:
$$4x^3 + b_2x2 + 2b_4x + b_6$$
does *not* divide
$$x^4 - b_4x^2 - 2b_6x - b_8$$
More more specifically, if the first divides the second then $\Delta = 0$. Henc, there is an $x_0 \in \overline{k}$ such that the first polynomial vanishes to higher order at x_0 than it does at the second. Now choose $y_0 \in \overline{k}$ so that $P_0 = (x_0, y_0) \in E$. Then the doubling formula gives
$$[2]P_0 = O$$
that is, we have that E has a nontrivial point P_0 of over 2. But then for any odd integers m it must be that:
$$[m]P_0 = P_0 \neq 0$$
and so $[m] \neq [0]$. Finally, if $\mathrm{char}(k) = 2$, Then we can find a point of order 3 (see exercise 1.11.1(3)), which can be used to give the desired result.
2. Suppose that $\varphi \in \mathrm{Hom}(E_1, E_2)$ and $m \in \mathbb{Z}$ such that
$$[m] \circ \varphi = [0]$$
Then
$$(\deg[m]) \deg(\varphi) = 0$$
Hence, either $m = 0$ in which case $\deg[m]0$ or else $m \neq 0$ where we then get from (10 that $\deg[m] \geq 1$ which must imply $\deg(\varphi) = 0$, i.e. $\varphi = [0]$. Thus $\mathrm{Hom}(E_1, E_2)$ is torsion-free.

3. We immediately get from (2) that $\mathrm{char}(\mathrm{End}(E)) = 0$. Indeed, if $\varphi, \psi \in \mathrm{End}(E)$ where $\varphi \circ \psi = [0]$. Then:
$$\deg(\varphi)\deg(\psi) = 0$$
from which we get one or the other is zero.

Another basic properties is that $\deg([m]) = m^2$; the proof delayed as it is a nice consequence of ref:HERE, however the reader prove this explicitly (thought it is a bit tedious). As had was demonstrated in the proof, there are torsion elements in elliptic curve group. Let us label these:

Definition 1.5.5: Torsion Subgroup of Elliptic Curve

Let E be an elliptic curve and $m \in \mathbb{Z}$ where $m \geq 1$. Then the *m-torsion subgroup* of E is:

$$E[m] := \{P \in E \mid [m]P = O\}$$

The *torsion subgroup* off E is defined as :

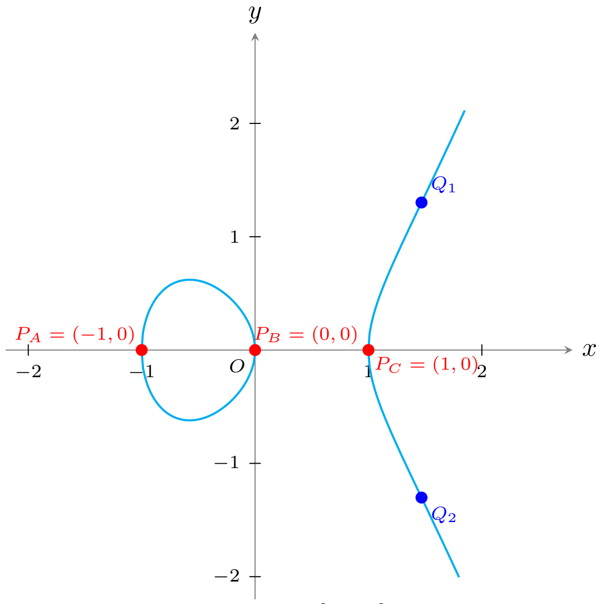
$$E_{\mathrm{tors}} = \bigcup_{m=1}^{\infty} E[m]$$

If E/k is an elliptic curve defined over k, then $E_{\mathrm{tors}}(k)$ denotes the point of finite order of E/k.

Example 1.5.2: Torsion Subgroup of Elliptic Curve

When considering the analytification of an elliptic curve and looking at the flat torus, the torsion points are the points z where $nz \in \Lambda$ (the lattice defining the elliptic curve). This shall impose strong conditions on the structure of the torsion group in the analytification, namely by some simple algebra the m-torsion points shall be isomorphic as a group to $(\mathbb{Z}/m\mathbb{Z})^2$. This generalizes to elliptic curves in general with a few notable exceptions, but the work is more involved (and will be one of the core results of this chapter).

Consider
$$E : y^2 = x^3 - x = x(x-1)(x+1)$$
When plotting the real points, it can be visualized as:



Elliptic Curve $y^2 = x^3 - x$

with real 2-torsion (red) and 3-torsion (blue) points

The 1-torsion point is always just O. The two-torsion points are those where $2P = O$, that is $P = -P$. Thus, they are points where $(x, y) = (x, -y)$. They are thus the points on the x-axis:

$$E[2] = \{O, (0, 0), (1, 0), (-1, 0)\}$$

The three torsion points are all points where $3P = O$, or $2P = -P$. Recall by Bézout's theorem that if T is a tangent line, then T has intersection multiplicity 3 with E if the point P is an *inflection point*. Thus, the 3-torsion points correspond to the inflection points!

The 4-torsion points $4P = O$ have all the 2 torsion points, and all the points where $2P = S$ where S is a 2-torsion point. This shall result in 16 points.

For the 5-torsion points, these points an be found by some judicious computation. In fact, due to the rather simple nature of an elliptic curve of $\text{char}(k) \neq 2, 3$, we convert $5P = O$ (more generally $nP = O$) into a polynomial whose roots correspond to torsion points (see ref:HERE).

Theorem 1.5.6: Galois Theory of Elliptic Function Fields

Let $\varphi : E_1 \to E_2$ be a nonzero isogeny.

1. For every $Q \in E_2$,
$$\#\varphi^{-1}(Q) = \deg_s \varphi$$
For every $P \in E_1$,
$$e_\varphi(P) = \deg_i \varphi$$

2. The map
$$\ker\varphi \xrightarrow{\sim} \mathrm{Gal}_{\varphi^*\overline{K}(E_2)}(\overline{K}(E_1)) \quad T \mapsto \tau_T^*$$
is an isomorphism where τ_T is the translation-by-T map and τ_T^* is the automorphism that τ_T induces on $\overline{K}(E_1)$.

3. Suppose that φ is separable. Then φ is unramified,
$$\#\ker\varphi = \deg\varphi$$
and $\overline{K}(E_1)$ is a Galois extension of $\varphi^*\overline{K}(E_2)$.

Proof **:**

1. From [1] we have $\#\varphi^{-1}(Q) = \deg_s(\varphi)$ for all but potentially finitely many $Q \in E_2$. Now, for any $Q, Q' \in E_2$, choose $R \in E_1$ where $\varphi(R) = Q' - Q$. Now as φ is a homomorphism, the translation τ_R is an injective map:
$$\varphi^{-1}(Q) \to \varphi^{-1}(Q') \qquad P \mapsto P + R$$
and by again by [1] we have these are the same size, and are thus in bijective correspondence. As Q, Q' were arbitrary we have:
$$\#\varphi^{-1}(Q) = \deg_s \varphi \qquad \forall Q \in E_2$$
Next, for the inseparable degree, choose $P, P' \in E_1$ where $\varphi(P) = \varphi(P') = Q$ and let $R = P' - P$. Then $\varphi(R) = O$, and so as shown in example 1.5.1 $\varphi \circ \tau_R = \varphi$. As τ_R is an isomorphism
$$e_\varphi(P) = e_{\varphi\circ\tau_R}(P) = e_\varphi(\tau_R(P))e_{\tau_R}(P) = e_\varphi(P')$$
Thus, each point $\varphi^{-1}(Q)$ has the same ramification index. But then:
$$\begin{aligned}(\deg_s\varphi)(\deg_i\varphi) = \deg\varphi &= \sum_{P\in\varphi^{-1}(Q)} e_\varphi(P) \\ &= (\#\varphi^{-1}(Q))e_\varphi(P) \\ &= (\deg_s\varphi)e_\varphi(P)\end{aligned}$$
Dividing by $\deg_s(\varphi)$ on both sides gives the final assertion.

2. Let show
$$\ker\varphi \xrightarrow{\sim} \mathrm{Gal}_{\varphi^*\overline{K}(E_2)}(\overline{K}(E_1)) \quad T \mapsto \tau_T^*$$

Starting by showing it is well-defined, let $T \in \ker\varphi$. Certinaly $\tau_T^* : \overline{K}(E_1) \to \overline{K}(E_1)$ is well-defined as it maps f to f^T where $f^T(P) = f(P+T)$. Let' show it fixes the base field. Let $f \in \overline{k}(E_2)$. Then as $\varphi \circ \tau_T = \varphi$ (as T is in the kernel):

$$\tau^*(\varphi^* f) = (\varphi \circ \tau_T)^* f = \varphi^* f$$

Hence, τ_T^* fixes $\varphi^*(\overline{k}(E_2))$, showing the map $T \to \tau_T^*$ is well-defined. Next, the map is a homomorphism since $T + S \mapsto \tau_{S+T} = \tau_S \circ \tau_T$ and as $\tau_{S+T} = \tau_{T+S}$, the group-structure is preserved. Next, from part (1) we have that $\#\ker\varphi = \deg_s(\varphi)$ and by Galois theory we have

$$\#\mathrm{Gal}_{\varphi^*(\overline{K}(E_2))}(\overline{K}(E_1)) \leq [\overline{K}(E_1) : \varphi^*(\overline{K}(E_2))] = \deg_s(\varphi)$$

Hence, if we show the map is injective, we have an isomorphism. If for the sake of contradiction τ_T^* fixes $\overline{k}(E_1)$ (so that it is in the kernel of the map, then every function on T_1 take the same value at T and O. But then it must be that $T = O$ (one approach is that the coordinate function x has a pole at O and no other poles, and as the take the same values the only possibility is $T = O$).

3. If φ is separable, then $\#\varphi^{-1}(Q) = \deg(\varphi)$ for all $Q \in E_2$ and hence φ is unramified. Letting $Q = O$, we get that :

$$\#\varphi^{-1}(O) = \#\ker\varphi = \deg\varphi$$

Then from (2), we have:

$$\#\mathrm{Gal}_{\varphi^*(\overline{k}(E_2))}(\overline{k}(E_1))[\overline{k}(E_1) : \varphi^*\overline{k}(E_2)]$$

showing the extension is Galois.

Corollary 1.5.7: Extending E_1-isogenies

Let

$$\varphi : E_1 \to E_2 \qquad \psi : E_1 \to E_3$$

be nonconstant isogenies and assume φ is separable. Then if

$$\ker\varphi \subseteq \ker\psi$$

there exists a unique isogeny:

$$\lambda : E_2 \to E_3$$

such that $\psi = \lambda \circ \varphi$, that is the following diagram commutes

$$\begin{tikzcd} & E_2 \arrow[dd, dashed, "\lambda"] \\ E_1 \arrow[ur, "\varphi"] \arrow[dr, "\psi"] & \\ & E_3 \end{tikzcd}$$

***Proof* :**
Sine φ is separable, by theorem 1.5.6 $\overline{k}(E_1)/(\varphi^*\overline{k}(E_2))$ is a Galois extension; let G be the galois group. The inclusion $\ker\varphi \subseteq \ker\psi$ and the isomorphism in theorem 1.5.6(2) implies that G fixes $\psi^*\overline{k}(E_3)$. Then by Galois theory, there are field inclusions:

$$\psi^*\overline{k}(E_3) \subseteq \varphi^*\overline{k}(E_2) \subseteq \overline{k}(E_1)$$

But then we have a field inclusion $\psi^*\overline{k}(E_3) \subseteq \overline{k}(E_1)$, and hence a map

$$\lambda : E_2 \to E_3$$

satisfying

$$\varphi^*(\lambda^*\overline{k}(E_3)) = \psi^*(E_3)$$

or geometrically, $\lambda \circ \varphi = \psi$. Finally, λ is an isogeny since:

$$\lambda(O) = \lambda(\varphi(O)) = \psi(O) = O$$

completing the proof.

From group theory, we know that (normal) subgroups of a group G characterize homomorphisms from G, namely $\varphi : G \to G/\ker\varphi$. When working with Elliptic curves, given a finite subgroup $K \subseteq E$, is there always an elliptic curve E' such that $\varphi : E \to E'$ has kernel K? This is not immediately obvious as taking the quotient does *not* usually preserve smoothness (the reader may recall from [4] the extra regularity conditions that needed to be added to insure the resulting quotient object is still a smooth manifold). As it turns out for elliptic curves, E' always exists:

Proposition 1.5.8: Every Kernel Characterized By Elliptic Curve

Let E be an elliptic curve and $\Phi \leqslant E$ be a finite subgroup of E. Then there is a unique elliptic curve E' and a separable isogeny

$$\varphi : E \to E'$$

where $\ker\varphi = \Phi$. The resulting elliptic curve is often denoted E/Φ.

***Proof* :**
By theorem 1.5.6(2), each $T \in \Phi$ gives rise to an automorphism τ_T^* of $\overline{k}(E)$. Let $L = \overline{k}(E)^\Phi$ be the subfield of $\overline{k}(E)$ fixed by Φ so that by Galois theory the $\overline{k}(E)$ is a galois extension of L with galois group isomorphic to Φ. Certainly L has transcendence degree 1 over $\overline{k}$, so there is a unique smooth curve $C/\overline{k}$ and a finite morhpism

$$\varphi : E \to C \qquad \varphi^*(\overline{k}(C)) = L$$

Let's first show φ is separable; it suffices to show it is unramified. If $P \in E$ and $T \in \Phi$, then every function $f \in \overline{k}(C)$ gives

$$f(\varphi(P+T)) = (\tau_T^* \circ \varphi^*)f(P) \stackrel{!}{=} (\varphi^* f)(P) = f(\varphi(P))$$

where $\stackrel{!}{=}$ comes from τ_T^* fixing every element of $\varphi^*(\overline{k}(C))$. As this is true for all f, including the identity, we have $\varphi(P+T) = \varphi(P)$. Now, choose $Q \in C$ and $P \in E$ where $\varphi(P) = Q$. Then:

$$\{P + T \mid T \in \Phi\} \subseteq \varphi^{-1}(Q)$$

By [1] we have

$$\#\varphi^{-1}(Q) \leq \deg \varphi = \#\Phi$$

with equality if and only if φ is unramified at all points in the inverse of $\varphi^{-1}(Q)$. And as the points $P+T$ are distinct as T ranges over Φ, φ is indeed unramified at all points in $\varphi^{-1}(Q)$. As Q was arbitrary, φ is unramified. Thus it is separable.

Finally, by Hurwitz genus formula theorem 0.2.20 as φ is unramified we get:

$$0 = 2\text{genus}(E) - 2 = (\deg \varphi)(2\text{genus}(C) - 2)$$

as $\deg \varphi \neq 0$, it must be that $\text{genus}(C) = 1$, and hence C must be an elliptic curve. By letting $\varphi(O)$ be the distinguished point on C, we see that φ is also an isogeny, completing the proof.

Corollary 1.5.9: Kernel over any field K

Let E/K be an elliptic curve over K and $\Phi \leqslant E$ be a finite subgroup of E/K (in particular, it is $\text{Gal}_K(\overline{K})$-invariant). Then there is a unique elliptic curve E' and a separable isogeny

$$\varphi : E \to E'$$

where $\ker \varphi = \Phi$.

Proof **:**
exercise

1.6 The Invariant Differential

Let E/K be an elliptic curve given by the Weierstrass equation

$$E : y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$$

Then we saw it has a nonvanishing holomorphic form:

$$\omega = \frac{dx}{2y + a_1x + a_3} \in \Omega_E$$

that is it has no zeros or poles; the form is called the *invariant differential* of E for the following reason

Theorem 1.6.1: Invariant Differential is Translation Invariant

Let E be an elliptic curve, ω the invariant differential, and $\tau_Q : E \to E$ a translation map given by Q. Then

$$\tau_Q^*\omega = \omega$$

This is more easily visualized when considering E as $\mathbb{C}/\Lambda$ (thought of as a torus in 3d-space), in this case the differential from $dx/2dy = dz$ is invariant under translation of the group law (similarly, it is the form dz on $\mathbb{C}$ that is invariant under translation by the lattice Λ, and hence is well-defined on $\mathbb{C}/\Lambda$).

***Proof* :**
One way of going about this is by direct computation, and another is judicious use of the results accumulated about differentials.

For the longer proof, let's say we have chosen a coordinate representation x, y and consider the translation of these functions $x(P+Q)$ and $y(P+Q)$. Write these in terms $x(P), x(Q), y(P), y(Q)$ using the addition formula, and then compute $dx(P+Q)$ and write it in terms of a rational function times $dx(P)$ (where $x(Q), y(Q)$ are constants). Then the reader would arrive it:

$$\frac{dx(P+Q)}{2y(P+Q)+a_1x(P+Q)+a_3} = \frac{dx(P)}{2y(P)+a_1x(P)+a_3}$$

For the other proof, recall that Ω_E is a one-dimension $\overline{k}(E)$-vector space. Hence given τ_Q* is a linear transformation on Ω_E, there exists an element a_Q such that

$$\tau_Q^* = a_Q\omega$$

Taking div of both sides and re-arrnging, we get:

$$\begin{aligned}\mathrm{div}(a_Q) &= \mathrm{div}(\tau_Q^*) - \mathrm{div}(\omega)\\ &\stackrel{!}{=} \tau_Q^*\mathrm{div}(\omega) - \mathrm{div}(\omega)\\ &= 0 \text{as } \mathrm{div}(\omega) = 0\end{aligned}$$

where $\stackrel{!}{=}$ comes from the fact that τ_Q is an isomorphism which preserves the roots/poles. As $\mathrm{div}(a_Q) = 0$, $a_Q \in \overline{k}$. Note that $Q \mapsto a_Q$ is a rational function (as the computations above can attest to), and $a_Q \neq 0$ since τ_Q is an isomorphism. Hence, we can define the rational map

$$f : E \to \mathbb{P}^1 \qquad Q \mapsto [a_Q : 1]$$

Now, this map is *not* surjective as it misses $[0:1]$ and $[1:0]$, and hence *must* be the constant map. But then a_Q does not depend on Q, so if we know a_Q for a particular value we're done. But certainly:

$$a_O = 1$$

as then $\tau_O^* = \mathrm{id}$, and so $\tau_Q^*\omega = \omega$, as we sought to show.

The invariant differential allows us to translate that rather complicated group operation on an elliptic curve to addition in a vector space:

Theorem 1.6.2: Pullback on forms Respects Group Law

Let E, E' be elliptic curves, ω the invariant differential from on E and

$$\varphi, \psi : E \to E'$$

two isogenies. Then:

$$(\varphi + \psi)^* \omega = \varphi^* \omega + \psi^* \omega$$

where the addition on the left hand side is addition in $\operatorname{Hom}(E', E)$ (which is fundamentally based on the group law as $\varphi + \psi(P) = \varphi(P) + \psi(P)$ which is addition in E') and the addition on the right hand side is addition in the vector space Ω_E.

To get a sense, recall that ω is invariant under translation, and that an isogeny will "wrap" the torus $\mathbb{C}/\Lambda$ a certain amount of times another one (like a covering map) in such a way that it preserves the group structure, so parallel flows will match up. Then we can think of the linearity by looking at the "intensity" of the shadow of the flow on E' cast by E, where the number of times it wraps gives the intensity. What this result says is that we can first add two isogenies and then pull back, or pull them back separately, and the effect will be the same.

1.6.1 Generalization to Abelian Varieties ome key parts of the proof rely on properties of elliptic curves, however the core structure is only reliant on how E has a group structure and it is derived from $\operatorname{Pic}^0(E)$. For this reason, this same idea generalizes to abelian varieties

Proof **:**
If $\varphi = [0]$ or $\psi = [0]$ then it is immediate. If $\varphi + \psi = [0]$ then as:

$$\psi^* = (-\varphi)^* = \varphi^* \circ [-1]^*$$

it suffices to show that:

$$[-1]^* \omega = -\omega$$

Indeed:

$$[-1](x, y) = (x, -y - a_1 x - a_3)$$

and so:

$$\begin{aligned} [-1]^* \left(\frac{dx}{2y + a_1 x + a - 3} \right) &= \frac{dx}{2(-y - a_1 x - a_3) + a_1 x + a_3} \\ &= -\frac{dx}{2y + a_1 x + a_3} \end{aligned}$$

giving us the desired result in this special case.

Now assume $\varphi, \psi, \varphi + \psi$ are nonzero. Let (x_1, y_1) and (x_2, y_2) be "independent" Weierstrass coordinates on E, i.e. they satisfy the given Weierstrass equation for E, but satisfy no other algebraic relations. Formally, we would say that:

$$([x_1, y_1, 1], [x_2, y_2, 1])$$

give coordinates for $E \times E$ sitting inside $\mathbb{P}^2 \times \mathbb{P}^2$[a]. Let

$$(x_3, y_3) = (x_1, y_1) + (x_2, y_2),$$

so x_3 and y_3 are rational combinations of x_1, x_2, y_1, y_2 given by the addition formula on E. Furthermore, for any (x, y), let $\omega(x, y)$ denote the corresponding invariant differential,

$$\omega(x, y) = \frac{dx}{2y + a_1 x + a_3}.$$

Then, using the addition formula and the standard rules of differentiation, express $\omega(x_3, y_3)$ in terms of $\omega(x_1, y_1)$ and $\omega(x_2, y_2)$:

$$\omega(x_3, y_3) = f(x_1, y_1, x_2, y_2)\omega(x_1, y_1) + g(x_1, y_1, x_2, y_2)\omega(x_2, y_2),$$

where f and g are rational functions of the indicated variables. In doing this calculation, remember that since x_i and y_i satisfy the given Weierstrass equation, the differentials dx_i and dy_i are related by

$$(2y_i + a_1 x_i + a_3)\, dy_i = (3x_i^2 + 2a_2 x_i + a_4 - a_1 y_i)\, dx_i.$$

In this way, $\omega(x_3, y_3)$ can be expressed as a $\overline{K}(x_1, y_1, x_2, y_2)$-linear combination of dx_1 and dx_2.

I claim that $f = g = 1$. This can be proven explicitly, but instead, let's rely on theorem 1.6.1. Fix values for x_2 and y_2, say by choosing some $Q \in E$ and setting

$$x_2 = x(Q) \quad \text{and} \quad y_2 = y(Q).$$

Then $dx_2 = dx(Q) = 0$ so $\omega(x_2, y_2) = 0$. By theorem 1.6.1:

$$\omega(x_3, y_3) = \tau_Q^* \omega(x_1, y_1) = \omega(x_1, y_1).$$

Substituting these into the expression for $\omega(x_3, y_3)$ we get

$$f(x_1, y_1, x(Q), y(Q)) = 1$$

as a rational function in $\overline{K}(x_1, y_1)$. Thus f does not depend on x_1 and y_1, so $f \in \overline{K}(x_2, y_2)$. But $f(x_2, y_2)$ satisfies $f(x(Q), y(Q)) = 1$ for every point $Q \in E$, so $f = 1$. The same arguments works for x_2 and y_2, hence $g = 1$. Thus, if $(x_3, y_3) = (x_1, y_1) + (x_2, y_2)$ (+ is addition on E) then $\omega(x_3, y_3) = \omega(x_1, y_1) + \omega(x_2, y_2)$ (+ is addition in Ω_E).

Now let (x', y') be Weierstrass coordinates on E' and set

$$(x_1, y_1) = \varphi(x', y'), \quad (x_2, y_2) = \psi(x', y'), \quad (x_3, y_3) = (\varphi + \psi)(x', y').$$

Substituting this into $\omega(x_3, y_3) = \omega(x_1, y_1) + \omega(x_2, y_2)$ gives

$$(\omega \circ (\varphi + \psi))(x', y') = (\omega \circ \varphi)(x', y') + (\omega \circ \psi)(x', y'),$$

which means:

$$(\varphi + \psi)^* \omega = \varphi^* \omega + \psi^* \omega$$

as we sought to show.

[a]Alternatively, Weil would say (x_1, y_1) and (x_2, y_2) are "independent generic points of E"

Corollary 1.6.3: m-multiplication Map On Invariant Differential

Let E be an elliptic curve, ω an invariant differential, and $m \in \mathbb{Z}$. Then:

$$[m]^*\omega = m\omega$$

Proof **:**
If $m = 0, 1$, the result is immediate. More generally, let $\varphi = [m]$ and $\psi = [1]$. Then by theorem 1.6.2 we get:

$$[m+1]^*\omega = [m]^*\omega + \omega$$

repeating inductively gets the desired result.

Corollary 1.6.4: multiplication map finite separable endormophism

Let E/K be an elliptic curve and $m \in \mathbb{Z}$ where $m \neq 0$ in K. Then $[m] : E \to E$ is a finite separable endomorphism

Proof **:**
Let ω be an invariant differential on E. Then by corollary 1.6.3 and that $m \neq 0$ in K we get:

$$[m]^*\omega = m\omega \neq 0$$

Hence $[m]$ is finite, and by 0.2.9 $[m]$ is separable.

Corollary 1.6.5: Detecting Separable Frobenius Map

Let E be an elliptic curve over $\mathbb{F}_q$, $\varphi : E \to E$ be the qth power Frobenius endomrophism, and $m, n \in \mathbb{Z}$. Then

$$m + n\varphi : E \to E$$

is separable if and only if $p \nmid m$. In particular, $1 - \varphi$ is separable

Proof **:**
Let ω be the invariant differential on E. Recall that $\psi : E \to E$ is inseprable if and only if $\psi^*\omega = 0$. Applying this to $\psi == +n\varphi$ and using corollary 1.6.3 and corollary 1.6.4 we get:

$$(m + n\varphi)^*\omega = m\omega = n\varphi^*\omega$$

As φ is inseparable, $\varphi^*\omega = 0$ (which can also be deduced via direct computation as $\varphi^*(dx) = qx^{q-1}dx = 0$). Thus

$$(m + n\varphi)^*\omega = [m]^* + [n]^* \circ \varphi^*\omega = m\omega$$

Finally, $m\omega = 0$ if and only if $p \big| m$, giving the desired result.

From this, we can deduce some important results about the endomorphism ring of an elliptic curve

Proposition 1.6.6: Properties of Endomorphism Ring

Let E/K be an elliptic curve and ω a nonzero invariant differential on E. Define the map:

$$\mathrm{End}(E) \to \overline{K} \qquad \varphi \mapsto a_\varphi \qquad \text{where} \qquad \varphi^*\omega = a_\varphi \omega$$

Then:

1. $\varphi \mapsto a_\varphi$ is a ring homomorphism
2. $\ker(\varphi \mapsto a_\varphi)$ is the set of inseparable endomorphisms of E
3. If $\mathrm{char}(K) = 0$, $\mathrm{End}(E)$ is a commutative ring.

***Proof* :**
Let's first show this map is well-defined (namely it must be that $a_\varphi \in \overline{K}$). Certainly $a_\varphi \in \overline{K}(E)$ (as Ω_E is a 1-dimensional $\overline{K}(E)$-vector space and φ^* becomes linear). We need $a_\varphi \in \overline{K}$. If $a_\varphi = 0$ we're done, so assume $a_\varphi \neq 0$. Then As $\mathrm{div}(\omega) = 0$ we have:

$$\mathrm{div}(a_\varphi) = \mathrm{div}(\varphi^*\omega) - \mathrm{div}(\omega) = \varphi^*\mathrm{div}(\omega) - \mathrm{div}(\omega) = 0$$

and so a_φ has no zeroes or poles, hence $a_\varphi \in \overline{K}$ giving a well-defined map.

1. Using theorem 1.6.2 we get

$$a_{\varphi+\psi}\omega = (\varphi + \psi)^*\omega = \varphi^*\omega + \psi^*\omega = a_\varphi\omega + a_\psi\omega = (a_\varphi + a_\psi)\omega$$

   Hence $a_{\varphi+\psi} = a_\varphi + a_\psi$. Similarly:

$$a_{\varphi\circ\psi}\omega = (\varphi \circ \psi)^*\omega = \psi^*(\varphi^*\omega) = \psi^*(a_\varphi\omega) = a_\varphi\psi^*(\omega) = a_\varphi a_\psi \omega$$

   giving $a_{\varphi\circ\psi} = a_\varphi a_\psi$

2. $$a_\varphi = 0 \qquad \Leftrightarrow \qquad \varphi^*\omega = 0 \qquad \Leftrightarrow \qquad \varphi \text{ is inseparable}$$

3. If $\mathrm{char}(K) = 0$, every endomorphism is separable, so $\mathrm{End}(E)$ injective into $\overline{K}^*$ meaning it *must* be commutative.

1.7 Dual Isogenies and Torsion Subgroups

The isogeny $\varphi : E_1 \to E_2$ was shown to be a group homomorphism by taking advantage of the induced map $\varphi_* : \mathrm{Pic}^0(E_1) \to \mathrm{Pic}^0(E_2)$. However there is the natural map φ_*:

$$\varphi_* : \mathrm{Pic}^0(E_2) \to \mathrm{Pic}^0(E_1)$$

Letting $\kappa_i : E_i \xrightarrow{\sim} \mathrm{Pic}^0(E_i)$ be the isomorphism linking the degree 0 Picard group and the elliptic curve, we get:

$$E_2 \xrightarrow{\kappa_2} \mathrm{Pic}^0(E_2) \xrightarrow{\varphi^*} \mathrm{Pic}^0(E_1) \xrightarrow{\kappa_1^{-1}} E_1$$

This map is certainly a group homomorphism, but it is not so obvious that it is a rational map (and hence an isogeny).

Theorem 1.7.1: Dual Isogeny

Let $\varphi : E_1 \to E_2$ be a nonconstant isogeny of degree m. Then there exists a unique isogeny

$$\hat{\varphi} : E_2 \to E_1$$

such that $\hat{\varphi} \circ \varphi = [m]$, where the composition

$$\hat{\varphi} : E_2 \xrightarrow{\kappa_2} \mathrm{Pic}^0(E_2) \xrightarrow{\varphi^*} \mathrm{Pic}^0(E_1) \xrightarrow{\kappa_1^{-1}} E_1$$

is given by $\hat{\varphi}(Q) = \hat{\varphi} \circ \varphi(P) = [\deg \varphi](P)$

Proof **:**
Let $\varphi : E_1 \to E_2$ be an isogeny and $\hat{\varphi}$ be given as above. If $\varphi = [0]$, then $\hat{\varphi} = [0]$ satisfies the requirements.

To show uniqueness, assume $\hat{\varphi}, \hat{\varphi}'$ are dual isogenies so that

$$(\hat{\varphi} - \hat{\varphi}') \circ \varphi = [m] - [m] = [0]$$

As φ is non-constant, we see that $\hat{\varphi} - \hat{\varphi}'$ must be constant, hence $\hat{\varphi} = \hat{\varphi}'$ (as $\hat{\varphi}(O) = O$).

Next, for the group homomorphism, suppoer $\psi : E_2 \to E_3$ is another nonconstant isogeny of, say, degree n. Suppose $\hat{\varphi}$ and $\hat{\psi}$ exist. Then:

$$\left(\hat{\varphi} \circ \hat{\psi}\right) \circ (\psi \circ \varphi) = \hat{\varphi} \circ [n] \circ \varphi = [n] \circ \hat{\varphi} \circ \varphi = [nm]$$

Thus

$$\hat{\varphi} \circ \hat{\psi} = \widehat{\varphi \circ \psi}$$

Now, if $\mathrm{char}(k) = 0$, then φ is separable, and if $\mathrm{char}(k) = p$ then by ref:HERE φ is the composition of a separable morphism and a Frobenius morphism. Thus, it suffices to prove the existence of $\hat{\varphi}$ when φ is either separable or the Frobenius morphism

1. (separable case) As φ has degree m, by theorem 1.5.6

   $$\ker \varphi = m$$

   hence, every element of $\ker \varphi$ has order dividing m; importantly:

   $$\ker \varphi \subseteq \ker[m]$$

   Hence by proposition 1.5.8 get

   $$\hat{\varphi} : E_2 \to E_1 \qquad \hat{\varphi} \circ \varphi = [m]$$

2. (Frobenius morphism case) If φ is the q^{th}-power Frobenius morphism with $q = p^e$, then φ is clearly the composition of the p^{th}-power Frobenius morphism with itself e times. Hence

it suffices to prove that $\hat{\varphi}$ exists if φ is the p^{th}-power Frobenius morphism, so in particular, $\deg \varphi = p$ from (section 0.2.1).

We look at the multiplication-by-p map on E. Let ω be an invariant differential. Then by corollary 1.6.3 and the fact that $\operatorname{char}(K) = p$, we see that

$$[p]^*\omega = p\omega = 0.$$

We conclude from (II.4.2c) that $[p]$ is not separable, and thus when we decompose $[p]$ as a Frobenius morphism followed by a separable map (section 0.2.1), the Frobenius morphism does appear. In other words,

$$[p] = \psi \circ \varphi^e$$

for some integer $e \geq 1$ and some separable isogeny ψ. Then we can take

$$\hat{\varphi} = \psi \circ \varphi^{e-1}$$

Showing that it exists and is unique, let us construct such a map and show it is of the desired form. Let $Q \in E_2$. Then the image of Q under the indicated composition is

$$\begin{aligned}
\operatorname{sum}(\varphi^*((Q) - (O))) &= \sum_{P \in \varphi^{-1}(Q)} [e_\varphi(P)]P - \sum_{T \in \varphi^{-1}(O)} [e_\varphi(T)]T && \text{by definition of } \varphi^* \\
&= [\deg_i \varphi] \left(\sum_{P \in \varphi^{-1}(Q)} P - \sum_{T \in \varphi^{-1}(O)} T \right) && \text{thm } 1.5.6 \\
&= [\deg_i \varphi] \circ [\#\varphi^{-1}(Q)]P && \text{for any } P \in \varphi^{-1}(Q) \\
&= [\deg \varphi]P && \text{thm } 1.5.6
\end{aligned}$$

But by construction,

$$\hat{\varphi}(Q) = \hat{\varphi} \circ \varphi(P) = [\deg \varphi]P,$$

as we sought to show.

Definition 1.7.2: Dual Isogeny

Let $\varphi : E_1 \to E_2$ be an isogeny. Then the *dual isogeny* $\hat{\varphi}$ is the unique map

$$\hat{\varphi} : E_2 \to E_1$$

given by theorem 1.7.1, where if $\varphi = [0]$ it will be defined that $\hat{\varphi} = [0]$.

Geometrically, if the degree of the isogeny gives you some notion of "winding" the torus, the dual isogeny gives you some notion of "unwinding" the torus I must re-think this because it is still very must a multiplication map!!, However it is still self-dual so there can be something here!.

Proposition 1.7.3: Properties of Dual Isogenies

Let $\varphi : E_1 \to E_2$ be an isogeny and $m = \deg \varphi$. Then

1. on E_1, E_2 respectively
$$\hat{\varphi} \circ \varphi = [m] \qquad \text{and} \qquad \varphi \circ \hat{\varphi} = [m]$$
2. If $\lambda : E_2 \to E_3$ is an isogeny, then:
$$\widehat{\lambda \circ \varphi} = \hat{\varphi} \circ \hat{\lambda}$$
3. If $\psi : E_1 \to E_2$ is an isogeny, then:
$$\widehat{\varphi + \psi} = \hat{\varphi} + \hat{\psi}$$
4. For all $m \in \mathbb{Z}$
$$\widehat{[m]} = [m] \qquad \text{and} \qquad \deg[m] = m^2$$
5. $\deg \hat{\varphi} = \deg \varphi$
6. $\hat{\hat{\varphi}} = \varphi$ (it is self-adjoint).

***Proof* :**
If φ is constant and either ψ, λ is constant, the results are trivial so assume otherwise.

1. The fact that $\hat{\varphi} \circ \varphi = [m]$ is by definition, and for the other notice
$$(\varphi \circ \hat{\varphi}) \circ \varphi = \varphi \circ [m] = [m] \circ \varphi$$
Hence as φ is not constant and surjective, it has a right-inverse and so $\varphi \circ \hat{\varphi} = [m]$
2. Let $n = \deg \lambda$. Then:
$$\left(\hat{\varphi} \circ \hat{\lambda}\right) \circ (\lambda \circ \varphi) = \hat{\varphi} \circ [n] \circ \varphi = [n] \circ \hat{\varphi} \circ \varphi = [nm]$$
Then by the uniqueness of the dual, we have:
$$(\hat{\varphi} \circ \hat{\lambda}) = \widehat{\lambda \circ \varphi}$$
3. That dualizing preserves the binary operation is more subtle. The proof shall be presented for characteristic 0 (in [8, exercise 3.31] it goes over for arbitrary characteristic)
Let $x_1, y_1 \in K(E_1)$ and $x_2, y_2 \in K(E_2)$ be Weierstrass coordinates. Start by looking at E_2 considered as an elliptic curve defined over the field $K(E_1) = K(x_1, y_1)$[a]. Then another way of saying that $\varphi : E_1 \to E_2$ is an isogeny is to note that $\varphi(x_1, y_1) \in E_2(K(x_1, y_1))$, and similarly for $\psi(x_1, y_1)$ and $(\varphi + \psi)(x_1, y_1)$. Now consider the divisor
$$D = ((\varphi + \psi)(x_1, y_1)) - (\varphi(x_1, y_1)) - (\psi(x_1, y_1)) + (O) \in \mathrm{Div}_{K(x_1,y_1)}(E_2).$$

The definition of $\varphi+\psi$ implies that D sums to O, so by corollary 1.4.5 D is linearly equivalent to 0. Thus there is a function

$$f \in K(x_1,y_1)(E_2) = K(x_1,y_1,x_2,y_2)$$

that, *when considered as a function of* x_2 *and* y_2, has divisor D.

Now, consider f as a function of x_1 and y_1, that is, treat f as a function on E_1 considered as an elliptic curve defined over $K(x_2,y_2)$. Suppose that $P_1 \in E_1(K(x_2,y_2))$ is a point satisfying $\varphi(P_1) = (x_2,y_2)$. Then examining D, specifically the term $-\mathrm{div}(\varphi(x_1,y_1))$, notice that f has a pole at P_1, namely the function $f(x_1,y_1;x_2,y_2)$ has a pole when x_1,y_1,x_2,y_2 satisfies $(x_2,y_2)=\varphi(x_1,y_1)$. Furthermore, by [1]:

$$\nu_{P_1}(f) = e_\varphi(P_1).$$

Similarly, f has a pole at P_1 if $(x_2,y_2)=\psi(P_1)$, and it has a zero at P_1 if $(x_2,y_2) = (\varphi+\psi)(P_1)$. It follows that as a function of x_1 and y_1, the divisor of f has the form

$$(\varphi+\psi)^*((x_2,y_2)) - \varphi^*((x_2,y_2)) - \psi^*((x_2,y_2)) + \sum n_i(P_i) \in \mathrm{Div}_{K(x_2,y_2)}(E_1),$$

where the P_i's are in $E_1(\bar{K})$, i.e., $\sum n_i(P_i) \in \mathrm{Div}_{\bar{K}}(E_1)$. Since this is the divisor of a function, it sums to O, so by theorem 1.7.1 the point point

$$(\varphi+\psi)(x_2,y_2) - \varphi(x_2,y_2) - \psi(x_2,y_2)$$

does not depend on (x_2,y_2), i.e., it is in $E_1(\bar{K})$. Putting $(x_2,y_2)=O$ shows that it is equal to O, which completes the proof that

$$\overline{\varphi+\psi} = \hat{\varphi}+\hat{\psi}.$$

4. If $m=0$ it is true by definition, and if $m=1$ it is true by (3). For $m \neq 0,1$ use induction and (3) with:
$$\widehat{[m+1]} = \widehat{[m]} + \widehat{[1]}$$
Then considering by definition $[\hat{m}] = [\deg[m]]$; letting $d = \deg[m]$ we get, by definition of dual and of what we just proved:
$$[d] = \widehat{[m]} \circ [m] = [m]\circ[m] = [m^2]$$
And as $\mathrm{End}(E)$ is torsion free, $d=m^2$.

5. let $m = \deg\varphi$. Then:
$$m^2 = \deg[m] = \deg(\varphi\circ\hat{\varphi}) = (\deg(\varphi))(\deg(\hat{\varphi})) = m(\deg(\hat{\varphi}))$$
hence $m = \deg\hat{\varphi}$

6. Let $m = deg\varphi$. Then:
$$\hat{\varphi}\circ\varphi = [m] = \widehat{[m]} = \widehat{\hat{\varphi}\circ\varphi} = \hat{\varphi}\hat{\hat{\varphi}}$$
and as φ is surjective
$$\varphi = \hat{\hat{\varphi}}$$

[a]This is where we use the characteristic 0 assumption, since all of our results on elliptic curves have assumed that the base field is perfect

These results show that $\mathrm{Hom}(E_1, E_2)$ and $\mathbb{Z}$ are tied together in some important way. The following definition gives the right concpet:

Definition 1.7.4: Quadratic Form

Let A be an abelian group. Then a function

$$d : A \to \mathbb{R}$$

is a *quadratic form* if it satisfies the following:

1. $d(\alpha) = d(-\alpha)$ for all $\alpha \in A$
2. The pairing
$$A \times A \to \mathbb{R} \qquad (\alpha, \beta) \mapsto d(\alpha + \beta) - d(\alpha) - d(\beta)$$
is bilinear

Furthermore, d is said to be *positive definite* if

1. $d(\alpha) \geq 0$ for all $\alpha \in A$
2. $d(\alpha) = 0$ if and only if $\alpha = 0$

Corollary 1.7.5: Degree Map is Positive Definite Quadratic Form

Let E_1, E_2 be elliptic curves. Then the degree map

$$\deg : \mathrm{Hom}(E_1, E_2) \to \mathbb{Z}$$

is a positive definite quadratic form

Proof **:**
Proposition 1.7.3 proves all except the bi linearity. Consider:

$$[-] : \mathbb{Z} \to \mathrm{End}(E_1)$$

and let

$$\langle \varphi, \psi \rangle = \deg(\varphi + \psi) - \deg(\varphi) - \deg(\psi)$$

Then:

$$\begin{aligned}
[\langle \varphi, \psi \rangle] &= [\deg(\varphi + \psi)] - [\deg(\varphi)] - [\deg(\psi)] \\
&= \left(\widehat{(\varphi + \psi)} \circ (\varphi + \psi) - \hat{\varphi} \circ \varphi - \hat{\psi}\psi \right) \\
&= \hat{\varphi} \circ \psi + \hat{\psi} \circ \varphi && \textit{proposition } 1.7.3(3)
\end{aligned}$$

Then by using proposition 1.7.3(3) a second time, the last expression is linear in both φ and ψ as we sought to show.

One of the great advantages is the ability to find the torsion subgroups of the elliptic curve:

Corollary 1.7.6: Torsion Subgroup of Elliptic Curve

Let E be an elliptic curve and let $m \in \mathbb{Z}$ with $m \neq 0$.

1. If $m \neq 0$ in K, i.e., if either $\operatorname{char}(K) = 0$ or $p = \operatorname{char}(K) > 0$ and $p \nmid m$, then
$$E[m] = \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$$
2. If $\operatorname{char}(K) = p > 0$, then one of the following is true:
   (a) $E[p^e] = \{O\}$ for all $e = 1, 2, 3, \ldots$
   (b) $E[p^e] = \frac{\mathbb{Z}}{p^e\mathbb{Z}}$ for all $e = 1, 2, 3, \ldots$

***Proof* :**

1. As $\deg[m] = m^2$ and by theorem 1.5.6(3) $\#\ker[m] = \deg[m]$ we have
$$\#E[m] = m^2$$
There are further condition's imposed on the orders of $E[m]$: for integer d dividing m, we must have (for the same rezoning)
$$\#E[d] = d^2$$
Then as finite abelian groups are classified, the only possibility is
$$E[m] = (\mathbb{Z}/m\mathbb{Z}) \times (\mathbb{Z}/m\mathbb{Z})$$
2. Let φ be that pth Frobenius morphism so that
$$\begin{aligned} \#E[p^e] &= \deg_s[p^e] && \textit{theorem } 1.5.6(1) \\ &= (\deg_s(\hat{\varphi} \circ \varphi))^e && \textit{proposition } 1.7.3(1) \\ &= (\deg_s \hat{\varphi})^e && \textit{section } 0.2.1 \end{aligned}$$
Then by proposition 1.7.3(5) we have:
$$\deg \hat{\varphi} = \deg \varphi = p$$
Now, either φ is separable or inseparable. If φ is inseparable then $\deg_s \hat{\varphi} = 1$ so
$$\#E[p^e] = 1 \qquad \forall e \in \mathbb{N}$$
If φ is separable then
$$\#E[p^e] = p^e \qquad \forall e \in \mathbb{N}$$
Then doing the same trick as before, namely using each $\#E[p^d] = p^d$ the only possiblity is:
$$E[p^e] = (\mathbb{Z}/p^e\mathbb{Z})$$
completing the proof

1.8 Tate Module

By corollary 1.7.6 we have abstractly classified the torsion subgroups:

$$E[m] \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$$

where $m \geq 2$ and is prime to $\mathrm{char}(k)$ if $\mathrm{char}(k) > 0$. However, $E[m] \subseteq E$ has structure structure to it as E is a curve. There are two important structures that is usually added to this group:

1. A *scheme structure*. More precisely, $E[m]$ is a *group scheme* and remembers more of the sheaf structure and the functions defined on theses points

2. Given $k \subseteq \overline{k}$, there is a natural *galois action* on these points.

The need to bring in the machinery of scheme theory is delegated to chapter ref:HERE. In this section, the galois action naturally defined on $E[m]$ will be explored. In particular, given $P \in E[m]$ so that $[m](P) = O$ we have for any $\sigma \in \mathrm{Gal}_k(\overline{k})$:

$$[m]\left(P^\sigma\right) = \left([m](P)\right)^\sigma = O^\sigma = O$$

Hence, the galois action is well-defined on $E[m]$ for each m:

$$\mathrm{Gal}_k(\overline{k}) \to \mathrm{Aut}(E[m]) \cong \mathrm{GL}_2(\mathbb{Z}/m\mathbb{Z})$$

These are useful, however the fact that the coefficients in the matrix representation have torsion makes these modules difficult to work with individually. This can be remedied by considering all $E[m^\ell]$ groups at once by taking the limit of these groups:

Definition 1.8.1: Tate Module

Let E be an elliptic curve and $\ell \in \mathbb{Z}$ be a prime. Then the (ℓ)-*adic Tate module* of E is the group:

$$T_\ell(E) = \varprojlim_n E[\ell^n]$$

via the diagram given by

$$E[\ell^{n+1}] \xrightarrow{[\ell]} E[\ell]$$

Naturally, $T_\ell(E)$ has a $\mathbb{Z}_\ell$-module structure, and as each map $[\ell]$ is surjective the limit topology on $T_\ell(E)$ is isomorphic to the ℓ-adic topology as a $\mathbb{Z}_\ell$-module. Furthermore, the galois action commutes with the multiplication by ℓ map, hence there is an action $\mathrm{Gal}_k(\overline{k}) \curvearrowright T_\ell(E)$. As the profinite group $\mathrm{Gal}_k(\overline{k})$ acts continuously on each discrete group $E[\ell^n]$, by construction it acts continuously on $T_\ell(E)$ giving us the continuous representation

Definition 1.8.2: ℓ-adic Representation

Let E be an elliptic curve and $T_\ell(E)$ be a Tate module. Then an ℓ-adic representation of $T_\ell(E)$ is a homomorphism

$$\rho_\ell : \mathrm{Gal}_k(\overline{k}) \to \mathrm{Aut}(T_\ell(E))$$

1.8.1 Convention From now on, ℓ will be a prime number that is different from the characteristic of the base field k.

Proposition 1.8.3: Tate Module Structure

Let E be an elliptic curve and $T_\ell(E)$ a Tate module. Then:

$$T_\ell(E) \cong_{\mathbb{Z}_\ell} \mathbb{Z}_\ell \times \mathbb{Z}_\ell \qquad \text{if} \qquad \ell \neq \mathrm{char}(k)$$

$$T_\ell(E) \cong_{\mathbb{Z}_\ell} \{0\} \text{ or } \mathbb{Z}_\ell \qquad \text{if} \qquad \ell = \mathrm{char}(k) > 0$$

***Proof* :**
This follows from the group-structure characterization of $E[m]$ in corollary 1.7.6.

With the given convention, this is a 2-dimension representation, name by choosing a $\mathbb{Z}_\ell$-basis for $T_\ell(E)$ and the inclusion $\mathbb{Z}_\ell \hookrightarrow \mathbb{Q}_\ell$:

$$\mathrm{Gal}_k(\overline{k}) \to \mathrm{GL}_2(\mathbb{Z}_\ell) \hookrightarrow \mathrm{Gal}_2(\mathbb{Q}_\ell)$$

or if avoiding choosing a basis we get the natural:

$$\rho_\ell : \mathrm{Gal}_k(\overline{k}) \to \mathrm{Aut}(T_\ell(E)) \hookrightarrow \mathrm{Aut}(T_\ell(E)) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$$

Tate-module shall be give “local” information about our isogenies. Notice that if φ is an isogeny:

$$\varphi : E_1 \to E_2$$

Then it induces a map for each $E[\ell^n]$:

$$\varphi_1 : E_1[\ell^n] \to E_2[\ell^n]$$

which naturally commutes with the $[\ell]$ map, and hence there is a map:

$$\varphi_\ell : T_\ell(E_1) \to T_\ell(E_2)$$

giving a map:

$$\mathrm{Hom}(E_1, E_2) \to \mathrm{Hom}(T_\ell(E_1), T_\ell(E_2))$$

and if $E_1 = E_2 = E$ then there is a map:

$$\mathrm{End}(E) \to \mathrm{End}(T_\ell(E))$$

Note that if φ was defined over a field k (so that it commutes with $\mathrm{Gal}_k(\overline{k})$ Then we can define

$$\mathrm{Hom}_k(T_\ell(E_1), T_\ell(E_2))$$

to be the group of $\mathbb{Z}_\ell$-linear maps from $T_\ell(E_1)$ to $T_\ell(E_2)$ that commute with the action fo $\mathrm{Gal}_k(\overline{k})$.

Theorem 1.8.4: Endomorphism To ℓ-adic is Injective

let E be an elliptic elliptic curve. Then for given ℓ,

$$\mathrm{End}(E) \hookrightarrow \mathrm{End}(T_\ell(E))$$

is injective

Proof **:**
This was left as an exercise by Silverman, do it!

A much more striking result is that "localizing" is an injective process!

Theorem 1.8.5: Localizing Isogenies

Let E_1, E_2 be elliptic curves and $\ell \in \mathbb{Z}$ by convention. Then the natural map

$$\mathrm{Hom}(E_1, E_2) \otimes \mathbb{Z}_\ell \hookrightarrow \mathrm{Hom}(T_\ell(E_1), T_\ell(E_2))$$

induced by $\varphi \mapsto \varphi_\ell$ is injective

This theorem is actually an isomorphism if k is the right field. This is *much* harder to prove. [8] refers to Tate's and flamings Proof. The theorem will be stated for the readers general education, but shall not be proven

Theorem 1.8.6: Isogeny Theorem

let $\ell \neq \mathrm{char}(k)$ be prime. Then if either:

1. k is a finite field
2. k is a number field

Then
$$\mathrm{Hom}_k(E_1, E_2) \otimes \mathbb{Z}_\ell \cong \mathrm{Hom}_k(T_\ell(E_1), T_\ell(E_2))$$

Proof **:**
see ref:HERE

Proof **:**
of theorem 1.8.4

Let us start with proving the following simpler claim:

Let $M \subseteq \mathrm{Hom}(E_1, E_2)$ be a finitely generated subgroup and

$$M^{\mathrm{div}} = \{\varphi \in \mathrm{Hom}(E_1, E_2) \mid [m] \circ \varphi \in M \text{ for some } m \in \mathbb{Z}_{\geq 1}\}$$

Then M^{div} is finitely generated.

To prove this statement,

FINISH HERE

Corollary 1.8.7: Rank of Hom Group

Let E_1, E_2 be elliptic curves. then $\text{Hom}(E_1, E_2)$ has rank at most 4

***Proof*:**
First, $\text{Hom}(E_1, E_2)$ is torsion-free by proposition 1.5.4. Then:

$$\text{rk}_{\mathbb{Z}}\text{Hom}(E_1, E_2) = \text{rk}_{\mathbb{Z}_\ell}\text{Hom}(E_1, E_2) \otimes \mathbb{Z}_ell$$

where if one is finite, the other must also be and these must be equal. Then by theorem 1.8.5 we can estimate

$$\text{rk}_{\mathbb{Z}_\ell}\text{Hom}(E_1, E_2) \otimes \mathbb{Z}_ell \leq \text{rk}_{\mathbb{Z}_\ell}\text{Hom}(T_\ell(E_1), T_\ell(E_2))$$

Finally, choosing a $\mathbb{Z}_\ell$-basis for the two Tate modules we get from proposition 1.8.3

$$\text{Hom}(T_\ell(E_1), T_\ell(E_2)) \cong M_2(\mathbb{Z}_\ell)$$

which has $\mathbb{Z}_\ell$-rank 4, giving the desired bound.

a comment on how the isogeny theorem has a homological interpretation

Another important theorem that ought to be stated for general education purposes but will not be proven is what is the image of ρ_ℓ. This question was answered for at least number fields by Serre:

Theorem 1.8.8: Image of Galois Representation

Let k be a number field and E/k an elliptic curve *without* complex multiplication[a]. Then

1. $\rho_\ell(\text{Gal}_k(\overline{k})$ has finite index in $\text{Aut}(T_\ell(E))$ for all prime ℓ
2. $\rho_\ell(\text{Gal}_k(\overline{k}) = \text{Aut}(T_\ell(E))$ for all but finitely many ℓ

[a]Recall that means $\text{End}(E) \cong \mathbb{Z}$

***Proof*:**
see cite:HERE

Silverman gives a comment that if E has CM then the image of the gal rep must be abelian! p.92

1.9 Weil Pairing and Natural Functionals

Let E/K be an ellitpic curve and $m \geq 2$ where $\gcd(\text{char}(K), m) = 1$ if $\text{char}(K) > 0$.

By corollary 1.7.6,

$$E[m] \cong_{\mathbf{Grp}} \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$$

These are points on the elliptic curve, or as seen on a flat torus they are the points corresponding to $(1/n)\Lambda/\Lambda$. There is thus a natural notion of area. In particular, what would be nice is to have a non-generate alternative bilinear map:

$$\det : E[m] \times E[m] \to \mathbb{Z}/m\mathbb{Z} \qquad \det(aT_1 + bT_2, cT_3 + dT_4) = ad - bc$$

However, this definition is not basis independent (though it is independent up to a multiple of $(\mathbb{Z}/m\mathbb{Z})$) and it is not invariant under the galois action. However, such a functional would be natural to define given the geometric properties of torsion points; we endeavor to "fix" this.

Recall by corollary 1.4.5 that D is a principal divisor if:

$$\sum_{P\in E} n_P = 0 \qquad \text{and} \qquad \sum_{P\in E} [n_P]P = O$$

Let $T \in E[m]$. As $m \geq 2$, there exists a $f \in \overline{k}(E)$ such that

$$\operatorname{div}(f) = m(T) - m(O)$$

Now, as $[m]$ is a surjective isogeny, consider T' where $[m]T' = T$ (note that mean $T' \in E[m^2]$). Then there is a function $g \in \overline{k}(E)$ such that

$$\operatorname{div}(g) = [m]^*(T) - [m]^*(O) = \sum_{R\in E[m]} ((T' + R) - (R))$$

(recall that $\#E[m] = m^2$ and that $[m^2]T' = O$). Notice that:

$$\operatorname{div}(f \circ [m]) = m\operatorname{div}(g)$$

Therefore, by choosing an appropriate element in $\overline{k}$ to multiply f we have that:

$$f \circ [m] = g^m$$

Now let $S \in E[m]$ be a torsion point (where possibly $S = T$). Then for any $X \in E$:

$$g(X + S)^m = f([m]X + [m]S) = f([m]X) = g(X)^m$$

Thus,

$$\frac{g(X + S)^m}{g(X)^M} = 1 \qquad \text{therefore} \qquad \frac{g(X + S)^m}{g(X)^M} \in \mu_m$$

Therefore, as a function of X, $g(X + S)/g(X)$ takes on only finitely many values (i.e. the mth roots of unity) and the morphism

$$E \to \mathbb{P}^1 \qquad X \mapsto \frac{g(X + S)}{g(X)}$$

is not surjective, and hence is constant. Therefore, the following is well-defined

Definition 1.9.1: Weil Pairing

Let E/K be an elliptic curve, $m \geq 2$ where $\gcd(\mathrm{char}(K), m) = 1$ if $\mathrm{char}(K) > 0$. Then:

$$e_m : E[m] \times E[m] \to \mu_m \qquad e_m(S,T) = \frac{g(X+S)}{g(X)}$$

where $X \in E$ is any point where $g(X+S)$ and $g(X)$ are both defined and nonzero.

Proposition 1.9.2: Properties of Weil Pairing

Let E/k be an elliptic curve, and e_m the Weil-pairings for each $m \geq 2$. Then

1. (bilinear)
$$e_m(S_1+S_2,T) = e_m(S_1,T)e_m(S_2,T)$$
$$e_m(S,T_1+T_2) = e_m(S,T_1)e_m(S,T_2)$$

2. (alternating)
$$e_m(T,T) = 1 \qquad \text{and} \qquad e_m(S,T) = e_m(T,S)^{-1}$$

3. (nondegenerate)
$$\mathrm{it}e_m(S,T) = 1 \quad \forall S \in E[m], \qquad \text{then } T = O$$

4. (Galois Invariant) For all $\sigma \in \mathrm{Gal}_k(\overline{k})$
$$e_m(S,T)^\sigma = e_m(S^\sigma, T^\sigma)$$

5. (compatibility)
$$e_{mm'}(S,T) = e_m([m']S,T) \qquad \forall S \in E[mm'],\ T \in E[m]$$

***Proof* :**

1. For the first factor:
$$e_m(S_1+S_2,T) = \frac{g(X+S_1+S_2)}{g(X)} = \frac{g(X+S_1+S_2)}{g(X+S_1)}\frac{g(X+S_1)}{g(X)}$$
$$= e_m(S_2,T)e_m(S_1,T).$$

   Note how any value of Y can be chosen in computing $e_m(S_2,T) = g(Y+S_2)/g(Y)$; in this case $Y = X + S_1$.

   For the second factor, let $f_1, f_2, f_3, g_1, g_2, g_3$ be the appropriate functions for the points T_1, T_2, and $T_3 = T_1 + T_2$. Choose a function $h \in K(E)$ with divisor
$$\mathrm{div}(h) = (T_1+T_2) - (T_1) - (T_2) + (O)$$
   Then
$$\mathrm{div}\left(\frac{f_3}{f_1 f_2}\right) = m\ \mathrm{div}(h)$$

so

$$f_3 = cf_1f_2h^m \text{ for some } c \in \overline{K}^*$$

Next, composing with $[m]$ and recalling that $f_i \circ [m] = g_i^m$, by taking the mth root:

$$g_3 = c' \cdot g_1 \cdot g_2 \cdot (h \circ [m]) \quad \text{for some } c' \in \overline{K}^*$$

Then:

$$\begin{aligned} e_m(S, T_1 + T_2) &= \frac{g_3(X+S)}{g_3(X)} \\ &= \frac{g_1(X+S)g_2(X+S)h([m]X+[m]S)}{g_1(X)g_2(X)h([m]X)} \\ &= e_m(S,T_1)e_m(S,T_2) && \text{as } [m]S = O. \end{aligned}$$

2. the rest can be found in Silverman ([8, chapter III.8])

Corollary 1.9.3: Surjectivity of Weil Pairing

Let E/k be an elliptic curve an e_m the Weil pairings for $m \geq 2$. Then there exists points $S, T \in E[m]$ such that $e_m(S,T)$ is a primitive mth root of unity. Thus,

$$\text{If } E[m] \subseteq E(k) \text{ then } \mu_m \subseteq k^*.$$

***Proof* :**
Let $\mathrm{im} e_m = \mu_d$ as e_m ranges over all $S, T \in E[m]$. Then:

$$1 = e_m(S,T)^d = e_m([d]S, T)$$

for all $S, T \in E[m]$. By nondegeneracy, $[d]S = O$. As $S \in E[m]$ is arbitrary, m must divide d but then $m = d$.

Next ,if $E[m] \subseteq E(k)$, the galois invariance of the e_m-pairing implies $e_m(S,T) \in k^*$ for all $S, T \in E[m]$. But then $\mu_m \subseteq k^*$, completing the proof.

The Weil pairing can also be used to show how φ and $\hat{\varphi}$ are adjoint to each other

Proposition 1.9.4: Weil Pairing and Isogeny-adjoint

Let $\varphi : E_1 \to E_2$ be an isogeny. Then for all $S \in E_1[m]$ and $T \in E_2[m]$

$$e_m(S, \hat{\varphi}(T)) = e_m(\varphi(S), T)$$

***Proof* :**
Let

$$\mathrm{div}(f) = m(T) - m(O) \quad \text{and} \quad f \circ [m] = g^m$$

be as usual. Then

$$e_m(\varphi S, T) = \frac{g(X + \varphi S)}{g(X)}$$

Choose a function $h \in \overline{K}(E_1)$ satisfying

$$\varphi^*((T)) - \varphi^*((O)) = (\hat{\varphi}T) - (O) + \mathrm{div}(h)$$

Such an h exists by theorem 1.7.1 which tells us that $\hat{\varphi}T$ is precisely the sum of the points of the divisor on the left-hand side of this equality. Now, observe that

$$\mathrm{div}\left(\frac{f \circ \varphi}{h^m}\right) = \varphi^*\mathrm{div}(f) - m\mathrm{div}(h) = m(\hat{\varphi}T) - m(O)$$

and

$$\left(\frac{g \circ \varphi}{h \circ [m]}\right)^m = \frac{f \circ [m] \circ \varphi}{(h \circ [m])^m} = \left(\frac{f \circ \varphi}{h^m}\right) \circ [m]$$

Then by directly using the definition of the e_m-pairing:

$$\begin{aligned} e_m(S, \hat{\varphi}T) &= \frac{(g \circ \varphi / h \circ [m])(X + S)}{(g \circ \varphi / h \circ [m])(X)} \\ &= \frac{g(\varphi X + \varphi S)}{g(\varphi X)} \cdot \frac{h([m]X)}{h([m]X + [m]S)} \\ &= e_m(\varphi S, T) \end{aligned}$$

as we sought to show.

We next extend these results to Tate-modules. Like in section 1.8, let ℓ be a prime number different from char(k). By combining each $e_{\ell^n} : E[\ell^n] \times E[\ell^n] \to \mu_{\ell^n}$ we get the ℓ-adic Weil pairing on the Tate-module:

$$e : T_\ell(E) \times T_\ell(E) \to T_\ell(\mu) \tag{1.4}$$

where $T_\ell(\mu)$ is formed by taking the limit over:

$$e : mu_{\ell^{n+1}} \xrightarrow{\zeta \mapsto \zeta^\ell} \mu_{\ell^n}$$

It must be shown that these pairing are compatible, namely that:

$$e_{\ell^{n+1}}(S, T)^\ell = e_{\ell^n}([\ell]S, [\ell]T) \qquad \forall S, T \in E[\ell^{n+1}]$$

Note by linearity that

$$e_{\ell^{n+1}}(S, T)^\ell = e_{\ell^{n+1}}(S, [\ell]T)$$

Then by applying proposition 1.9.2(5) to S and $[\ell]T$ with $m = \ell^n$ and $m' = \ell$, we get the desired compatibility showing that e from equation (1.4) is well-defined.

Proposition 1.9.5: Properties of Tate-module Weil Pairing

Given the above, the map:

$$e : T_\ell(E) \times T_\ell(E) \to T_\ell(\mu)$$

is bilinear, alternating, nondegenerate, and galois invariant. If $\varphi : E_1 \to E_2$ is an isogeny, the nφ and $\hat{\varphi}$ are adjoints for the pairing:

$$e(\varphi S, T) = e(S, \hat{\varphi} T)$$

Proof **:**
exercise

Finally, we use the Weil pairing to show that we can use the determinant and the trace to compute the degree of an isogeny.

Proposition 1.9.6: Degree, Determinant, and Trace

Let $\varphi \in \mathrm{End}(E)$ and $\varphi_\ell : T_\ell(E) \to T_\ell(E)$ the induced map. Then

$$\det(\varphi_\ell) = \deg(\varphi) \qquad \mathrm{tr}(\varphi_\ell) = 1 + \deg(\varphi) - \deg(1 - \varphi)$$

Proof **:**
Let $\{v_1, v_2\}$ be a $\mathbb{Z}_\ell$-basis for $T_\ell(E)$ and write

$$\varphi_\ell(v_1) = av_1 + bv_2, \quad \varphi_\ell(v_2) = cv_1 + dv_2$$

so the matrix of φ_ℓ relative to this basis is

$$\varphi_\ell = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Using properties of the Weil pairing:

$$\begin{aligned}
e(v_1, v_2)^{\deg \varphi} &= e([\deg \varphi] v_1, v_2) && \text{bilinearity of } e, \\
&= e(\hat{\varphi}_\ell \varphi_\ell v_1, v_2) && \textit{theorem } 1.7.1, \\
&= e(\varphi_\ell v_1, \varphi_\ell v_2) && \textit{proposition } 1.9.5,\ \textit{proposition } 1.7.3 \\
&= e(av_1 + bv_2, cv_1 + dv_2) \\
&= e(v_1, v_2)^{ad - bc} && \text{since } e \text{ is bilinear and alternating} \\
&= e(v_1, v_2)^{\det \varphi_\ell}
\end{aligned}$$

Since e is nondegenerate, we conclude that $\deg \varphi = \det \varphi_\ell$. Finally, for any 2×2 matrix A, a trivial calculation yields

$$\mathrm{tr}(A) = 1 + \det(A) - \det(1 - A)$$

1.10 Endomorphism Ring

Let us characterize $\mathrm{End}(E)$ further. From proposition 1.5.4 we have that $\mathrm{End}(E)$ has characteristic 0, has no divisors, and rank at most 4 as a $\mathbb{Z}$-module. It also has an anti-involution $\varphi \mapsto \hat{\varphi}$ with the property that $\varphi\hat{\varphi} = [m]$ (where $\varphi\hat{\varphi} = [0]$ if and only if $\varphi = 0$). As it turns out, rings satisfying this conditions can be classified with some basic algebraic objects, namely we will say that are certain *orders* as defined in [5, chapter 22.7]

Definition 1.10.1: Order

Let $\mathcal{K}$ be a (not necessarily commutative) $\mathbb{Q}$-algebra that is finitely generated over $\mathbb{Q}$. Then an *order* $\mathcal{R} \subseteq \mathcal{K}$ is a subring that is a finitely generated $\mathbb{Z}$-module and $\mathcal{R} \otimes \mathbb{Q} = \mathcal{K}$

As a reminder, if $K/\mathbb{Q}$ is a quadratic imaginary field, $\mathcal{O}_K$ the corresponding ring of integers, and $f \geq 1$, then $\mathbb{Z} + f\mathcal{O}_K$ is an order of $K/\mathbb{Q}$ (in [5], it was an exercise to show these are all orders). In fact, the orders we are interesting in shall arise via quadratic algebras. This was defined in [5, chapter 23])

Definition 1.10.2: Quaternion Algebras

A *quaternion algebra* over $\mathbb{Q}$ is a central-simple algebra $\mathbb{Q}$ that has dimension 4. The *definite* or *non-split* quaternion $\mathbb{Q}$-algebra of the form:

$$\mathcal{K} = \mathbb{Q} + \alpha\mathbb{Q} + \beta\mathbb{Q} + \alpha\beta\mathbb{Q}$$

where

$$\alpha^2, \beta^2 \in \mathbb{Q} \qquad \alpha^2, \beta^2 < 0 \qquad \beta\alpha = -\alpha\beta$$

Theorem 1.10.3: Characterizing Endomorphism Ring of Elliptic Curve

Let $\mathcal{R}$ be a ring with characteristic 0 with no zero divisors (a domain) such that

1. $\mathcal{R}$ has rank at most 4 as a $\mathbb{Z}$-module
2. $\mathcal{R}$ has an anti-involution $\alpha \mapsto \hat{\alpha}$ satisfying

$$\widehat{\alpha + \beta} = \hat{\alpha} + \hat{\beta} \qquad \widehat{\alpha\beta} = \hat{\alpha}\hat{\beta} \qquad \hat{\hat{\alpha}} = \alpha$$

   and if $a \in \mathbb{Z} \subseteq \mathcal{R}$, $\hat{a} = a$

3. For all $\alpha \in \mathcal{R}$, $\alpha\hat{\alpha} \in \mathbb{N}$ and $\alpha\hat{\alpha} = 0$ if and only if $\alpha = 0$.

Then $\mathcal{R}$ is isomorphic to one of three rings:

1. $\mathbb{Z}$
2. order in an imaginary quadratic extension of $\mathbb{Q}$
3. order of a quaternion algebra over $\mathbb{Q}$

***Proof* :**
Let $\mathcal{K} = \mathcal{R} \otimes \mathbb{Q}$ and extend the anti-involution in the natural way. If $\mathcal{K} = \mathbb{Q}$, then by algebra $\mathcal{R} = \mathbb{Z}$. Otherwise, by definition of order, it suffices to prove $\mathcal{K}$ a quadratic field extension, or a quaternion algebra over $\mathbb{Q}$. Denote the norm and the trace of the anti-involution by:

$$N(\alpha) = \alpha\hat{\alpha} \qquad T(\alpha) = \alpha + \hat{\alpha}$$

It is quick to see that the trace is $\mathbb{Q}$-linear (as the involution fixes $\mathbb{Q}$), and if $\alpha \in \mathbb{Q}$, $T(\alpha) = 2\alpha$. Via some arithmetic's, it could be verified that:

$$T(\alpha) = 1 + N(\alpha) + N(\alpha - 1)$$

And if $T(\alpha) = 0$ then

$$0 = 0 \cdot (\alpha - \hat{\alpha}) = (\alpha - \alpha)(\alpha - \hat{\alpha}) = \alpha^2 - T(\alpha)\alpha + N(\alpha) = \alpha^2 + N(\alpha)$$

giving

$$N(\alpha) = -\alpha^2$$

Thus, we get that:

$$\alpha \neq 0 \quad \text{and} \quad T(\alpha) = 0 \implies \alpha^2 \in \mathbb{Q} \quad \text{and} \quad \alpha^2 < 0$$

As it is assumed that $\mathcal{K} \neq \mathbb{Q}$, choose some $\alpha \in \mathcal{K} \setminus \mathbb{Q}$. Replacing α with $\alpha - \frac{1}{2}T(\alpha)$ we get that $T(\alpha) = 0$, so $\alpha^2 \in \mathbb{Q}$ and $\alpha^2 < 0$, implying $\mathbb{Q}(\alpha)$ is a quadratic imaginary field. If $\mathcal{K} = \mathbb{Q}(\alpha)$, we're done.

Suppose not. Then there exists an element $\beta \in \mathcal{K} \setminus \mathbb{Q}(\alpha)$. Replace β with:

$$\beta - \frac{1}{2}T(\beta) - \frac{T(\alpha\beta)}{2\alpha^2}\alpha$$

This is done since, as $T(\alpha) = 0$ and $\alpha^2 \in \mathbb{Q}^*$

$$T(\beta) = T(\alpha\beta) = 0$$

so $\beta^2 \in \mathbb{Q}$ and $\beta^2 < 0$. Now, as the trace of $\alpha, \beta, \alpha\beta$ is 0, using the norm we get:

$$\alpha = -\hat{\alpha}, \ \beta = -\hat{\beta}\alpha\beta = -\hat{\beta}\hat{\alpha}$$

substituting we get

$$\alpha\beta = -\beta\alpha$$

Thus,

$$\mathbb{Q}[\alpha, \beta] = \mathbb{Q} + \alpha\mathbb{Q} + \beta\mathbb{Q} + \alpha\beta\mathbb{Q}$$

is a quaternion algebra. What's left to show is $\mathcal{K} = \mathbb{Q}[\alpha, \beta]$. As $\mathbb{Q}[\alpha, \beta] \subseteq \mathcal{K}$, If $1, \alpha, \beta, \alpha\beta$ are linearly independent, by corollary 1.8.7 it can be concluded that $\mathbb{Q}[\alpha, \beta] = \mathcal{K}$. To that end, consider the $\mathbb{Q}$-linear combination:

$$w + x\alpha + y\beta + z\alpha\beta = 0$$

By taking the trace, we get $2w = 0$, so $w = 0$. Then multiplying by α on the left hand side and β on the right hand side we get:

$$(x\alpha^2)\beta + (y\beta^2)\alpha + z\alpha^2\beta^2 = 0$$

Then since $1, \alpha, \beta$ are $\mathbb{Q}$-linearly independent (as $\alpha \in \mathbb{Q}$, $\beta \in \mathbb{Q}(\alpha)$), we get:

$$x\alpha^2 = y\beta^2 = z\alpha^2\beta^2 = 0$$

hence $x = y = z = 0$, giving linear independence, hence $\mathbb{Q}[\alpha, \beta]$ is dimension 4, hence $\mathcal{K} = \mathbb{Q}[\alpha, \beta]$ completing the proof.

Corollary 1.10.4: Classifying Endomorphism Ring in Char 0

Let E/K be an elliptic curve. Then $\mathrm{End}_K(E)$ is either isomorphic to:

1. $\mathbb{Z}$
2. an order of a quadratic imaginary field $\mathbb{Q}(\alpha)$
3. an order of a quaternion algebra $\mathbb{Q}[\alpha, \beta]$

If $\mathrm{char}(K) = 0$, only the first two are possible

Proof **:**
Theorem 1.10.3 gives the classification. If $\mathrm{char}(K) = 0$, then proposition 1.6.6 proved $\mathrm{End}_K(E)$ must be commutative, hence it cannot be an order of a quaternion algebra.

If $K = \mathbb{F}_q$, it shall be shown that $\mathrm{End}(E)$ will always be larger than $\mathbb{Z}$, and furthermore there are always elliptic curve $E/\mathbb{F}_{p^2}$ such that $\mathrm{End}(E)$ is a quaternion algebra.

It shall be useful for number-theoretical purposes to give the following two definitions:

Definition 1.10.5: Split Quaternion Algebra

Let p be prime or ∞, $\mathbb{Q}_p$ the p-adics if p is finite and $Q_\infty = \mathbb{R}$. A quaternion algebra is said to *split* if

$$\mathcal{K} \otimes_{\mathbb{Q}} \mathbb{Q}_p \cong M_2(\mathbb{Q}_p)$$

otherwise, $\mathcal{K}$ is said to be *ramified* at p. The *invariant* of $\mathcal{K}$ at p is given by:

$$\mathrm{inv}_p(\mathcal{K}) = \begin{cases} 0 & \mathcal{K} \text{ splits at } p \\ \frac{1}{2} & \mathcal{K} \text{ ramifies at } p \end{cases}$$

Theorem 1.10.6: Numeric Properties of Quaternion Algebra

Let $\mathcal{K}$ be a quaternion algebra. Then

1. $\text{inv}_p(\mathcal{K}) = 0$ for all but finitely many p (including ∞) and
$$\sum_p \text{inv}_p(\mathcal{K}) \in \mathbb{Z}$$
2. $\mathcal{K} \cong_{\mathbb{Q}} \mathcal{K}'$ as $\mathbb{Q}$-algebras if and only if $\text{inv}_p(\mathcal{K}) = \text{inv}_p(\mathcal{K}')$ for all p

This is a special case of the classification of Central Simple Algebras By the Brauer group (see [2])

***Proof* :**
This proof is given in detail in [2] namely we have the short exact sequence

$$0 \to \text{Br}(\mathbb{Q}) \to \bigoplus_p \text{Br}(\mathbb{Q}_p) \xrightarrow{\sum_p \text{inv}_p} \frac{\mathbb{Q}}{\mathbb{Z}} \to 0$$

where

$$\text{Br}(\mathbb{Q}_p) \xrightarrow{\cong} \begin{cases} \mathbb{Q}/\mathbb{Z} & p \neq \infty \\ \{0, \frac{1}{2}\} & p = \infty \end{cases}$$

Then the result is given by considering that quaternions correspond to elements of order 2 in $\text{Br}(\mathbb{Q})$.

1.11 Automorphism Group

As compared to the endomorphism ring, the automorphism group is simpler to calculate

Theorem 1.11.1: Classification of Automorphism Group of Elliptic Curve

Let E/K be an elliptic curve. Then $\text{Aut}(E)$ is a finite automorphism group whose order divides 24. In particular, its order can be summarized by the following:

$\#\,\text{Aut}(E)$	$j(E)$	$\text{char}(K)$
2	$j(E) \neq 0, 1728$	$-$
4	$j(E) = 1728$	$\text{char}(K) \neq 2, 3$
6	$j(E) = 0$	$\text{char}(K) \neq 2, 3$
12	$j(E) = 0 = 1728$	$\text{char}(K) = 3$
24	$j(E) = 0 = 1728$	$\text{char}(K) = 2$

***Proof* :**
Let $\text{char}(K) \neq 2, 3$. Then E is given by

$$E : y^2 = x^3 + ax + b$$

Every automorphism of E has the form

$$x = u^2 x' \qquad y = u^3 y'$$

for some $u \in \overline{K}^*$. This substitution is an automorphism of E if and only if

$$u^{-4} a = a \qquad u^{-6} b = b$$

if $ab \neq 0$ (i.e. $j(E) \neq 1728$), then the only possibility if $u = \pm 1$. IF $b = 0$, then $j(E) = 1728$ and $u^4 = 1$ and if $a = 0$ and $j(E) = 0$ and $a^6 = 1$. Thus, $\mathrm{Aut}(E)$ has order 2, 4, or 6 depending on a, b. A similar argument holds for $\mathrm{char}(K) = 2, 3$ with more care towards the given Weierstrass equation.

As seen in section 1.8, $\mathrm{Aut}(E)$ has a natural $\mathrm{Gal}_k(\overline{k})$-action on it (at least when $\mathrm{char}(k) \neq 2, 3$). This gives the following:

Corollary 1.11.2: Group of Unity and Aut(E)

Let E/K be an elliptic curve, $\mathrm{char}(K) \neq 2, 3$, and

$$n = \begin{cases} 2 & j(E) \neq 0, 1728 \\ 4 & j(E) = 1728 \\ 6 & j(E) = 0 \end{cases}$$

Then there is a natural isomorphism of $\mathrm{Gal}_k(\overline{k})$-modules (letting $G = \mathrm{Gal}_k(\overline{k})$

$$\mathrm{Aut}(E) \cong_G \mu_n$$

where μ_n are the roots of unity.

***Proof* :**
The map

$$[-] : \mu_n \to E \qquad [\zeta](x, y) = (\zeta^2 x, \zeta^3 y)$$

is an isomorphism of abstract groups. Certainly the map commutes with the G-action, hence it is a G-module isomorphism with the action of $\mathrm{Aut}(E)$, completing the proof.

Exercise 1.11.1

1. (Node and Cusp) (move over the hw here)
2. let $f_1, ..., f_n \in K(C)$ where $K(C)$ is a function field of a curve, $p \in C$, and $\nu_p(f_k) = -k$ (or more generally, each have a different valuation at the point p. Show that $f_1, ..., f_n$ are linearly linearly independent. Show this is no longer necessary if there exists two terms f_i, f_j that have the same valuation at p and are maximal.
3. Find a point of order 3 on E

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Formal Group of Elliptic Curve

This is like the Lie Algebra. It is fundamentally tied to K-theory and chromatic homotopy theory so I really want to get back to this!

2.1 Expansion around O

To study better the origin, it is nicer if in coordinate representation $O = [0 : 0 : 1]$ or $O = (0, 0)$. To that end, let

$$z = \frac{-x}{y},\ w = \frac{-1}{y} \qquad \text{hence} \qquad x = \frac{z}{w},\ y = \frac{-1}{w}$$

Then the Weierstrass equation becomes

$$w = z^3 + a_1 zw + a_2 z^2 w + a_3 w^2 + a_4 z w^2 + a_6 w^3 = f(z, w)$$

To write w as an algebraic function of z is impossible, but by recursively plugging in the right hand side into each w, "at the limit" we get a power series:

$$\begin{aligned}
w &= z^3 + (a_1 z + a_2 z^2)w + (a_3 + a_4 z)w^2 + a_6 w^3 \\
&= z^3 + (a_1 z + a_2 z^2)[z^3 + (a_1 z + a_2 z^2)w + (a_3 + a_4 z)w^2 + a_6 w^3] \\
&\quad + (a_3 + a_4 z)[z^3 + (a_1 z + a_2 z^2)w + (a_3 + a_4 z)w^2 + a_6 w^3]^2 \\
&\quad + a_6[z^3 + (a_1 z + a_2 z^2)w + (a_3 + a_4 z)w^2 + a_6 w^3]^3 \\
&\vdots \\
&= z^3 + a_1 z^4 + (a_1^2 + a_2)z^5 + (a_1^3 + 2a_1 a_2 + a_3)z^6 \\
&\quad + (a_1^4 + 3a_1^2 a_2 + 3a_1 a_3 + a_2^2 + a_4)z^7 + \cdots \\
&= z^3(1 + A_1 z + A_2 z^2 + \cdots)
\end{aligned}$$

where each $A_n \in \mathbb{Z}[a_1, ..., a_6]$ is a polynomial in the coefficients of E.

Proposition 2.1.1: Properties of Power Series Expansion

Let E/K be an elliptic curve over K. Then:

1. The above procedure gives a power series
$$w(z) = z^3(1 + A_1 z +_2 z^2 + \cdots) \in \mathbb{Z}[a_1, ..., a_6]\,[[z]]$$
2. The series $w(z)$ is the unique power series in $\mathbb{Z}[a_1, ..., a_6][[z]]$ satisfying
$$w(0) = 0 \qquad \text{and} \qquad w(z) = f(z, w(z))$$
3. If $\mathbb{Z}[a_1, ..., a_6]$ is given a graded ring structure with weights given by $\mathrm{wt}(a_i) = i$, then A_n is a homogeneous polynomial of weight n.

The key result is a modification of Hensel's lemma (see [2] for the usual Hensel's lemma for lifting polynomials factored mod $\mathfrak{p}$). Recall that a corollary of Hensel's lemma is that if $f \in R[x]$ where R is a DVR, then if $\overline{f} \in \kappa[x]$ has a simple root, the root can be lifted. This result is true even if the root if the root "starts" at $\mathfrak{p}^n$. Furthermore, as long as R is complete with respect to an ideal $I \subseteq R$, with some mild extra condition's Hensel's lemma can be recovered:

Theorem 2.1.2: Hensel's Lemma – Augmented

Let R be a ring, complete with respect to to $I \subseteq R$, and let $F(w) \in R[w]$. Suppose that there exists an integer $n \geq 1$ and an element $a \in R$ st

$$F(a) \in I^n \qquad \text{and} \qquad F'(a) \in R^*$$

Then for any $\alpha \in R^*$ satisfying $\alpha = F'(a) \bmod I$, the sequence:

$$w_0 = a \qquad w_{n+1} = w_m - \frac{F(w_m)}{\alpha}$$

converges to an element $b \in R$ satisfying

$$F(b) = 0 \qquad \text{and} \qquad b = a(\bmod I^n)$$

if R is an integral domain, these conditions determine b uniquely.

Proof **:**
First, note that $\alpha \in F'(a) + I \subset R^* + I = R^*(1 + I) = R^*$, so α is a unit. To make the recurrence more clear, replace $F(w)$ by $F(w+a)/\alpha$, which gives the recurrence

$$w_0 = 0, \quad F(0) \in I^n, \quad F'(0) \equiv 1 \pmod{I}, \quad w_{m+1} = w_m - F(w_m).$$

Since $F(0) \in I^n$,

$$w_m \in I^n \implies w_m - F(w_m) \in I^n,$$

from which it follows that

$$w_m \in I^n \text{ for all } m \geq 0.$$

Now, by induction:

$$w_m \equiv w_{m+1} \pmod{I^{m+n}} \text{ for all } m \geq 0.$$

Indeed, for $m = 0$, this just says that $F(0) \equiv 0 \pmod{I^n}$, which is one of the initial assumptions. Assume now that the desired congruence is true for all integers strictly smaller than m. Let X and Y be new variables and factor

$$F(X) - F(Y) = (X - Y)(F'(0) + XG(X,Y) + YH(X,Y)),$$

where G and H are polynomials in $R[X,Y]$. Then

$$\begin{aligned} w_{m+1} - w_m &= (w_m - F(w_m)) - (w_{m-1} - F(w_{m-1})) \\ &= (w_m - w_{m-1}) - (F(w_m) - F(w_{m-1})) \\ &\stackrel{!}{=} (w_m - w_{m-1})(1 - F'(0) - w_m G(w_m, w_{m-1}) \\ &\qquad\qquad - w_{m-1} H(w_m, w_{m-1})) \in I^{m+n}. \end{aligned}$$

where $\stackrel{!}{=}$ follows from the induction hypothesis and the assumptions that $F'(0) \equiv 1 \pmod{I}$ and $w_m, w_{m-1} \in I^n$. Hence:

$$w_m - w_{m+1} \in I^{m+n} \text{ for all } m \geq 0.$$

Now, as R is complete with respect to I, it follows that the sequence w_m converges to an element $b \in R$, and since every $w_m \in I^n$, $b \in I^n$. Furthermore, taking the limit of the relation $w_{m+1} = w_m - F(w_m)$ as $m \to \infty$ yields $b = b - F(b)$, so $F(b) = 0$.

Finally, if R is an integral domain, to show uniqueness suppose that $c \in I^n$ satisfies $F(c) = 0$. Then

$$0 = F(b) - F(c) = (b - c)(F'(0) + bG(b, c) + cH(b, c)).$$

If $b \neq c$, then $F'(0) + bG(b, c) + cH(b, c) = 0$, which would imply that

$$F'(0) = -bG(b, c) - cH(b, c) \in I.$$

This contradicts the assumption that $F'(0) \equiv 1 \pmod{I}$. Hence $b = c$.

***Proof* :**
of proposition 2.1.1

1. Use Hensel's lemma on

$$R = \mathbb{Z}[a_1, ..., a_6]\,[[z]] \qquad I = (z)$$

2. Use Hensel's lemma on

$$F(w) = f(z, w) - w \qquad a = 0 \qquad \alpha = -1$$

3. Define:

$$f_1(z, w) = f(z, w) \qquad f_{m+1}(z, w) = f_m(z, f(z, w))$$

Then by what we've shown:

$$w(z) = \lim_{m \to \infty} f_m(z, 0)$$

Assign the weights:

$$\mathrm{wt}(z) = -1 \qquad \text{and} \qquad \mathrm{wt}(w) = -3$$

Then $f(z, w)$ is homogeneous of weight -3 in $\mathbb{Z}[a_1, ..., a_6, z, w]$, and using induction we can see that $f_m(z, w)$ is homogeneous of weight -3 for every $m \geq 1$. Then in particular:

$$f_m(z, 0) = z^3(1 + B_1 z + B_2 z^2 + \cdots + B_N z^N)$$

is homogeneous of weight -3, so each $B_n \in \mathbb{Z}[a_1, ..., a_6]$ is homogeneous of weight n. Hence, A'_n must ave the same property as $f_m(z, 0)$ converges to $w(z)$ as $m \to \infty$.

Recalling that $x(z) = z/w(z)$ and $y(z) - 1/w(z)$, we get:

$$x(z) = \frac{z}{w(z)} = \frac{1}{z^2} - \frac{a_1}{z} - a_2 - a_3 z - (a_4 + a_1 a_3) z^2 - \cdots \tag{2.1}$$

$$y(z) = -\frac{1}{w(z)} = -\frac{1}{z^3} + \frac{a_1}{z^2} + \frac{a_2}{z} + a_3 + (a_4 + a_1 a_3) z + \cdots \tag{2.2}$$

And the invariant differential can be expressed as:

$$\begin{aligned}\omega(z) &= \frac{dx(z)}{2y(z)+a_1x(z)+a_3} \\ &= (1+a_1z+(a_1^2+a_2)z^2+(a_1^3+2a_1a_2+2a_3)z^3 \\ &\quad +(a_1^4+3a_1^2a_2+6a_1a_3+a_2^2+2a_4)z^4+\cdots)dz\end{aligned}$$

Certainly, the coefficients of $x(z)$ and $y(z)$ are in $\mathbb{Z}[a_1,...,a_6]$. The same can be deduced for $\omega(z)$, by looking at:

$$\begin{aligned}\frac{\omega(z)}{dz} &= \frac{dx(z)/dz}{2y+a_1x+a_3} = \frac{-2z^{-3}+\cdots}{-2z^{-3}+\cdots} \in \mathbb{Z}\left[\frac{1}{2},a_1,\ldots,a_6\right][[z]] \\ \frac{\omega(z)}{dz} &= \frac{dy(z)/dz}{3x^2+2a_2x+a_4-a_1y} = \frac{-3z^{-4}+\cdots}{-3z^{-4}+\cdots} \in \mathbb{Z}\left[\frac{1}{3},a_1,\ldots,a_6\right][[z]]\end{aligned}$$

showing the denominator is simultaneously a power of 2 and a power of 3, which is only possible for for 1, giving that the coefficients of $\omega(z)$ are in $\mathbb{Z}[a_1,...,a_6]$.

The key take-away from the construction of $x(z)$ and $y(z)$ is that the satisfy the Weierstrass equation:

$$E: y^2+a_1xy+a_3y = x^3+a_2x^2+a_4x+a_6$$

or said differently, they provide a formal solution to the Weierstrass equation. Given a field K, the usual next step is to find which values of K satisfy the above equation, however it is not necessarily the case that for each $\alpha \in K$ that $x(\alpha)$ or $y(\alpha)$ are well-defined; that is it is possible that they diverge. A special exception to this is when K is a complete nonarchemedian local field. In this case, given R is the ring of integers and $\mathfrak{m}$ the maximal ideal of R, and that $a_1,...,a_6 \in R$, the power series $x(z)$ and $y(z)$ converge for any $z \in \mathfrak{m}$[1]. There is thus an injective map

$$\mathfrak{m} \mapsto E(K) \qquad z \mapsto (x(z), y(z))$$

with inverse $(x,y) \mapsto -x/y$. This shall be important when addressing the solution of elliptic curves over local fields in chapter 4.

Though points may not be well-defined, the construction of the group-law of an elliptic curve can be done relying only on the properties of these equations and of equations of lines. Consider z_1, z_2 to be formal independent indeterminates, and $w_1 = w(z_1)$, $w_2 = w(z_2)$. Then considering a (z,w)-plane review here how to think of this, the line connection (z_1,w_1) and (z_2,w_2) has slope:

$$\lambda = \lambda(z_1,z_2) = \frac{w_2-w_1}{z_2-z_1} = \sum_{n=3}^{\infty} A_{n-3}\frac{z_2^n-z_2^n}{z_2-z_1} \in \mathbb{Z}[a_1,...,a_6][[z_1,z_2]]$$

This slope is not a number, notice even that λ has no constant or linear term. Now let:

$$\nu = \nu(z_1,z_2) = w_1 - \lambda z_1 \in \mathbb{Z}[a_1,...,a_6][[z]]$$

Then the connecting line has the simple equation:

$$\omega = \lambda z + \nu$$

[1] see [2]

Finally, substituting this into the Weierstrass equation gives a cubic in z, two of whose roots are z_1 and z_2. By looking at the quadratic term, the third root z_3 can be expressed as a power series of z_1 and z_2:

$$finish here!$$

2.2 Formal Groups

here

3

Elliptic Curve Over Finite Fields

3.1 Number of Rational Points

Let $E/\mathbb{F}_q$ be an elliptic curve over a finite field. A fundamental question of interest is to have an idea of $\#(E/\mathbb{F}_q)$. As E is defined over a finite field, the Weierstrass equation will be of the form

$$E : y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6 \qquad (x, y) \in \mathbb{F}_q^2$$

By fixing x, it can immediately seen that each $x \in \mathbb{F}_q$ can yield at most 2 values of y as this becomes a quadratic equation, given an immediate upper bound of:

$$\#(E/\mathbb{F}_q) \leq 2q + 1$$

Then by doing some field theory, notice that an arbitrary quadratic equation over $\mathbb{F}_q$ (corresponding to a quadratic extension) has a 50% of being equal to $\mathbb{F}_q$, that is the equation is being solvable over $\mathbb{F}_q$ [1]. Thus, it is expected that this bound ought to be "closer" in magnitude to q rather than $2q$. This result is exactly what E. Artin conjectured and was proving by Hasse in the 1930

Theorem 3.1.1: Hasse's Theorem On Elliptic Curves

Let $E/\mathbb{F}_q$ be an elliptic curve defined over a finite field. Then

$$|\#(E/\mathbb{F}_q) - q - 1| \leq 2\sqrt{q}$$

[1] Look at the possible discriminants of the quadratic

Proof :
Let $E/\mathbb{F}_q$ be an elliptic curve over $\mathbb{F}_q$ and let choose some Weierstrass equation to represent E. Then

$$\varphi : E \to E \qquad (x, y) \mapsto (x^q, y^q)$$

is the Frobenius endomorphism. As $\mathrm{Gal}_{\mathbb{F}_q}(\overline{\mathbb{F}}_q)$ is (topologically) generated by the qth-power map on $\overline{\mathbb{F}}_q$, we get that for any $p \in E(\overline{\mathbb{F}}_q)$

$$p \in E(\mathbb{F}_q) \qquad \text{if and only if} \qquad \varphi(p) = p$$

Thus

$$E(\mathbb{F}_q) = \ker(1 - \varphi)$$

Thus by corollary 1.6.5 and theorem 1.5.6(3)

$$\#E(\mathbb{F}_q) = \# \ker(1 - \varphi) = \deg(1 - \varphi)$$

where it is key that $1 - \varphi$ is separable.

Now, recall that $\mathrm{End}(E)$ is a positive definite quadratic form (corollary 1.7.5). $\deg(\varphi) = q$. Positive-definite quadratic forms have a Cauchy-Schwarz inequality given by:

$$|\langle \alpha, \beta \rangle|^2 \leq \deg(\alpha) \deg(\beta)$$

Since the degree is a positive definite quadratic form, we can express $\deg(1 - \varphi)$ as:

$$\deg(1 - \varphi) = \deg(1) + \deg(\varphi) - 2\langle 1, \varphi \rangle = 1 + q - 2\langle 1, \varphi \rangle$$

By the Cauchy-Schwarz inequality:

$$|\langle 1, \varphi \rangle|^2 \leq \deg(1) \cdot \deg(\varphi) = 1 \cdot q = q$$

This gives us:

$$|\langle 1, \varphi \rangle| \leq \sqrt{q}$$

Therefore:

$$1 + q - 2\sqrt{q} \leq \deg(1 - \varphi) \leq 1 + q + 2\sqrt{q}$$

Since $\#E(\mathbb{F}_q) = \deg(1 - \varphi)$, we have:

$$q + 1 - 2\sqrt{q} \leq \#E(\mathbb{F}_q) \leq q + 1 + 2\sqrt{q}$$

re-ordering we get:

$$|\#(E/\mathbb{F}_q) - q - 1| \leq 2\sqrt{q}$$

as we sought to show.

Note that though this give an upper bound on the number of points, there is no head-way in giving an algorithm for computing $\#E(\mathbb{F}_q)$. More about this shall be cover in ref:HERE.

The following is an interesting consequence. Consider the cubic

$$f(x) = ax^3 + bx^2 + cx + d \in K[x]$$

be a cubic with *distinct* roots in $\overline{\mathbb{F}}_q$. Define

$$\chi : \mathbb{F}_q^* \to \{\pm 1\}$$

where $\chi(a) = 1$ if and only if a is a square in $\mathbb{F}_q^*$. Extend the map to $\mathbb{F}_q$ by mapping $\chi(0) = 0$. Then χ can be used to cont the $\mathbb{F}_q$-rational points of of the elliptic curve

$$E : y^2 = f(x)$$

Then each $x_0 \in \mathbb{F}_q$ will give either 0, 1, or 2 points of the form $(x_0, y_0) \in E(\mathbb{F}_q)$, depending on whether $f(x_0)$ a non-square, equal to 0, or a square in $\mathbb{F}_q$ (respectively). Thus, combinatorially we can decompose $\#E(\mathbb{F}_q)$ as (remembering the point at infinity):

$$\#E(\mathbb{F}_q) = 1 + \sum_{a \in \mathbb{F}_q} (1 + \chi(f(a))) = 1 + q + \sum_{a \in \mathbb{F}_q} \chi(f(a))$$

But then from theorem 3.1.1 we get:

Corollary 3.1.2: Hasse Theorem With Quadratic Character

Let $E/\mathbb{F}_q$ be an elliptic curve and χ be the quadratic character defined above. Then:

$$\left| \sum_{a \in \mathbb{F}_q} \chi(f(a)) \right| \leq 2\sqrt{q}$$

***Proof* :**
comes from above discussion.

Silverman comments on how the cubic $f(x)$ tends to be equally distributed between square and non-square; interesting remark but maybe I"ll put it somewhere else (p.139a)

3.2 The Weil Conjectures

The above result can is in fact a consequence of the *Weil Conjectures* and is in some book proceeded after the Weil conjecture is covered. In the 19th, Emil Artin and André Weil were working on the problem of understanding the sets $\#E(\mathbb{F}_{q^n})$ for all n. In the process of research, Weil creating generating functions with coefficients $\#E(\mathbb{F}_{q^n})$. This lead to the discovery of the close connection between the Riemann hypothesis and the following set of power series:

Definition 3.2.1: Zeta Function For Varieties

Let $V/\mathbb{F}_q$ be a variety of the field $\mathbb{F}_q$. Then The *zeta function* of $V/\mathbb{F}_q$ is the power series

$$Z(V/\mathbb{F}_q; t) = \exp\left(\sum_{n=1}^{\infty} (\#V(\mathbb{F}_{q^n}) \frac{t^n}{n} \right) = e^{\left(\sum_{n=1}^{\infty} (\#V(\mathbb{F}_{q^n}) \frac{t^n}{n}\right)}$$

For future reference, if $F(t) \in \mathbb{Q}[[t]]$ has no constant terms, then

$$\exp(F(t)) = \sum_{k \geq 0} \frac{F(t)^k}{k!}$$

To recover the coefficients, we can do:

$$\#V(\mathbb{F}_{q^n}) = \frac{1}{(n-1)!} \frac{d^n}{dt^n} \log \left(Z(V/\mathbb{F}_q; t)\right)\big|_{t=0}$$

Example 3.2.1: Zeta Function of Projective Space

Let $V = \mathbb{P}^N$ be projective space of dimension N. Then $[x_0 : \cdots : x_N] \in \mathbb{P}(\mathbb{F}_{q^n})^N$ are points $x_i \in \mathbb{F}_{q^n}$ where at least one point is not zero and they are equivalent up to a constant multiple $u_i \in \mathbb{F}_{q^n}^*$. Then:

$$\#\mathbb{P}^N(\mathbb{F}_{q^n}) = \frac{q^{n(N+1)} - 1}{q^n - 1} = \sum_{i=0}^{N} q^{ni}$$

Thus

$$\log Z(\mathbb{P}^N/\mathbb{F}_q; t) = \sum_{n=1}^{\infty} \left(\sum_{i=1}^{N} q^{ni} \right) \frac{t^n}{n} = \sum_{i=0}^{N} -\log(1 - q^i t)$$

Thus:

$$Z(\mathbb{P}^N/\mathbb{F}_q; t) = \frac{1}{(1-t)(1-qT)\cdots(1-q^N t)}$$

Notice the zeta function is rational; this shall in fact be one of the central reasons to consider the exponential of the series.

Naturally, this is quickly generelizable as the key property was that $\#V(\mathbb{F}_{q^n})$ has the form:

$$\#V(\mathbb{F}_q) = \pm\alpha_1^n \pm \alpha_2^n \pm \cdots \pm \alpha_r^n$$

for $\alpha_1, ..., \alpha_r \in \mathbb{C}$; in such cases $Z(V/\mathbb{F}_q; t)$ is rational.

The above may be a simple coincidence as $\mathbb{P}^n$ is *the* smooth projective variety. The following theorem, first conjectured by Weil in 1949 and proven piece-wise over the course of 24 years shows this is in fact a very deep result:

Theorem 3.2.2: Weil Conjecture

Let $V/\mathbb{F}_q$ be a smooth projective variety of dimension N. Then:

1. (rationality) The zeta-function of $V/\mathbb{F}_q$ is rational:
$$Z(V/\mathbb{F}_q;t) \in \mathbb{Q}(t)$$
2. (functional equation) There exists an integer $\epsilon \in \mathbb{Z}$ called the *Euler characteristic* of V such that
$$Z(V/\mathbb{F}_q;1/q^N t) \pm q^{N \cdots \frac{\epsilon}{2}} t^{\epsilon} Z(V/\mathbb{F}_q;t)$$
3. (Riemann Hypothesis) The zeta- function factors as:
$$Z(V/\mathbb{F}_q;t) = \frac{P_1(t)P_3(t)\cdots P_{2N-1}(t)}{P_0(t)P_2(t)\cdots P_{2N}(t)}$$
where $P_i(t) \in \mathbb{Z}[t]$ are defined as: $P_0(t) = 1 - t$, $P_{2N}(t) = 1 - q^N t$ and
$$P_i(t) = \prod_{j=1}^{b_i}(1 - \alpha_{ij} t) \qquad |\alpha_{ij}| = q^{i/2}$$
where the quantity b_i of $P_i(t)$ is called the i^{th} *Betti number* of V.[a]

[a] For curves, the sum of the b_i are equal to $2g$ where g is the genus

Proof **:**
see [7]; or for a paper closer to the historical original in English see [6]. Historically, Weil first proved in 1949 the result for abelian varieties and curves. The general rationality with given by Dwork in 1960 via p-adic functional analysis, and then M. Artin, Grothendeick, among other mathematicians developed ℓ-adic cohomology which gives another proof of rationality as well as the providing the functional equations. Finally, Deligne in 1973 proved the Riemann Hypothesis portion of the Weil conjecture, officially closing off the conjecture.

The case for singular (projective) varieties is a little more complicated and doesn't quite follow, but there are partial results.

It was later discovered that all of these can be proven using the Riemann-Roch Theorem.

For the purposes of this textbook, the result shall be proved for elliptic curves. The key idea is that the Frobenius endomorphism can be used to generated a polynomial equation for each $\#(\mathbb{F}_{q^n})$:

Proposition 3.2.3: Properties of Frobenius Endomorphism

Let $E/\mathbb{F}_q$ be an elliptic curve and

$$\varphi : E \to E \qquad (x, y) \mapsto (x^q, y^q)$$

be the qth-power Frobenius endomorphism. Let

$$a = q + 1 - \#E(\mathbb{F}_q)$$

Then

1. if $\alpha, \beta \in \mathbb{C}$ are roots for the polynomial $T^2 - aT + q$, then α, β are complex conjugates satisfying $|\alpha| = |\beta| = \sqrt{q}$, and for every $n \geq 1$

$$\#E(\mathbb{F}_{q^n}) = q^n + 1 - \alpha^n - \beta^n$$

2. The Frobenius endomorphism satisfies

$$\varphi^2 - a\varphi + q = 0$$

in $\mathrm{End}(E)$

***Proof* :**
As computed in theorem 3.1.1:

$$\#E(\mathbb{F}_q) = \deg(1 - \varphi)$$

Then by choosing an appropriate ℓ and using proposition 1.9.6 we get:

$$\begin{aligned} \det(\varphi_\ell) &= \deg(\varphi) = q \\ \mathrm{tr}(\varphi_\ell) &= 1 + \deg(\varphi) - \deg(1 - \varphi) = 1 + q - \#(\mathbb{F}_q) = a \end{aligned}$$

Thus, the characteristic polynomial of φ_ℓ is:

$$\det(T - \varphi_\ell) = T^2 - \mathrm{tr}(\varphi_\ell)T + \det(\varphi) = T^2 - aT + q$$

1. As the characteristic polynomial of φ_ℓ has $\mathbb{Z}$-coefficents, over $\mathbb{C}$ we get:

$$\det(T - \varphi_\ell) = T^2 - aT + q = (T - \alpha)(T - \beta)$$

So then for every $m/n \in \mathbb{Q}$:

$$\det(\frac{m}{n} - \varphi_\ell) = \frac{m}{n}^2 - a\frac{m}{n} + q = \frac{\det(m - n\varphi_\ell}{n^2} = \frac{\deg(m - n\varphi}{n^2} \geq 0$$

Thus, the polynomials is non-negative for all $T \in \mathbb{R}$, so it is either has complex conjugate roots or it has a double root. In both cases, it must be that $|\alpha| = |\beta|$ and

$$\alpha\beta = \det \varphi_\ell = \deg \varphi = q$$

and so it must be that $|\alpha| = |\beta| = q$

For higher powers, it is immediate to see that for each $n \geq 1$, the q^nth Frobenius endomorphism satisfies:

$$\#(\mathbb{F}_{q^n}) = \deg(1 - \varphi^n)$$

It follows by using the JCF that the characteristic polynomial of φ_ℓ^n given by

$$\det(T - \varphi_\ell^n) = (T - \alpha^n)(T - \beta^n)$$

(note that the JCF is upper triangular with α, β on the diagonal). Then:

$$\begin{aligned} \#E(\mathbb{F}_{q^n}) &= \deg(1 - \varphi^n) \\ &= \det(1 - \varphi_\ell^n) \\ &= 1 - \alpha^n - \beta^n + q^n \end{aligned}$$

giving the desired equation

2. By The Cayley-Hamilton theorem, φ_ℓ satisfies its characteristic polynomial, thus $\varphi_\ell^2 - a\varphi_\ell + q = 0$. Thus

$$\deg(\varphi^2 - a\varphi + q) = \det(\varphi_\ell^2 - a\varphi_\ell + q) = \det(0) = 0$$

Thus $\varphi^2 - a\varphi + q \equiv 0 \in \mathrm{End}(E)$.

Theorem 3.2.4: Weil Conjecture For Elliptic Curves

Let $E/\mathbb{F}_q$ be an elliptic curve. Then there is an $a \in \mathbb{Z}$ such that

$$Z(E/\mathbb{F}_q; T) = \frac{1 - aT + qT^2}{(1 - T))\,(1 - qT)}$$

Furthermore:

$$Z\left(E/\mathbb{F}_q; \frac{1}{qT}\right) = Z(E/\mathbb{F}_q; T)$$

and

$$1 - aT + qT^2 = (1 - \alpha T)(1 - \beta T)$$

where $|\alpha| = |\beta| = \sqrt{q}$.

***Proof* :**
By proposition 3.2.3, this becomes a matter of compuation:

$$\begin{aligned} \log Z(E/\mathbb{F}_q; T) &= \sum_{n=1}^{\infty} \frac{\#E(\mathbb{F}_{q^n}}{n} T^n \\ &= \sum_{n=1}^{\infty} \frac{1 - \alpha^n - \beta^n + q^n}{n} T^n \\ &= -\log(1 - T) + \log(1 - \alpha T) + \log(1 - \beta T) - \log(1 - qT) \end{aligned}$$

Thus:

$$Z(E/\mathbb{F}_q;T) = \frac{(1-\alpha T)(1-\beta T)}{(1-T)(1-qT)}$$

where α, β are complex conjugate with absoluet value $\sqrt{q}$ that satisfy

$$a = \alpha + \beta = \mathrm{tr}(\varphi_\ell) = 1 + q - \deg(1-\varphi) \in \mathbb{Z}$$

The functional equation is immediate with $\epsilon = 0$.

To link this to the usual Riemann Hypothesis and Riemann-zeta function, let $T = q^{-s}$ so that:

$$\zeta_{E/\mathbb{F}_q}(s) = Z(E/\mathbb{F}_q; q^{-s}) = \frac{1 - aq^{-s} + q^{1-2s}}{(1-q^{-s})(1-q^{1-s})}$$

which then gives:

$$\zeta_{E/\mathbb{F}_q}(s) = \zeta_{E/\mathbb{F}_q}(1-s)$$

giving the usual functional equation for the Riemann-zeta function, albeit with a trivial epsilon-factor. Note that if $\zeta_{E/\mathbb{F}_q}(s) = 0$, then $|q^s| = \sqrt{q}$ (as $|\alpha| = |\beta| = \sqrt{q}$), so $\mathrm{Re}(s) = \frac{1}{2}$, showing that the famous the Riemann Hypothesis is true over elliptic curves.

3.3 Endomorphism Ring

Theorem 3.3.1: Endomorphism Ring for Finite Characteristic

Let K be a field of characteristic p, and let E/K be an elliptic curve. Then for each integer $r \geq 1$, let

$$\varphi_r : E \to E^{(p^r)} \qquad \text{and} \qquad \hat{\varphi}_r : E^{(p^r)} \to E$$

be the p^r-power Frobenius map and its dual. Then:

1. the following are equivalent
   - ($\Rightarrow$) $E[p^r] = 0$ for one (all) $r \geq 1$
   - ($\Rightarrow$) $\hat{\varphi}_r$ is (purely) inseparable for one (all) $r \geq 1$
   - ($\Rightarrow$) The map $[p] : E \to E$ is purely inseparable and $j(E) \in \mathbb{F}_{p^2}$
   - ($\Rightarrow$) $\mathrm{End}(E)$ is an order in a quaternion algebra
   - ($\Rightarrow$) The formal group $\hat{E}/K$ assocaited to E has height 2
2. If the equivalent conditions in (1) do not hold, then:
   - (a) $E[p^r] = \mathbb{Z}/p^r\mathbb{Z}$ for all $r \geq 1$
   - (b) The formal group $\hat{E}/K$ has height 1
   - (c) If $j(E) \in \overline{\mathbb{F}}_p$, then $\mathrm{End}(E)$ is an order of a quadratic imaginary field[a]

[a] Silverman sates that for the case where $j(E)$ is transcendental over $\mathbb{F}_p$, exercise V.5.8

***Proof* :**
Silverman p.145; will get back to it!

Definition 3.3.2: Hasse Invariant: Super-Singular and Ordinary

Let E have satisfy the properties in the equivalence given in theorem 3.3.1. Then E is said to be *super-singular* and E has *Hasse invariant* 0. Otherwise, E is said to be *ordinary* and has *Hasse invariant* 1.

The terminology here is confusing as it pre-dates modern nomenclature. The term "singular" and "supersingular" are not related, namely elliptic curves are by definition nonsingular. The term originates in the study of elliptic curves over $\mathbb{C}$ whose endomorphism ring are larger than $\mathbb{Z}$. These curves were originally called "singular" as in "unusual" or "rare". Naturally, this terminology does not hold anymore as then any elliptic curve $E/\overline{\mathbb{F}}_p$ would be called singular. As shall be discussed in section 3.4, "most" elliptic curves have endomorphism rings which are orders of imaginary quadratic fields as opposed to quaternion algebras, making the latter "supersingular".

Silverman mentions there are characterization of supersingular elliptic curves via sheaf cohomology and residue of differentials; this feels like it will be important. Silverman p.145

3.4 Calculating Hasse Invariant

Silverman p.148

(at the end, Silverman states the following really interesting open problem: assume E has no CM. then

$$\#\left\{p < x \mid E/\mathbb{F}_p \text{ is supersingular}\right\} \sim \frac{c\sqrt{x}}{\log(x)}$$

Though this is not known, it has been proven that there are at least infinitely many primes p for which $E/\mathbb{F}_p$ is supersingular)

4

Elliptic Curves Over Local Fields

- K is a local field, complete with respect to a discrete valuation ν
- $R = \{x \in K \mid \nu(x) \geq 0\}$ is the ring of integers
- $R^* = \{x \in K \mid \nu(x) = 0\}$ is the group of units of R
- $\mathcal{M} = \{x \in K \mid \nu > 0\}$ is the maximal ideal of R
- π is a uniformizer of R so that $\mathcal{M} = \pi R$
- $k = R/\mathcal{M}$ is the residue field of R

It will be assumed that ν is normalized so that $\nu(\pi) = 1$ and by convention $\nu(0) = \infty$. As always, K, k will be assumed to be perfect fields.

4.1 Minimal Weierstrass Equation

Let E/K be an elliptic curve so that in coordinates the Weierstrass equation is of the form:

$$E : y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$$

The substitution $(x, y) \mapsto (u^{-2}x, u^{-3}y)$ gives a new isomorphic elliptic curve where the Weierstrass equation has a_i replaced by u^ia_i, so choose u divisible by sufficiently large power of π so that all the coefficients are in R. Thus, the discriminant can be take to be non-negative:

$$\nu(\Delta) \geq 0$$

As ν is discrete as well, it the notion of a *minimal* discriminant, and hence a minimal Weierstrass equation is well-posed:

Definition 4.1.1: Minimal Weierstrass Equation

Let E/K be an elliptic curve. Then a Weierstrass equation for E is called *minimal* and is a *minimal Weierstrass equation* for E at ν if $\nu(\Delta)$ is minimized with the conditions that $a_1, ..., a_6 \in R$. The discriminant that minimizes ν is called the *minimal discriminant* of E at ν and the minimal value of $\nu(\Delta)$ is called the *valuation* of the minimal discriminant.

To find the minimal discriminant, there are some useful tricks. First, as each $a_i \in R$, $\Delta \in R$. If the equation is not minimal, then by looking at proposition 1.1.3 we see that the coordinate change will effect the discriminant by:

$$\Delta' = u^{-12}\Delta \in R$$

Thus, if each $\alpha_i \in R$ and $\nu(\Delta) < 12$, the equation must be minimal. Furthermore, as $c_4' u^{-4} c$ and $c_6' = u^{-6} c_5$, if either "$a_i \in R$ and $\nu(c_4) < 4$" or $a_i \in R$ and $\nu(c_6) < 6$, the equation is minimal. If $\text{char}(K) \neq 2, 3$, the converse holds as well (see exercise 4.2.1(1)).

Example 4.1.1: Minimal Weierstrass Equation

Consider

$$E : y^2 + xy + y = x^3 + x^2 + 22x - 9$$

over $\mathbb{Q}_p$. Then $\Delta = -2^{15}5^2$ and $c_4 = -5 \cdot 211$. Then as $\nu(c_4) < 4$ for each prime $p \in \mathbb{Z}$, it is the minimal Weierstrass equation for each $\mathbb{Q}_p$.

Proposition 4.1.2: Properties of Minimal Weierstrass Equation

1. Every elliptic curve has a minimal Weierstrass equation
2. A minimal Weierstrass equation is unique up to change of coordinates

$$x = u^2 x' + r \qquad y = u^3 y' + u^2 s x' + t$$

for $u \in r^*$, $r, s, t \in R$.

3. The invariant differential associated to a minimal Weierstrass equation

$$\omega = \frac{dx}{2y + a_1 x + a_3}$$

is unique up to multiplication by an element of R^*.

4. Given a Weierstrass equation for E/K whose coefficients are in R, any change of coordinates

$$x = u^2 x' + r \qquad y = u^3 y' + u^2 s x' + t$$

used to produce a minimal Weierstrass equation satisfies $u, r, s, t \in R$

Proof **:**

1. immediate as ν is discrete
2. Recall from proposition 1.4.2(2) that E/K is unique up to change of coordinates with $u \in K^*$ and $r, s, t \in K$. Suppose that the given and new equation are both minimal. Then by minimality it must be that:
$$\nu(\Delta') = \nu(\Delta)$$
hence applying the transformation formulas given in proposition 1.1.3 it must be that
$$u^{12}\Delta' = \Delta$$
and so $u \in R^*$ (as $\nu(u) = 0$). Similarly for the b_6, b_8 formulas, we get $4r^3, 3r^4$ respectively, showing $r \in R$, and doing the same for a_2, a_6 gives that $s, t \in R$ respectively.
3. Immediate from (2), namely $\omega' = u\omega$
4. By minimality, we know $\nu(\Delta') \leq \nu(\Delta)$, and $u^12\Delta' = \Delta$. Thus $\nu(u) \geq 0$, so $u \in R$. Then repeating the same argument as in (2) we manipulate formulas to conclude $r, s, t \in R$.

4.2 Reduction Modulo π

The advantage of working over local fields is keeping track of information of a single prime. This exaction can be achieved by modding. Let
$$R \to k(= R/\pi R) \qquad t \mapsto \bar{t}$$
Note that the bar notation is used both for the algebraic closure of a field and for the equivalence class in the canonical projection; bear the distinction in mind. Then given a minimal Weierstrass equation E/K, we reduce its coefficients modulo π to obtain a (possibly singular!) curve over k:
$$\overline{E} : y^2 + \overline{a}_1xy + \overline{a}_3y = x^3 + \overline{a}_2x^2 + \overline{a}_4x + \overline{a}_6$$
The curve $\overline{E}$ is called the *reduction of E modulo π*. By using the minimal equation of E, the Weierstrass equation $\overline{E}$ of is unique up to standard change of coordinates in k.

As mentioned, reducing can produce singular points:

Example 4.2.1: Singular After Reduction
Consider
$$E : y^2 = x^3 - x$$
over $\mathbb{Q}_2$ so that $\pi = 2$. Its discriminant is $\Delta = 64$, so reducing modulo 2 gives $\nu(\Delta) = 0$ in $\mathbb{Z}_2/2\mathbb{Z}_2 \cong \mathbb{F}_2$.

It is thus interesting to understand these singular points. Recall that the set of nonsingular points, E_{ns}, form a group (proposition 1.3.7). Thus, define:
$$E_0(K) = \left\{P \in E(K) \mid \overline{P} \in \overline{E}_{\text{ns}}(k)\right\}$$
$$E_1(K) = \left\{P \in E(K) \mid \overline{P} = \overline{O}\right\}$$

Note that these sets do not depend on the Weierstrass equation by proposition 4.1.2. Furthermore, E_1 must be the set of singular points if it contains more than 1 point (more than O) as a nonsingular points cannot have higher intersection multiplicity. The notation E_1 comes from the fact that the above can be seen as a π-filtration. In fact, we have the following

Theorem 4.2.1: Exact Sequence of π-reduction

Let E/K be an elliptic curve and $E_1(K)$, $E_0(K)$ defined as above. Then there exists a short exact sequence

$$0 \to E_1(K) \to E_0(K) \to \overline{E}_{\text{ns}}(k) \to 0$$

Proof **:**

First, given surjective map $E_0(K) \to \tilde{E}_{\text{ns}}(k)$ exists and that the reduction map $E_0(K) \to \tilde{E}_{\text{ns}}(k)$ is a homomorphism, exactness follow as $E_1(K)$ is the kernel. To show $E_0(K) \to \overline{E}_{\text{ns}}(k)$ is surjective; this is an application of Hensel's lemma. Let:

$$f(x, y) = y^2 - a_1xy + a_3y - x^3 - a_2x^2 - a_4x - a_6 = 0$$

be a minimal Weierstrass question for E, and $\tilde{f}(x, y)$ be the corresponding polynomial modulo π. Let $\tilde{P} = (\tilde{\alpha}, \tilde{\beta}) \in \tilde{E}_{\text{ns}}(k)$ be a point. As $\tilde{P}$ is nonsingular:

$$\frac{\partial \tilde{f}}{\partial x}(\tilde{P}) \neq 0 \qquad \text{or} \qquad \frac{\partial \tilde{f}}{\partial y}(\tilde{P}) \neq 0$$

Without loss of generality, let's say $\frac{\partial \tilde{f}}{\partial x}(\tilde{P}) \neq 0$. Then choose some $y_0 \in R$ such that $\tilde{y}_0 = \tilde{\beta}$. Then the equation

$$\tilde{f}(x, y_0) = 0$$

has $\tilde{\alpha}$ as a simple root. Thus, by Hensel's lemma, the root $\tilde{\alpha}$ can be lifted to a root $x_0 \in R$. Then letting $P = (x_0, y_0)$ gives the point that maps to $\tilde{P}$, showing surjectivity.

To show it is a homomorphism, we must show compatibility of the group laws. Let $L \subseteq \mathbb{P}^2_K$ be a line given by

$$L : Ax + By + Cz = 0$$

where $A, B, C \in R$ and at least one $A, B, C \in R^*$. The reduction of L to $\tilde{L}$ is given by:

$$\tilde{L} : \tilde{A}x + \tilde{B}y + \tilde{C}z = 0$$

Now, consider $P_1, P_2 \in E_0(K)$ and $P_3 \in E(K)$ such that $P_1 + P_2 + P_3 = O$. We must show $P_3 \in E_0(K)$ and $\tilde{P}_1 + \tilde{P}_2 + \tilde{P}_3 = \tilde{O}$. This is a lot of case-work depending on the multiplicity of the points and the multiplicity of their reduction, so some of it will be left as an exercise see ref:HERE. This proof will cover the case where all the redcued points are distinct, and two of the reduced points are equal.

For the first, if L is a line intersecting E at P_1, P_2, P_3, then when counting appropriate multiplicities, then the line $\tilde{L}$ intersecting $\tilde{E}$ will intersect $\tilde{P}_1, \tilde{P}_2, \tilde{P}_3$ with correct multiplicities. Let us first suppose $\tilde{P}_1, \tilde{P}_2, \tilde{P}_3$ are distinct. Then $\tilde{L} \cap \tilde{E} = \{\tilde{P}_1, \tilde{P}_2, \tilde{P}_3\}$ and by assumption $\tilde{P}_1, \tilde{P}_2 \in \tilde{E}_{\text{ns}}(k)$. Then as $\tilde{E}_{\text{ns}}$ is an abelian group (1.3.5), $\tilde{P}_3 \in \tilde{E}_{\text{ns}}(k)$, by definition $P_3 \in E_0(K)$ and reducing gives $\tilde{P}_1 + \tilde{P}_2 + \tilde{P}_3 = \tilde{O}$.

For the second, let us prove that if $P, Q \in E_0(K)$ are distinct points where $\tilde{P} = \tilde{Q}$, and L is a line passing through P, Q, then $\tilde{L}$ is tangent to $\tilde{E}$ and $\tilde{P}$. Let's prove the case where $\tilde{P} \neq \tilde{O}$, as $\tilde{P} = \tilde{O}$ is similar but slightly different. Let

$$P = (\alpha, \beta) \in E(K) \qquad \text{and} \qquad Q = (\alpha + \mu, \beta + \lambda) \in E(K)$$

As $\tilde{P} = \tilde{Q} \neq \tilde{O}$, $\alpha, \beta \in R$ and $\mu, \lambda \in \mathfrak{m}$. As $P \in E_0(K)$, either:

$$\frac{\partial \tilde{f}}{\partial x}(\tilde{P}) \neq 0 \qquad \text{or} \qquad \frac{\partial \tilde{f}}{\partial y}(\tilde{P}) \neq 0$$

this time let's assume $\frac{\partial \tilde{f}}{\partial y}(\tilde{P}) \neq 0$. Since $f(P) = f(Q) = 0$, we compute the Taylor expansion of $f(x, y)$ around Q:

$$\begin{aligned} 0 &= f(\alpha + \mu, \beta + \lambda) \\ &= f(\alpha, \beta) + \frac{\partial f}{\partial x}(\alpha, \beta)\mu + \frac{\partial f}{\partial y}(\alpha, \beta)\lambda + a\mu^2 + b\mu\lambda + c\lambda^2 \qquad\qquad a, b, c \in R \\ &= \frac{\partial f}{\partial x}(\alpha, \beta)\mu + \frac{\partial f}{\partial y}(\alpha, \beta)\lambda + a\mu^2 + b\mu\lambda + c\lambda^2. \end{aligned}$$

As $(\partial \tilde{f}/\partial y)(\tilde{P}) \neq 0$, $(\partial f/\partial y)(\alpha, \beta) \in R^*$, so

$$v(\lambda) = v\left(\frac{\partial f}{\partial y}(\alpha, \beta)\lambda\right) = v\left(\frac{\partial f}{\partial x}(\alpha, \beta)\mu + a\mu^2 + b\mu\lambda + c\lambda^2\right) \geq v(\mu).$$

Thus $\lambda/\mu \in R$, so dividing the Taylor expansion by μ and reducing modulo π gives:

$$\frac{\partial f}{\partial x}(P) + \frac{\partial f}{\partial y}(P) \cdot \frac{\lambda}{\mu} \equiv 0 \pmod{\mathcal{M}}.$$

Thus, the slope of the tangent line to $\tilde{E}$ at the point $\tilde{P}$ is

$$\frac{dy}{dx}(\tilde{P}) = -\frac{(\partial \tilde{f}/\partial x)(\tilde{P})}{(\partial \tilde{f}/\partial y)(\tilde{P})} = \widetilde{\lambda/\mu}.$$

Now, the line L through P and Q is given by the equation

$$L : y - \beta = \frac{\lambda}{\mu}(x - \alpha).$$

As $\lambda/\mu \in R$, the reduction of L is the line through $\tilde{P}$ having slope $\widetilde{\lambda/\mu}$. But then $\tilde{L}$ is tangent to $\tilde{E}$ at $\tilde{P}$.

With this proven, let us assume $P_1, P_2 \in E_0(K)$ where $P_3 \in E(K)$ such that $P_1 + P_2 + P_3 = 0$ and $\tilde{P}_1 = \tilde{P}_2 \neq \tilde{P}_3$. Then $\tilde{L}$ is tangent to $\tilde{E}$ at $\tilde{P}_1$, hence $\tilde{P}_3 \in \tilde{L}$ and

$$2\tilde{P}_1 + \tilde{P}_3 = \tilde{O}$$

Thus, $P_3 \in E_0(K)$, giving the desired result.

The rest of the cases all follow via similar reasoning and shall be left as a exercise.

If $\nu(\Delta) = 0$, $\tilde{\Delta} \neq 0$, and $\tilde{E}_{\text{ns}} = \tilde{E}$, so $E_0(K) = E(K)$:

$$0 \to E_1(K) \to E(K) \to \tilde{E}(k) \to 0$$

Notice that if K has a finite residue field, then $\tilde{E}(k)$ is finite. For $E_1(K)$, we have alraedy analyzed this group in chapter 2

Proposition 4.2.2: $E_0(K)$ and Formal Group

Let E/K be given be a minimal Weierstrass equation, let $\hat{E}/R$ be the formal group associated to E, and let $w(z) \in R[[z]]$ be the power series associated to E/K. Then:

$$\hat{E}(\mathfrak{m}) \xrightarrow{\sim} E_1(K) \qquad z \mapsto \left(\frac{z}{w(z)}, -\frac{1}{w(z)}\right)$$

is a group isomorphism

Note that by definition of the power series $x(z)$ and $y(z)$ equations 2.1 in section 2.1 this is:

$$z \mapsto (x(z), y(z))$$

showing that this is the equivalent of the *lie algebra* for elliptic curve.

***Proof* :**
By definition the power series $w(z)$ satisfies:

$$w(z) = z^3(1 + \cdots) \in R[[x]]$$

Then certainly $w(z)$ converges for each $z \in \mathfrak{m}$, hence $(z/w(z), -1/w(z)) \in E(K)$. Since:

$$\nu\left(\frac{-1}{w(z)}\right) = -3\nu(z) < 0$$

then these points are at the point at infinity when reducing, hence

$$(z/w\widetilde{(z), -1}/w(z)) = \widetilde{O}$$

meaning $(z/w(z), -1/w(z)) \in E_1(K)$. Thus the map

$$\hat{E}(\mathfrak{m}) \xrightarrow{\sim} E_1(K) \qquad z \mapsto \left(\frac{z}{w(z)}, -\frac{1}{w(z)}\right)$$

is well-defined. As the group law of $\hat{E}$ is given by the group law of E on the (z, w)-plane, this map is a homomorphism. As $w(z) = 0$ if $z = 0$, this map is injective. We'll show this map has an inverse, giving us it is an isomorphism. Let $(x, y) \in E_1(K)$. Then as (x, y) reduces to a point at infinity modulo π, $\nu(x) < 0$ and $\nu(y) < 0$. From the Weierstrass equation (i.e $y^2 + \cdots = x^3 + \cdots$), we get

$$3\nu(x) = 2\nu(y) = -6n$$

for some $n \in \mathbb{N}_{>0}$ Then as $\nu(x/y) < 0$, $x/y \in \mathfrak{m}$, so we can define the map:

$$E_1(K) \to \hat{E}(\mathfrak{m})(x, y) \mapsto -\frac{x}{y}$$

which is, again, certainly a homomorphism, and is injective only O can map to 0 since x, y cannot be zero (or else the valuation would be infinite, not negative). Then this is the inverse map, hence the map in the statement is an isomorphism, as we sought to show.

Exercise 4.2.1

1. Let $\text{char}(K) \neq 2, 3$. Show that the equation is minimal if and only if $a_i \in R$ and $\nu(\Delta) < 12$ or $\nu(c_4) < 4$.

4.3 Points of Finite Order

Theorem 4.3.1: Elliptic Curve Over Local Field: Finite Order Points

Let E/K be an elliptic curve over a local field K and $n \in \mathbb{N}_{>0}$ where $\gcd(n, \text{char}(k)) = 1$. Then

1. $E_1(K)$ has no nontrivial points of order n
2. If $\tilde{E}/k$ is nonsingular, then the reduction map:

$$E(K)[n] \hookrightarrow \tilde{E}(k)$$

   is injective.

Proof **:**

1. As $E_1(K) \cong \hat{E}(\mathfrak{m})$, by ref:HERE it has no non-trivial torsion points of order m.
2. If $\tilde{E}/k$ is non-singular, $E_0(K) = E(K)$ and $\tilde{E}_{\text{ns}}(k) = \tilde{E}(k)$. Then by theorem 4.2.1 we get the short exact sequence

$$0 \to E_1(K) \to E(K) \to \tilde{E}(k) \to 0$$

   and as $E_1(K)$ has no n-torsion, it is an exercise in group theory to show that $E(K)[n]$ must inject into $\tilde{E}(k)$, completing the proof.

This result shall soon be incredibly indispensable for us: if K is any number field, K_ν the localization at a place, then certainly $E(K) \hookrightarrow E(K_\nu)$, hence the n-torsion subgroup of $E(K)$ must injective in each k_ν:

$$E(K)[n] \hookrightarrow \tilde{E}(k_\nu) \qquad \forall \nu \in M_K$$

where M_K is the set of places.

Example 4.3.1: Torsion of Rational Elliptic Curve

1. Let $E/\mathbb{Q}$ be the elliptic curve

$$E : y^2 + y = x^3 - x + 1$$

Its discriminant is $-611 = -13 \cdot 47$, so $\tilde{E}$ is nonsingular mod 2. Then by directly plugging in we see that $\tilde{E}(\mathbb{F}_2) = \{O\}$, and $E(\mathbb{Q})[2] = \{O\}$ since all the 2-torsion points have non-rational coordinate points. Thus, by theorem 4.3.1, $E(\mathbb{Q})$ has *no* (nonzero) torsion points!

2. Let $E/\mathbb{Q}$ be the elliptic curve

$$E : y^2 = x^3 + 1$$

Its discriminant is $-2^4 \cdot 3^5$, hence $\tilde{E}$ is nonsingular for $p \geq 5$. Computing for $\mathbb{F}_5$ and $\mathbb{F}_7$ (the first two primes ≥ 5):

$$\#\tilde{E}(\mathbb{F}_5) = 6 \qquad \text{and} \qquad \#\tilde{E}(\mathbb{F}_7) = 13$$

then the only subgroup that is possible must be the trivial subgroup, hence $E(\mathbb{Q})$ has no torsion points! Notice that $(1, 2) \in E(\mathbb{Q})$, and hence it must have infinite order, and hence $E(\mathbb{Q})$ must be infinite (notice how this is non-trivial!)

3. Let $E/\mathbb{Q}$ be the elliptic curve

$$E : y^2 = x^3 + x$$

Its discriminant is $\Delta = -64$. By inspection, we see immediately that $(0, 0) \in E(\mathbb{Q})$ has order 2. By computing:

$$\#\tilde{E}(\mathbb{F}_3) = 4 \qquad \#\tilde{E}(\mathbb{F}_3) = 4 \qquad \#\tilde{E}(\mathbb{F}_3) = 8$$

and more generally verify that $4 \big| \#\tilde{E}(\mathbb{F}_p)$ for $p \geq 3$ (exercise 4.3.1(1)).

Note that the group structure itself gives us information: take

$$\tilde{E}(\mathbb{F}_3) = O, (0, 0), (2, 1), (2, 2) \cong \mathbb{Z}/4\mathbb{Z}$$
$$\tilde{E}(\mathbb{F}_5) = O, (0, 0), (2, 0), (3, 0) \cong (\mathbb{Z}/2\mathbb{Z})^2.$$

Then since the torsion subgroup of $E(\mathbb{Q})[n]$ for $\neq 2^k$ would be of the form $\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$, we must conclude that the only nonzero torsion point is $(0, 0)$.

Theorem 4.3.2: Bound On Denominator of Torsion Subgroup

Let K be a local field, $p = \text{char}(k) > 0$, and E/K an elliptic curve with Weierstrass equation:

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$$

where $a_i \in R$ (note we are not assuming it must be minimal!). Let $P \in E(K)$ be a point of order m where $m \in \mathbb{N}_{>0}$. Then:

1. If $m \neq p^n$ for some n, then $x(P), y(P) \in R$
2. if $m = p^n$, then

$$\pi^{2r}x(P), \pi^{3r}y(P) \in R \qquad \text{where} \qquad r = \left[\frac{\nu(p)}{p^n - p^{n-1}}\right]$$

   where $[t]$ is the greatest integer in t, and $\nu(p) = 1$ if ν is normalized.

***Proof* :**
Silverman p. 193

Example 4.3.2: Best Case Boudning
Silverman p.194

4.3.1 The Action of Inertia

The injectivity of the torsion group can also be deduced in terms of the action of the galois group on the torsion points. Let K^{nr} be the maximal unramified extension of K, I_ν the inertia group of $\text{Gal}_K(\overline{K})$; for brevity $G_{\overline{K}/K}$. Then from local field theory ([2]), we have the commutative diagram (since we are assuming K/k are perfect):

$$\begin{tikzcd}
1 \arrow[r] & G_{\overline{K}/K^{\text{nr}}} \arrow[r] \arrow[d, "\sim"] & G_{\overline{K}/K} \arrow[r] & G_{K^{\text{nr}}/K} \arrow[r] \arrow[d, "\sim"] & 1 \\
& I_\nu & & G_{k^{\text{nr}}/k} &
\end{tikzcd}$$

where the top row is a short exact sequence. The inertia group I_ν is precisely the subgroup that acts trivially on the residue field $\overline{k}$.

Definition 4.3.3: Unramified Set At Valuation

Let Σ be a set an $G_{\overline{K}/K} \curvearrowright \Sigma$ be an action. Then Σ is said to be *unramified at* ν if I_ν acts trivially on Σ.

Recall from section 1.8 that if E/K is an elliptic curve then $G_{\overline{K}/K}$ acts on $E[m]$ and on $T_\ell(E)$

Proposition 4.3.4: Unramified sets and Torsion Subgroup

Let E/K be a elliptic curve over a local field K and assume $\tilde{E}/k$ is nonsingular. Then

1. Given $m \in \mathbb{N}_{>0}$, $\gcd(m, \mathrm{char}(k)) = 1$ (so $\nu(m) = 0$). Then $E[m]$ is unramified at ν
2. Let ℓ be a prime where $\ell \neq \mathrm{char}(k)$. Then $T_\ell(E)$ is unramified at ν.

Proof **:**
The result for (2) follows immediately from (a) as the definition of $T_\ell(E)$ is the limit of $E[\ell^n]$.

Notice that $E[m]$ need not in $E(K)$[a], so Let K'/K be a field extension where $E[m] \subseteq E(K')$ (exercise 4.3.1(2)). Let $R', \mathfrak{m}', k', \nu'$ be the ring of integers of K', maximal ideal of R', residue field of R', and valuation of K' respectively. As E is nonsingular in reduction, the minimal Weierstrass equation of E at ν has discriminant that satisfies $\nu(\Delta) = 0$. As ν' is a multiple of ν, $\nu'(\Delta) = 0$, and so $\tilde{E}/k'$ is nonsingular. Thus, by theorem 4.3.1 $E[m] \hookrightarrow \tilde{E}(k')$ is injective.

Let $\sigma \in I_\nu$, $P \in E[m]$. We need to show that $P^\sigma = P$. By definition of I_ν, the elements σ acts trivially on $\tilde{E}(k')$. Thus:

$$\widetilde{P^\sigma - P} = \tilde{P}^\sigma - \tilde{P} = \tilde{P} - \tilde{P} = \tilde{O}$$

As $P^\sigma - P \in E[m]$ (as P^σ is certainly still m-torsion and $E[m]$ is a subgroup), and the map $E[m] \hookrightarrow \tilde{E}(k')$ is injective, $P^\sigma - P = O$, i.e. $P^\alpha = P$, as we sought to show.

[a] This should not be too surprising, we saw that $E(\mathbb{Q})$ often has almost no torsion points

Exercise 4.3.1

1. Let $E : y^2 = x^3 + x$. Show that $4 \# \tilde{E}(\mathbb{F}_p)$ for all $p \geq 3$.
2. Let E/K be an elliptic curve. Show that there exists a finite extension K'/K such that $E[m] \subseteq E(K')$ (hint: recall that $E(\overline{K})$ always contains $E[m]$)

4.4 Good and Bad Reduction

A general Weierstrass is either nonsingular, or has a singularity in the form of a node or cusp. If E/K is an elliptic curve, its reduction $\tilde{E}$ can be one of these three:

Definition 4.4.1: Good and Bad Reduction

Let E/K be an elliptic curve over a local field K, and $\tilde{E}$ the reduction $\bmod\ \mathfrak{m}$ of a minimal Weierstrass equation for E. Then

1. E has a *good* or *stable* reduction if $\tilde{E}$ is nonsingular
2. E has a *multiplicative* or *semistable* reduction if $\tilde{E}$ has a node
3. E has an *additive* or *unstable* reduction if $\tilde{E}$ has a cusp.

If E have Multiplicative or additive reduction, E is said to have *bad reduction*. If E has multiplicative reduction, then the reduction is *split* if the slopes of the tangent lines at the node are in k, and it is *nonsplit* otherwise.

The reduction type is imminently deducible by examining the discriminant and c_4 as defined in section 1.1 (recall proposition 1.1.3):

Proposition 4.4.2: Good and Bad Reduction using Discriminant

Let E/K be an elliptic curve given by a minimal Weierstrass equation

$$E : y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$$

and let Δ be the discriminant of E, c_4 is the standard value

1. E has good reduction if and only if $v(\Delta) = 0$, i.e. $\Delta \in R^*$. In this case $\tilde{E}/k$ is an elliptic curve.
2. E has multiplicative reduction if and only if $v(\Delta) > 0$ and $v(c_4) = 0$, i.e., $\Delta \in \mathfrak{m}$ and $c_4 \in R^*$. In this case $\tilde{E}_{\text{ns}}$ is the multiplicative group
$$\tilde{E}_{\text{ns}}(\bar{k}) \cong \bar{k}^*$$
3. E has additive reduction if and only if $v(\Delta) > 0$ and $v(c_4) > 0$, i.e., $\Delta, c_4 \in \mathfrak{m}$. In this case $\tilde{E}_{\text{ns}}$ is the additive group
$$\tilde{E}_{\text{ns}}(\bar{k}) \cong \bar{k}^+$$

Proof **:**
Immediate from proposition 1.1.3 and proposition 1.3.7.

Example 4.4.1: Good, Multiplicative, Additive Reduction
Let $p \geq 5$ be a prime and let :

$$\begin{aligned} E_1 : y^2 =& x^3 + px^2 + 1 \\ E_2 : y^2 =& x^3 + x^2 + p \\ E_3 : y^2 =& x^3 + p \end{aligned}$$

Then over $\mathbb{Q}_p$, E_1 has good reduction over $\mathbb{Q}_p$, E_24 has (split) multiplicative reduction, and E_3

has additive reduction. Note that for E_3, in $Q(\sqrt[6]{p})$, we get a rood reduction by doing:

$$x \mapsto \sqrt[3]{p}x' \qquad y \mapsto \sqrt[p]{y}'$$

which gives a minimal Weierstrass equation with good reduction. On the other hand, E_2 has multiplicative reduction over every extension of $\mathbb{Q}_p$.

This motivates the stable/semi-stable/unstable terminology: extending the ground field of an elliptic curve can turn an additive curve to a good or multiplicative reduction, while those two won't change (proposition 4.4.4). Silverman notes that there is a more precise definition in terms of stability of points in modulo space, which I will ref:HERE once I learn it.

The above example gives motivation to the following definition:

Definition 4.4.3: Potential Good Reduction

Let E/K be an elliptic curve. Then E/K has *potential good reduction* if there is a finite extension K'/K such that E has a good reduction over K'.

Example 4.4.2: Motivating Complex Multiplication

Let E/K where $K/\mathbb{Q}_p$ is a finite extension. Then if E has complex multiplication (CM), then E has potential good reduction. return to this, I want this exercise down here.

Proposition 4.4.4: Semistable Reduction Theorem

Let E/K be an elliptic curve.

1. Let K'/K be an unramified extension. Then the reduction of type E over K is the same as the reduction type of E over K'
2. Let K'/K be a finite extension. Then if E has either good or multiplicative reduction type, then it has the same reduction type over K'
3. There exists a finite extension K'/K such that E has either good or (split) multiplicative reduction over K'

***Proof* :**

1. The case for arbitrary characteristic is longer and Silverman left it to his next book . He proves it here for $\text{char}(p) \geq 5$. Then let E have the minimal Weierstrass equation

$$E : y^2 = x^3 + Ax + B$$

Let R' be the ring of integers of K', ν' the extension of ν for K. Let

$$x = (u')^2 x' \qquad y = (u')^3 y'$$

be the change of coordinates to the minimal equation for E over K'. Since K'/K is unramified, $\text{im}\nu = \text{im}\nu' = \mathbb{Z}$, so given $\nu'(u') = n$, we can find an $u \in K$ such that $\nu'(u) = \nu(u) = n$ (as ν' extends ν), so we have $u/u; \in (R')^*$. Thus, the substitution:

$$x = u^2 x' \qquad y = u^3 y'$$

also gives a minimal equation for E/K' (as u, u' have the same valuation, $\nu'(u^{-12}\Delta) = \nu'((u')^{-12}\Delta)$). Since now the coefficients are in R, by minimality of the original equation over K it must be that $\nu(u) = 0$; thus the original equation is *also* minimal over K'. From this we conclude: since $\nu(\Delta) = \nu'(\Delta)$ and $\nu(c_4) = \nu'(c_4)$, the reduction type stays the same.

2. We are no longer assuming the char$(p) \geq 5$ restriction. Let E/K be an elliptic curve with a minimal Weierstrass equation with corresponding Δ, c_4, and let R', ν' be the ring of integers and extension of K' and ν. Let

$$x = u^2x' + r \qquad y = u^3y' + su^2x' + t$$

be the change of coordinates of a giving a minimal Weierstrass equation for E/K'. Then the quantities $\Delta' c_4$ corresponding to the new minimal Weierstrass equation satisfy:

$$0 \leq \nu'(\Delta' 0 \nu(u^{-12}\Delta) \qquad \text{and} \qquad 0 \leq \nu'(c_4') = \nu'(u^{-4}c_4)$$

By proposition 4.1.2(4), $u \in R'$, so

$$0 \leq \nu'(u) \leq \min\left\{\frac{1}{12}\nu'(\Delta), \frac{1}{4}\nu'(c_4)\right\}$$

Then by proposition 4.4.2 a good (resp. multiplicative) reduction has $\nu(\Delta) = 0$ (resp. $\nu(c_4) = 0$), and so it must be that in both cases $\nu'(u) = 0$, thus:

$$\nu'(\Delta') = \nu'(\Delta) \qquad \text{and} \qquad \nu'(c_4') = \nu'(c_4)$$

and similarly for reduction over K'

3. This time assume char$(k) \neq 2$ just for computational reasons. Take a finite extension so that E/K can be put in Legendre form:

$$E : y^2 = x(x-1)(x-\lambda)$$

where $\lambda \neq 0, 1$. Then:

$$\Delta = 16\lambda^2(\lambda - 1)^2 \qquad \text{and} \qquad c_4 = 16(\lambda^2 - \lambda + 1)$$

Then:

(a) ($\lambda \in R$, $\lambda \not\equiv 0, 1 \bmod \mathfrak{m}$) Then $\Delta \in R^*$, hence we have good reduction

(b) ($\lambda \in R$, $\lambda \equiv 0, 1 \bmod \mathfrak{m}$) Then $\Delta \in \mathfrak{m}$ and $c_4 \in R^*$, hence we have multiplicative reduction

(c) ($\lambda \notin R$) Then there exits a $r \geq \mathbb{N}_{>0}$ such that $\pi^r\lambda \in R^*$. Replacing K by $K(\sqrt{\pi})$ if necessary, substitute $x = \pi^{-r}x'$ and $y = \pi^{-3r/2}y'$. Then:

$$(y')^2 = x'(x' - \pi^r)(x' - \pi^r\lambda)$$

Then E has integral coefficients, $\Delta' \in \mathfrak{m}$, and $c_4' \in R^*$, so E has multiplicative reduction.

Proposition 4.4.5: Potential Good Reduction and J-invariant

Let E/K be an elliptic curve. Then E has potential good reduction if and only if its j-invariant $j(E) \in R$

Proof **:**
Assume $\text{char}(k) \neq 2$ and if necessary take a finite extension K'/K so that E has a Weierstrass equation of the form:

$$E : y^2 = x(x-1)(x-\lambda) \qquad \lambda \neq 0, 1$$

First, suppose $j(E) \in R$. By proposition 1.1.8 j and λ have the relation:

$$256(1-\lambda(1-\lambda))^3 - j\lambda^2(1-\lambda)^2 = 0$$

As j is integral, we get from this equation that:

$$\lambda \in R \qquad \text{and} \qquad \lambda \neq 0, 1 \bmod \mathfrak{m}$$

Thus by proposition 4.4.4 the Legendre equation as integral coefficients and good reduction.

Conversely, suppose E has potential good reduction. Then K'/K is a finite extension where E has a good reduction. Let Δ' and c_4' be the value corresponding to a minimal equation for E over K'. Then $(\Delta') \in (R')^*$ and

$$j(E) = \frac{(c_4')^3}{\Delta'} \in R'$$

But $j(E) \in K$ (recall E is defined over K), so it must be that $j(E) \in R$, as we sought to show.

4.5 The group E/E0

Silverman shows $E(K)/E_0(K)$ is finite

Theorem 4.5.1: Kodoria-Néron Theorem

Let E/K be an elliptic curve. If E has split multiplicative reduction over K, then $E(K)/E_0(K)$ is a cyclic group of order

$$\nu(\Delta) = -\nu(j)$$

In all other cases, $E(K)/E_0(K)$ is finite and has order at most 4

Proof **:**
a lot of case work, or a more direct proof from Néron model; these shall be covered later its in vol 2 of Silverman

Corollary 4.5.2: Finite Index of E0

The subgroup $E_0(K) \leqslant E(K)$ has finite index

Proof **:**
ibid

4.5.1 **$E(\mathbb{Q}_p)$ not finitely generated** Let $K = \mathbb{Q}_p$ and consider $E/\mathbb{Q}_p$. Then the above shows that $E(\mathbb{Q}_p)$ has a subgroup of finite index isomorphic to $\mathbb{Z}_p$, but $\mathbb{Z}_p$ is *not* finitely generated, so $E(\mathbb{Q}_p)$ is not finitely generated. This shall contrast to $E(\mathbb{Q})$ which by the Modell-Weil Theorem *is* finitely generated (see section 5.3)

4.6 Néron–Ogg–Shafarevich Criterion

Theorem 4.3.1 with proposition 4.3.4 showed that if E/K has good reduction, and $\gcd(m, \mathrm{char}(k)) = 1$, $E[m]$ injects into $\overline{E}(k)$, equivalently $E[m]$ is an unramified set at ν. There are many partial converses to this result, the following is one that shall be pertinent for us when studying global fields:

Theorem 4.6.1: Criterion of Néron-Ogg-Shafarevich

Let E/K be an elliptic curve. Then the following are equivalent:

1. E has good reduction
2. $E[m]$ is unramified at ν for all $m \in \mathbb{N}_{>0}$ that are relatively prim to $\mathrm{char}(k)$
3. The Tate module $T_\ell(E)$ is unramified at ν for some (or all) ℓ where $\ell \neq \mathrm{char}(k)$
4. $E[m]$ is unramified for infinitely many $m \in \mathbb{N}_{>0}$ that are relatively prime to $\mathrm{char}(k)$

Proof **:**
Theorem 4.3.1 proves (1) $\Rightarrow$ (2) and (2) $\Rightarrow$ (3) $\Rightarrow$ (4) immediately, so we show (4) $\Rightarrow$ (1).

finish, too tired rn to focus

Corollary 4.6.2: Isogenies and Good Reduction

Let $E_1/K, E_2/K$ be two elliptic curves over K and φ an isogeny between the two. Then E_1/K has good reduction if and only if E_2/K has good reduction.

Proof **:**
Let $\varphi : E_1 \to E_2$ be the isogeny and let $m \in \mathbb{N}_{>0}$ be relatively prime to $\mathrm{char}(k)$ and $\deg\varphi$. Then we get the induced map $\varphi : E_1[m] \xrightarrow{\sim} E_2[m]$ which is an isomorphism of $G_{\overline{K}/K}$-modules. Thus either both $E_1[m]$ and $E_2[m]$ are ramified or unramified by ν, which by theorem 4.6.1 gives the

desired result.

Corollary 4.6.3: Potential good reduction and Inertia group on Tate module

Let E/K be an elliptic curve. Then E has potential good reduction if and only if the inertia group I_v acts on the Tate module $T_\ell(E)$ through a finite quotient for some (all) prime(s) $\ell \neq \operatorname{char}(k)$.

Proof **:**
Silverman p.202; will return when less tired

5

Elliptic Curves over Global Number Fields

So far, we manged to understand the behaviour over the complex numbers (the "ideal" space to find solutions), finite field, and use that with some extra tricks to find the solution over local fields. With quadratics of the form $f(x, y)$, this information fully determined its solutions over global fields. In this chapter, exactly why this is not the case for elliptic curves will be studied. The main objective of this chapter is to prove the *Mordell-Weil Theorem* that states that;

$$E(K) \cong \mathbb{Z}^r \times E(K)_{\text{tors}}$$

where $E(K)_{\text{tors}}$ is finite. By section 4.6, computing $E(K)_{\text{tors}}$ is well-understood and relatively algorithmic. On the other hand, The rank r is much more difficult to understand, and in fact as of writing this book there is no known procedure to find r for any global field K: the *Birch–Swinnerton-Dyer conjecture* (one of the Millennium Problems) stipulates that the rank r of $E(K)$ is the order of the simple pole of the Hasse-Weil L function $L(E, s)$ at $s = 1$.

In this chapter, there is a lot of notation to keep track of. Let:

K be a number field

$G_{L/K}$ is the galois group of L/K.

M_K be a set choices of representatives of all places of K (absolute value for each place)

M_K^∞ be the infinite places in M_K

M_K^0 be the finite place sin M_K

$\nu(x) = -\log|x|_\nu$ for an absolute value $\nu \in M_K$

ord_ν be the *normalized* valuation for $\nu \in M_K^0$ so that $\text{ord}_\nu(K^*) = \mathbb{Z}$

R be the ring of integers of K, i.e. $R = \{x \in K \mid \nu(x) \geq 0 \ \forall \nu \in M_K^0\}$

R^* be the ring units of K, i.e. $R = \{x \in K \mid \nu(x) = 0 \ \forall \nu \in M_K^0\}$

K_ν be the completion of K at ν

R_ν be the ring of integers of K_ν

$\mathfrak{m}_\nu$ be the maximal ideal of R_ν

k_ν be the residue field of K_ν i.e. $R_\nu/\mathfrak{m}_\nu$

5.1 Weak Mordell-Weil Theorem

The first part of this chapter is dedicated to proving the following:

If E/K is an elliptic curve and $m \in \mathbb{N}_{>0}$, then

$$E(K)/mE(K)$$

is finite.

In this entire section, we shall assume the same E/K and m, which we shall reference in each proof as "Take $E/K, m$".

Lemma 5.1.1: Finiteness Preserved Finite Extension

Take $E/K, m$. Let L/K be a galois extension. Then if $E(L)/mEL(L)$ is finite, $E(K)/mE(K)$ is finite.

Proof **:**
The map $E(K) \hookrightarrow E(L)$ induces a map $E(K)/mE(K) \to E(L)/mEL(L)$ with kernel:

$$\frac{E(K) \cap mE(L)}{mE(K)}$$

So for each $\overline{P} \in \ker$ chose a point $Q_P \in mE(L)$ such that $[m]Q_P = P$. Now fore each P, we can define the map

$$\lambda_P : G_{L/K} \to E[m] \qquad \lambda_P(\sigma) : Q^\sigma - Q$$

which is well-defined as:

$$[m](Q^\sigma - Q) = [m]Q^\sigma - [m]Q = P^\sigma - P = P - P = O$$

Note this map is not in general a homomorphism. The reader familiar with some group cohomology will recognize this is a 1-cocycle; this shall be elaborated on more in section 5.1.1. Now suppose $P, P \ in \ker$ satisfy $\lambda_P \lambda_{P'}$. Then:

$$(Q_P - Q_{P'})^\sigma = Q_P - Q_{P'} \qquad \forall \sigma \in G_{L/K}$$

Hence $Q_p - Q_{P'} \in E(K)$, so:

$$P - P' = [m]Q_P - [m]Q_{P'} = [m](Q_P - Q_{P'}) \in mE(K)$$

Hence $P \equiv P' \bmod mE(K)$, showing the map

$$l\varphi : \ker \to \mathrm{Hom}_{\mathbf{Set}}(G_{L/K}, E[m]) \qquad \varphi(P) = \lambda_P$$

is injective. Then as $G_{L/K}, E[m]$ are finite, ker must be finite. But now we get the exact sequence[a]:

$$0 \to \ker \to E(K)/mE(K) \to E(L)/mE(L)$$

showing $E(K)/mE(K)$ is an extension of finite groups, hence $E(K)/mE(K)$ is finite, as we sought to show.

[a]which is not a short exact sequence as $E(K)/mE(K)$ need not surject

Thus, is $E[m] \not\subseteq E(K)$, we can just add the points by taking a sufficiently large finite extension L/K. Thus, from now on we shall assume $E[m] \subseteq E(K)$.

The next step is to translate the finiteness of $E(K)/mE(K)$ to a statement about field extensions. This shall be tied a familiar concept. Recall in [5, chapter 18] that the Kummer pairing showed there was a duality between the cyclic extensions with enough roots of unity and $K^*/(K^*)^n$:

$$K^*/(K^*)^n \times G_{K'/K} \to \mu_m \qquad (a, \sigma) \mapsto \frac{\sqrt[m]{a}^{\sigma}}{\sqrt[m]{a}}$$

where K' is the extension with all the roots of unity. A similar function can be defined for elliptic curves:

Definition 5.1.2: Kummer Pairing

Take $E/K, m$. Then the *Kummer pairing* is a map:

$$\kappa : E(K) \times G_{\overline{K}/K} \to E[m]$$

where given $P \in E(K)$, choose $Q \in E(\overline{K})$ such that $[m]Q = P$ (recall $[m] : E(\overline{K}) \to E(\overline{K})$ is a surjective map). Then:

$$\kappa(P, \sigma) = Q^{\sigma} - Q$$

Proposition 5.1.3: Properties of Kummer Pairing

Take $E/K, m$ and let κ be the Kummer pairing. Then:

1. κ is well-defined
2. κ is bilinear
3. The left-kernel of κ is $m(E)$
4. The right-kernel f κ is $G_{\overline{K}/L}$ where

$$L = K([m]^{-1}E(K))$$

that is, L is the composition of all fields $K(Q)$ as Q ranges over the points in $E(\overline{K})$ satisfying $[m]Q \in E(K)$.

Hence, the Kummer pairing induces a perfect bilinear map:

$$\frac{E(K)}{mE(K)} \times G_{L/K} \to E[m]$$

***Proof* :**
These can be derived using group cohomology, but it is insightful to prove the directly.

really good proof! I'll write down soon!

Thus, the finiteness of $E(K)/mE(K)$ follows from the finiteness of L/K, thus let's understand this extension. We need one definition before the next proposition:

Definition 5.1.4: Good and Bad Reduction Over Global Field

Take E/K, and let νM_K^0 be a discrete valuation. Then E has *good* (resp. *bad*) *reduction at* ν if E has good (resp. bad) reduction over K_ν.

Note that it is not always possible to take a minimal Weierstrass equation for E over K_ν for each $\nu \in M_K^0$, however ref:HERE shows that it is possible when $K = \mathbb{Q}$ (Silverman VIII §8). Note that if

$$E : y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$$

and Δ is its discriminant, then for all but finitely many $\nu \in M_K^0$:

$$\nu(a_i) \geq 0 \qquad \text{and} \qquad \nu(\Delta) = 0$$

For any ν satisfying these conditions, $\tilde{E}_\nu/k_\nu$ has good reduction (i.e. is nonsingular). 1

Proposition 5.1.5: Weak Mordell and Special Field Extension

Let

$$L = K([m]^{-1}E(K))$$

Then

1. The extension L/K is abelian and has exponent m (every element of $G_{L/K}$ has order dividing m)
2. Let
$$S = \left\{\nu \in M_K^0 \,\middle|\, \nu \text{ has bad reduction}\right\} \cup \left\{\nu \in M_K^0 \,\middle|\, \nu(m) \neq 0\right\} \cup M_K^\infty$$
Then L/K is unramified outside of S.

Proof **:**

1. Follows immediately from proposition 5.1.3 as there is an injection:
$$G_{L/K} \to \mathrm{Hom}(E(K), E[m]) \qquad \sigma \to \kappa(-, \sigma)$$
2. will return to it

We've now translated the problem into one about the finite extension of L/K where it is unramified outside a set S. If you have gone through Class Field Theory ([2]), then this results follows as the ray class group is finitely generated, and a quotient of it is isomorphic to $\mathrm{Gal}_K(L)$, which is equal to $[L : K]$. But this can also be proven in a more direct manner with the added assumption given on L by proposition 5.1.5.

Proposition 5.1.6: Maximal Unramified is A Finite Extension

Let K be a number field, $S \subseteq M_K^0$ a finite set of places that contain M_K^∞, and let $m \in \mathbb{N}_{\geq 2}$. Let L/K be the maximal abelian extension of K having exponent m that is unramified outside of S. Then L/K is a finite extension.

Proof **:**
silverman p.213

Theorem 5.1.7: Weak Mordell Theorem

Let K be a number field, E/K an elliptic curve, $m \in \mathbb{N}_{\geq 2}$. Then:

$$\frac{E(K)}{mE(K)}$$

is finite.

***Proof* :**
Let $L = K([m]^{-1}E(K))$. Since $E[m]$ is finite, the perfect pairing given by proposition 5.1.3 shows $E(K)/mE(K)$ is finite if and only if $G_{L/K}$ is finite. Then proposition 5.1.5 along with the fact that there are only finitely many bad reductions gives the right properties on L to use proposition 5.1.6 to show L/K is a finite extension, hence $G_{L/K}$ is finite, as we sought to show.

Silverman give an alternative proof approach using that there are finitely many extensions of K of bounded degree that are unramified outside S, which I'll skip for now.

(important comment about computability!)

5.1.1 Cohomological Perspective of Kummer Pairing

It is worth understand how to understand Kummer pairing cohomological. There is a natural short exact sequence:

$$0 \to E[m] \to E(\overline{K}) \xrightarrow{[m]} E(\overline{K}) \to 0$$

Then the $G_{\overline{K}/K}$-cohomology yields a long exact sequence:

$$\begin{tikzcd} 0 \arrow[r] & E(K)[m] \arrow[r] & E(K) \arrow[r, "{[m]}"] & E(K) \arrow[dll, rounded corners, "\delta"'] \\ & H^1(G_{\overline{K}/K}, E[m]) \arrow[r] & H^1(G_{\overline{K}/K}, E(K)) \arrow[r, "{[m]}"] & H^1(G_{\overline{K}/K}, E(K)) \end{tikzcd}$$

From the middle of this exact sequence we extract the following short exact sequence, which is called the Kummer sequence for E/K:

$$0 \longrightarrow \frac{E(K)}{mE(K)} \xrightarrow{\delta} H^1(G_{\overline{K}/K}, E[m]) \longrightarrow H^1(G_{\overline{K}/K}, E(\overline{K}))[m] \longrightarrow 0.$$

this stuff is really cool! I will write this down real soon

5.2 Descent Procedure

Theorem 5.2.1: Descent Theorem

Let A be an abelian group. Suppose there exists a function:

$$h : A \to \mathbb{R}$$

such that the following 3 properties hold:

1. Choose $Q \in A$. Then there exists a $C_1 \in \mathbb{R}$ depending on A, Q st
$$h(P+Q) = 2h(P) + C_1 \qquad \forall P \in A$$
2. There exits an integer $m \in \mathbb{N}_{\geq 2}$ and a constant C_2 depending on A such that
$$h(mP) \geq m^2 h(P) - C_2 \qquad \forall P \in A$$
3. For every $C_3 \in \mathbb{R}$:
$$\{P \in A \mid h(P) \leq C_3\}$$
is finite

Then if A/mA is finite, A is finitely generated.

Proof **:**
Let $Q_1, ..., Q_r \in A$ be representatives for the cosets of A/mA[a], and let $P \in A$ be an arbitrary element. We shall show that P can be re-written as a linear combination of Q_i along with a point P_n that will have a bounded height independent of P, and hence there are only finitely many such P_n independent of P, showing A is finitely generated.

First, since $P \in A$, it is in some coset so:

$$P = mP_1 + Q_{i_1} \qquad 1 \leq i_1 \leq r$$

Now P_1 is in some coset, so we can find $P_1 = mP_2 + Q_{i_2}$ for some $1 \leq i_2 \leq r$. Repeat this process n times (where how we pick n will be evident soon) to have a list:

$$\begin{aligned} P &= mP_1 + Q_{i_1} \\ P_1 &= mP_2 + Q_{i_2} \\ &\vdots \\ P_{n-1} &= mP_n + Q_{i_n} \end{aligned}$$

Then for any index j:

$$h(P_j) \leq \frac{1}{m^2}(h(mP_j) + C_2)(2)$$
$$= \frac{1}{m^2}h(P_j - Q_{i_j} + C_2)$$
$$\leq \frac{1}{m^2}2h(P_{j-1}) + C_1' + C_2) \qquad (1)$$

where C_1' is the maximum among the constants for (1) for $Q \in \{-Q_1, ..., -Q_r\}$. Importantly, C_1', C_2 do not depend on P. Then we can use this inequality starting form P_n to get:

$$\begin{aligned} h(P_n) &\leq \left(\frac{2}{m}\right)^n h(P)\left(\frac{1}{m^2} + \frac{2}{m^4} + \frac{4}{m^8} + \cdots + \frac{2^{n-1}}{m^{2n}}\right)(C_1 + C_2) \\ &< \left(\frac{2}{m}\right)^n h(P) + \frac{C_1' + C_2}{m^2 - 2} \\ &\leq \frac{1}{2^n}h(P) + \frac{1}{2}(C_1' + C_2) && \text{as } m \geq 2 \end{aligned}$$

Now, as $h(P)$ is fixed, choose n sufficiently large so that:

$$h(P_n) \leq 1 + \frac{1}{2}(C_1' + C_2)$$

Now, P can be written as a linear combination:

$$P = m^n P_n + \sum_j^n m^{j-1} Q_{i_j}$$

hence P can be written as a linear combination of:

$$\{Q_1, ..., Q_r\} \cup \left\{Q \in A \;\middle|\; h(Q) \leq 1 + \frac{1}{2}(C_1' + C_2)\right\}$$

Thus, by (3) the later set is finite, hence A is finitely generated as we sought to show.

[a] Note r need not equal r, ex $A = \mathbb{Z}^2$ and $m = 2$

an important word on compatibility! Most of these constants are find able, the problem is finding generators for $E(K)/mE(K)$

5.3 Mordell-Weil Theorem Over Rationals

Fix a Weierstrass equation:

$$E : y^2 = x^3 + Ax + B \qquad A, B\mathbb{Z}$$

By theorem 5.1.7 $E(\mathbb{Q})/2E(\mathbb{Q})$ is finite, hence we need to find the required height function on $E(\mathbb{Q})$.

Definition 5.3.1: Height Function On Rationals

Let $t = p/q \in \mathbb{Q}$. Then the *height* of t is:

$$H(t) = \max\{|p|, |q|\}$$

Definition 5.3.2: Logarithmic Height On Elliptic Curve

The (logarithmic) height function on $E(\mathbb{Q})$ relative to a given Weierstrass equation is:

$$h_x : E(\mathbb{Q}) \to \mathbb{R} \qquad h_x(P) = \begin{cases} \log H(x(P)) & P \neq O \\ 0 & P = O \end{cases}$$

Not that h_x is always nonnegative.

Lemma 5.3.3: Height Function On Rational Elliptic Curve

Let $E/\mathbb{Q}$ be an elliptic curve with a Weierstrass equation:

$$E : y^2 = x^3 + Ax + B \qquad A, B \in \mathbb{Z}$$

1. Let $P_0 \in E(\mathbb{Q})$. There is a constant C_1 that depends on P_0, A, and B such that
$$h_x(P + P_0) \leq 2h_x + C_1 \quad \text{for all } P \in E(\mathbb{Q}).$$
2. There is a constant C_2 that depends on A and B such that
$$h_x([2]P) \geq 4h_x - C_2 \quad \text{for all } P \in E(\mathbb{Q}).$$
3. For every constant C_3, the set
$$\{P \in E(\mathbb{Q}) : h_x \leq C_3\}$$
is finite.

Proof **:**
here

5.4 Mordell-Weil Theorem over Number Fields

here

5.4.1 Heights on Projective Space

Consider $\mathbb{P}^n(\mathbb{Q})$. Then as $\overline{\mathbb{Q}}^{\mathbb{Z}} = \mathbb{Z}$ is a Principal Ideal Domain (PID), then any $P \in \mathbb{P}^n(\mathbb{Q})$ can be written as:

$$P = [x_0 : \cdots : x_n]$$

where $x_0, ..., x_n \in \mathbb{Z}$, $\gcd(x_0, ..., x_n) = 1$. Then we can define the height function to be:

$$H(P) = \max\{|x_0|, ..., |x_n|\}$$

Then certainly for any C, there are at most $2C + 1$ points less than less than C.

Definition 5.4.1: Standard Absolute Value

The set of *standard absolute values on* $\mathbb{Q}$, which we denote by $M_{\mathbb{Q}}$, consists of the following:

1. $M_{\mathbb{Q}}$ contains one Archimedean absolute value, defined by

$$|x|_\infty = \text{usual absolute value} = \max\{x, -x\}.$$

2. For each prime $p \in \mathbb{Z}$, the set $M_{\mathbb{Q}}$ contains one nonarchimedean (p-adic) absolute value defined by

$$\left|p^n \frac{a}{b}\right|_p = p^{-n} \quad \text{for } a, b \in \mathbb{Z} \text{ satisfying } p \nmid ab.$$

The set of *standard absolute values on a number field* K, denoted by M_K, is the set of all absolute values on K whose restriction to $\mathbb{Q}$ is one of the absolute values in $M_{\mathbb{Q}}$.

Definition 5.4.2: Local Degree

Let $\nu \in M_K$. The *local degree* at ν, denoted by n_ν, is

$$n_\nu = [K_\nu : \mathbb{Q}_\nu],$$

where K_ν and $\mathbb{Q}_\nu$ denote the completions of K and $\mathbb{Q}$ with respect to the absolute value ν.

With these, recall the following two results from [2]:

Theorem 5.4.3: Tower Formula For Number Fields

Let $L/K/\mathbb{Q}$ be number fields and $\nu \in M_K$. Then:

$$\sum_{\substack{w \in M_L \\ w | \nu}} n_w = [L : K] n_\nu$$

Proof **:**
see [2]

Theorem 5.4.4: Product Formula

Let $x \in K^*$. Then:

$$\prod_{\nu \in M_K} |x|_\nu = 1$$

Proof **:**
see [2]

Definition 5.4.5: Height Function On Projective Space

Let $P \in \mathbb{P}^n(K)$. Then the *height* relative to K is:

$$H_K(P) = \prod_{\nu \in M_K} \max\{|x_0|_\nu, ..., |x_n|_\nu\}^{n_\nu}$$

Proposition 5.4.6: Properties of Height Function On Projective Space

Let $P \in \mathbb{P}^n(K)$ where K is a number field. Then:

1. $H_K(P)$ is well-defined (independent of representative of P)
2. $$H_K(P) \geq 1$$
3. If L/K is a finite extension. Then:
$$H_L(P) = H_K(P)^{[L:K]}$$

Proof **:**

1. let $P = [x_0 : \cdots x_n] = [\lambda x_0 : \cdots \lambda x_n]$ where $\lambda \in K^*$. Then:
$$\begin{aligned}\prod_{\nu \in M_K} \max\{|\lambda x_0|_\nu, ..., \lambda |x_n|_\nu\}^{n_\nu} &= \prod_{\nu \in M_K} |\lambda|^{n_\nu} \max\{|x_0|_\nu, ..., |x_n|_\nu\}^{n_\nu} \\ &= \prod_{\nu \in M_K} \max\{|x_0|_\nu, ..., |x_n|_\nu\}^{n_\nu}\end{aligned}$$
2. Any point P can be multiplied by $\lambda \in K^*$ such that one of the coordinates is 1, hence every factor in the product $H_K(P)$ is at least 1

3. Computing:

$$\begin{aligned}
H_L(P) &= \prod_{w\in M_L} \max\{|x_i|_w\}^{n_w} \\
&= \prod_{\nu\in M_K} \prod_{\substack{w\in M_L \\ w|\nu}} \max\{|x_i|_\nu\}^{n_w} && x_i \in K \\
&= \prod_{\nu\in M_K} \max\{|x_i|_\nu\}^{[L:K]n_\nu} && \text{theorem 5.4.4} \\
&= H_K(P)^{[L:K]}
\end{aligned}$$

This matches with the simpler height function defined on $\mathbb{Q}$. Since any point P can be written with integer coefficients where $\gcd(x_0, ..., x_n) = 1$ (so there is at least one term without a p-factor for some prime p), it must be that $|x_i|_\nu \leq 1$ for all x_i and $|x_i|_\nu = 1$ for at least 1 i. Thus, all the nonarchimedean places contribute nothing, leaving the Archimedean place:

$$H_{\mathbb{Q}}(P) = \max\{|x_0|_\infty, ..., |x_n|_\infty\}$$

For convenience, it is sometimes better to work with a height function that is independent of number field:

Definition 5.4.7: Absolute Height Function

Let $P \in \mathbb{P}^n(\overline{\mathbb{Q}})$. Then the *absolute height* of P. Then choosing a field K such that $P \in P^n(K)$;

$$H(P) = H_K(P)^{\frac{1}{[K:\mathbb{Q}]}}$$

By proposition 5.4.6, $H(P)$ is well-defined.

Theorem 5.4.8: Height and Morphisms

Let $F : \mathbb{P}^n \to \mathbb{P}^m$ be a morphism of degree d. Then there exists positive constants C_1, C_2 dependent on F such that

$$C_1 H(P)^d \leq H(F(P)) \leq C_2 H(P)^d \qquad \forall P \in P^n(\overline{\mathbb{Q}})$$

***Proof* :**
here

comment on C_1 having bad estimates usually with the Nullstellensatz, and there are efforts for good estimates.

Corollary 5.4.9: Height and Automorphisms

Let $A \in \mathrm{GL}_{n+1}(\overline{\mathbb{Q}})$ and let $A : \mathbb{P}^n \to \mathbb{P}^n$ be the corresponding automorphism. Then there exists positive constants C_1, C_2 dependent on A such that

$$C_1 H(P) \leq H(F(P)) \leq C_2 H(P) \qquad \forall P \in P^n(\overline{\mathbb{Q}})$$

Proof **:**
immediate

Let's now see the relation between height and the roots of polynomials

Definition 5.4.10: Affine Height Function

Let $x \in \overline{\mathbb{Q}}$. Then define:

$$H(x) = H([x : 1])$$

and if $x \in K$ then

$$H(x) = H_K([x : 1])$$

Theorem 5.4.11: Height and Roots

Let

$$f(X) = a_0 X^d + a_1 X^{d-1} + \cdots + a_d = a_0(X - \alpha_1) \cdots (X - \alpha_d) \in \overline{\mathbb{Q}}[X]$$

Then:

$$2^{-d} \prod_{j=1}^{d} H(\alpha_j) \leq H([a_0 : \cdots : a_d]) \leq 2^{d-1} \prod_{j=1}^{d} H(\alpha_j)$$

Proof **:**
Silverman p.230

Finally, let us show the height is invariant to the galois action, making the concept properly "geometric":

Theorem 5.4.12: Height and Invariance To Galois Action

Let $P \in \mathbb{P}^n(\overline{\mathbb{Q}})$ and $\sigma \in G_{\overline{\mathbb{Q}}/\mathbb{Q}}$. Then:

$$H(P^\sigma) = H(P)$$

Proof **:**
Let $P \in \mathbb{P}^n$, and $K/\mathbb{Q}$ be a field such that $P \in \mathbb{P}^n(K)$. The field K may not be Galois over $\mathbb{Q}$, but in any case σ gives an isomorphism $\sigma : K \xrightarrow{\sim} K^\sigma$, and σ identifies the sets of absolute

values of K and K^σ

$$\sigma : M_K \xrightarrow{\sim} M_{K^\sigma}, \quad \nu \mapsto \nu^\sigma.$$

If $x \in K$ and $\nu \in M_K$, then the associated absolute value ν^σ satisfies $|x^\sigma|_{\nu^\sigma} = |x|_\nu$. Then σ also induces an isomorphism $K_\nu \xrightarrow{\sim} K_{\nu^\sigma}$, so the local degrees satisfy $n_\nu = n_{\nu^\sigma}$. What's left is to compute:

$$\begin{aligned} H_{K^\sigma}(P^\sigma) &= \prod_{w \in M_{K^\sigma}} \max\{|x_i^\sigma|_w\}^{n_w} \\ &= \prod_{\nu \in M_K} \max\{|x_i^\sigma|_{\nu^\sigma}\}^{n_{\nu^\sigma}} \\ &= \prod_{\nu \in M_K} \max\{|x_i|_\nu\}^{n_\nu} \\ &= H_K(P) & [K:\mathbb{Q}] = [K^\sigma : \mathbb{Q}] \end{aligned}$$

as we sought to show.

Theorem 5.4.13: Finitely Many Points of Constant Height

Let $C, d > 0$ be constants, $d \in \mathbb{N}_{>0}$. Then:

$$\left\{P \in \mathbb{P}^n(\overline{Q}) \;\middle|\; H(P) \leq C \text{ and } [\mathbb{Q}(P) : \mathbb{Q}] \leq d\right\}$$

is finite

***Proof* :**
Sivlerman p.233

5.4.2 Heights on Elliptic Curves

Let f, g be real valued-functions on a set S. Then

$$f = g + O(1)$$

if there exists constants C_1, C_2 such that

$$C_1 \leq f(P) - g(P) \leq C_2 \qquad \forall p \in S$$

If only the lower equality is satisfied, $g + O(1) \leq f$, and if the upper equality is satisfeid then $f \leq g + O(1)$.

lots of super interesting build-up!!

Theorem 5.4.14: Mordell-weil Theorem

Let $E/\mathbb{Q}$ be an elliptic curve. Then $E(\mathbb{Q})$ is finitely generated

Proof **:**
here!

5.5 Torsion Points

Recall in theorem 4.3.2 that we have a bound on how far away the torsion points are from the ring R of the local field K. As any global field can be interpreted as injecting into a local field, we immediately get:

Theorem 5.5.1: Integrality of Torsion Points

Take $E/K, m$, with Weierstrass equation

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6,$$

and assume that $a_1, \dots, a_6$ are all in the ring of integers R of K. Let $P \in E(K)$ be a torsion point of exact order $m \geq 2$.

1. If m is not a prime power, then

$$x(P), y(P) \in R.$$

2. If $m = p^n$ is a prime power, then for each $\nu \in M_K^0$ we let

$$r_\nu = \left\lceil \frac{\operatorname{ord}_\nu(p)}{p^n - p^{n-1}} \right\rceil,$$

where $\lceil \cdot \rceil$ denotes the greatest integer. Then

$$\operatorname{ord}_\nu(x(P)) \geq -2r_\nu \quad \text{and} \quad \operatorname{ord}_\nu(y(P)) \geq -3r_\nu.$$

In particular, if $\operatorname{ord}_\nu(p) = 0$, then $x(P)$ and $y(P)$ are ν-integral.

Proof **:**
Immediate from theorem 4.3.2 by taking $K \hookrightarrow K_\nu$ for some nonarchimedean place ν.

We can in fact have some slightly weaker divisibility conditions when considering $K = \mathbb{Q}$ return to this

Corollary 5.5.2: Integrality of Torsion Points Over $\mathbb{Q}$

Let $E/\mathbb{Q}$ with Weierstrass equation

$$y^2 = x^3 + Ax + B \qquad A, B \in \mathbb{Z}$$

and let $P \in E(\mathbb{Q})$ be a nonzero torsion point. Then:

1. $x(P), y(P) \in \mathbb{Z}$
2. Either $[2]P = O$ or $y(P)^2 \big| 4A^3 + 27B^3$ (i.e. $y(P)^2 \big| \Delta/16$)

***Proof* :**
Silverman p.241

Example 5.5.1: Torsion Points using Both Methods
Let

$$E : y^2 + x^3 - 43x + 166$$

Then:

$$4A^3 + +27B^2 = 425984 = 2^15 \cdot 13$$

Thus, the y-coordinate of any torsion point in $E(\mathbb{Q})$ is of the form:

$$\{0, \pm 1, \pm 2, \pm 4, \pm 8, \pm 16, \pm 32, \pm 64, \pm 128\}$$

By some computations, it can be determined that we get points:

$$\{(3, \pm 8), (-5, \pm 16), (11, \pm 32)\}$$

next, as E has good reduction modulo 3, $E_{\text{tors}}(\mathbb{Q}) \hookrightarrow \tilde{E}(\mathbb{F}_3)$, and by direct computation we have $\#\tilde{E}(\mathbb{F}_3) = 7$, giving us a necessary condition for the size of the group. Now, using the doubling point formula on $(3, 8)$, we get:

$$x(P) = 3 \qquad x([2]P) = -5 \qquad x([4]P) = 11, \qquad x([8]P) = 3$$

Hence, $[8]P = \pm p$, and so P must have exact order 7 or 9 (and cannot have 3 since $x(P) \neq x([2]P)$). But we know that the order must be 7, $E_{\text{tors}}(\mathbb{Q})$ is a cyclic group of order 7 given by the above six points along with O.

notice that we had a torsion point of order 7. We can ask more generally if given a prime p does there exist an elliptic curve $E/\mathbb{Q}$ such that $E_{\text{tors}}(\mathbb{Q})$ has a point of order 7. As it turns out, there answer is no for all but finitely many primes! This is a rather surprising result, as we saw that over $\mathbb{C}$ we have primes of every order.

Theorem 5.5.3: Mazur's Classification of Torsion Subgroup Over $\mathbb{Q}$

Let $E/\mathbb{Q}$ be an elliptic curve. Then $E_{\text{tors}}(\mathbb{Q})$ is isomorphic to one of the following:

1. $\mathbb{Z}/n\mathbb{Z}$ where $1 \leq n \leq 10$ or $n = 11$
2. $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2n\mathbb{Z}$ where $1 \leq n \leq 4$

***Proof* :**
see cite:HERE.

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